



Psyche Mission Description and Design Rationale

Carol A. Polanskey¹ · Linda T. Elkins-Tanton^{2,3} · James F. Bell III² · Eleanor K. Alonge¹ · Sarah H. Bairstow¹ · Richard P. Binzel⁴ · Abhijit Biswas¹ · Luke Bury^{1,5} · Ernest Cisneros² · Dongsuk Han¹ · Insoo Jun¹ · William M. Klipstein¹ · David J. Lawrence⁶ · Timothy J. McCoy⁷ · Nickolaos Mastrodemos¹ · José M.G. Merayo⁸ · Sarah K. Noble⁹ · David Y. Oh¹ · Rona Oran⁴ · Carolyn A. Ortega¹ · Ryan S. Park¹ · Patrick N. Peplowski⁶ · Thomas H. Prettyman¹⁰ · Marc D. Rayman¹ · Jodie B. Ream⁴ · Thomas Roatsch¹¹ · Timothy M. Weise¹ · Daniel D. Wenkert¹ · Benjamin P. Weiss⁴ · Maria T. Zuber⁴

Received: 17 April 2025 / Accepted: 6 September 2025 / Published online: 14 October 2025
© The Author(s) 2025

Abstract

The Psyche spacecraft launched on October 13, 2023 to journey to the asteroid of the same name. Psyche is the largest M-class asteroid and possibly the remanent core of an early differentiated planetesimal that was disrupted by collisions. The Psyche mission will test that hypothesis as the 14th mission in NASA's Discovery Program. An alternative hypothesis is that the asteroid is unmelted primordial material. We describe the proposal competition process leading to selection of the mission and its context with other small body missions. This paper will briefly introduce the three science instruments, gravity science investigation, and Deep Space Optical Communications technology demonstration, leading into a detailed explanation of the science mission architecture. The orbital science phase is divided into a series of circular mapping orbits at four distinct altitudes, each selected to address specific science objectives. The requirements and objectives for each orbit are accompanied by an assessment of the effectiveness of each phase. We discuss the structure of the Psyche team during the operations phase along with the roles and responsibilities of the science and flight operations teams. Key elements of mission operations that are unique to the Psyche mission are provided. The Science Data Center manages and archives the Psyche mission data. The contents of the archive data sets for each instrument are outlined as well as the interfaces between the Science Data Center, the instrument teams, and the Planetary Data System.

Keywords Asteroids · (16) Psyche · Psyche mission · Science operations · Mission operations

1 Introduction

This paper provides an overview of the Psyche mission, a detailed exploration of the science implementation and mission operations, and context for the remaining articles in this special collection. Section 1 looks back to how the mission began, places it in context with other missions to asteroids, provides a description of the target asteroid (16) Psyche, and outlines

the mission science objectives. Section 2 provides a brief overview of the science instruments and investigations along with references to the Psyche instrument papers. Section 3 describes the approach to science planning, the driving constraints, and other considerations that influence the architecture of the science mission. Details of the Psyche science orbits and observation designs are expanded in Sect. 4. Section 5 describes the Psyche science and flight operations teams along with details of science and engineering operations, while Sect. 6 outlines Psyche science data archiving.

1.1 Mission Synopsis

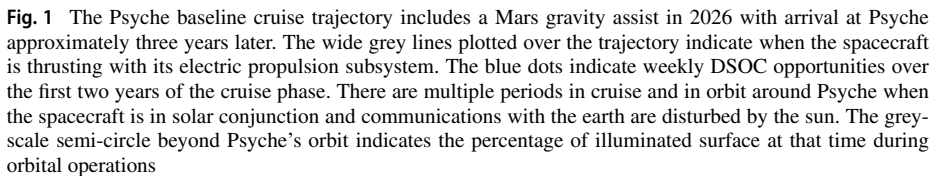
The Psyche spacecraft, launched on October 13, 2023, will visit the metal-rich, main-belt asteroid (16) Psyche (Dibb et al. 2024). The orbital science phase at Psyche, beginning in August 2029, follows a 5.6-year interplanetary cruise.

The spacecraft carries three science instruments: a Magnetometer, two redundant Multispectral Imagers, and a Gamma-Ray and Neutron Spectrometer (GRNS). Gravity science will be accomplished with the X-band high gain antenna (HGA) that also provides communications for the spacecraft, as well as the three low gain antennas (LGAs). A brief description of the instruments and science investigations is provided in Sect. 2 and in other articles in this collection (Weiss et al. 2023; Bell et al. 2025; Lawrence et al. 2025, and Zuber et al. 2022). The spacecraft also carries a high-rate laser technology demonstration, the Deep Space Optical Communications (DSOC) instrument (Biswas et al. 2024), which has already proven the feasibility of optical communications in the deep space environment at the distance of Mars and beyond. DSOC is a separate NASA project and was developed independently but in coordination with the Psyche project. DSOC is described further in Sect. 2.5.

The Psyche spacecraft launched from the Kennedy Space Center on a Falcon Heavy rocket provided by SpaceX. Figure 1 shows the interplanetary trajectory from launch through orbital operations. Each of the spacecraft subsystems and instruments were checked out and calibrated during the first 100 days of the mission. The solar electric propulsion system will continue to operate for most of the cruise phase until rendezvous with the asteroid at approximately 2.7 astronomical units (AU) solar range (see Sect. 5.3). The DSOC technology demonstration has periodic opportunities to communicate with their ground tracking stations during their prime mission, the first two years of Psyche cruise.

The spacecraft trajectory includes a gravity assist from Mars in May 2026. Prior to approaching Mars, there will be approximately 10 months when it is optimal for the trajectory to pause thrusting and coast. The only planned thrusting during this time is for a Mars Trajectory Correction Maneuver and two potential Mars Trim Maneuvers, which adjust the spacecraft's trajectory to ensure optimal arrival at the asteroid using the gravity assist. These maneuvers occur during the first half of the Mars gravity assist (MGA) phase, which spans from 80 days before the spacecraft's closest approach to Mars until 2 days after. The science instruments will be active during the coasting period around Mars, performing routine calibrations and acquiring data during Mars closest approach. None of the instrument activities during this period are required for mission success, but they will serve as a dry run for multi-instrument operations in orbit around Psyche and provide calibration data from a well-characterized target body.

The orbital phase is divided into four science orbit altitudes that accomplish specific mission objectives. Figure 2 illustrates the relationship between the science orbits. The highest altitude phase, Orbit A, provides the first opportunity to characterize Psyche. Imaging campaigns in Orbit B are designed to provide complete geologic and topographic maps of

 Springer

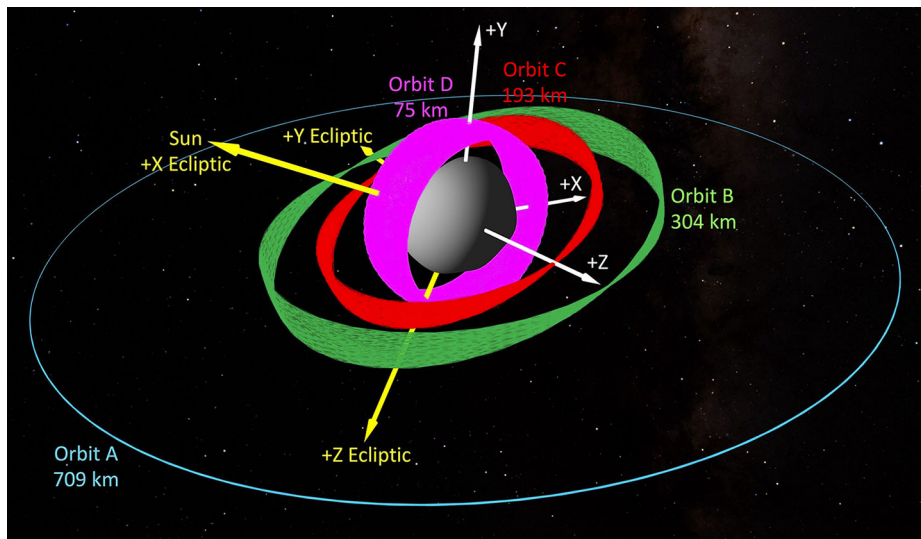


Fig. 2 The Psyche mission's four science orbit altitudes are designed to address the specific objectives of each of the four science investigations. The geometrical relationship between the sun, Psyche's spin axis, and the spacecraft orbital plane has important implications for placement of imaging campaigns on the mission timeline. Psyche's rotation axis (+Z) is tipped nearly into its orbital plane. At the time of this simulation, sunlight is coming from the left, and Psyche is at an equinox, so all parts of its surface are illuminated each rotation around its spin axis. At solstice, only half of the surface is illuminated. Orbits A, B, and C are polar orbits and Orbit D is inclined at 160° for stability. The three polar orbits enable global ground track coverage to support the mapping campaigns, although the extent of imaging coverage depends on Psyche's orbital season

1.2 Origins of the Psyche Mission

1.2.1 Origins

In 2011, Linda Elkins-Tanton and Erik Asphaug received emails from Bruce Bills and Daniel Wenkert at the Jet Propulsion Laboratory (JPL), asking whether they were interested in thinking about a mission proposal to test the hypotheses of a paper Elkins-Tanton, Benjamin Weiss, and Maria Zuber had published that year (Elkins-Tanton et al. 2011). Motivated by paleomagnetic measurements of the Allende carbonaceous chondrite from Weiss' group (Carpenter et al. 2011), the paper proposed the idea of partial differentiation for planetesimals: some planetesimals, in the first 2 Myr of the solar system, may have melted and differentiated into a metal core and rocky mantle in their interiors, but retained a colder, unmelted, primitive crust. The resulting planetesimal structure allows chondrites (primitive unmelted meteorites) to originate on internally differentiated parent bodies, from which they can record a magnetic field from the core dynamo (Weiss and Elkins-Tanton 2013).

Over the rest of 2011 and well into 2012, we discussed possible targets for this mission. We sought asteroid families, with the goal of detecting mantle cumulates, metallic core, and chondritic lid elements in the bodies, but no such family had been definitively identified, and the mission operations challenges of moving from one to another body in a family were too great. In 2012 we settled with some certainty upon the asteroid (16) Psyche, thought to be the largest metal asteroid by far.

Part of the challenge of proposing a mission is convincing a NASA Center to be your partner. We wanted to be included in the JPL portfolio for the upcoming Discovery proposal call, and we negotiated that process until well into 2013. The cost of supporting a proposal is significant, and Centers know they can only adequately support a certain number; strategically, choosing late when more information on each is available and proposals are more developed makes sense from the point of view of the Center but is risky for the proposal team, who has probably lost their chance to court another Center by the time the decision is made.

1.2.2 Preparation for Step 1 Proposal

From 2012 through 2013, the team participated in a series of Innovation Foundry exercises at JPL. The Foundry is JPL's team for ideating new concepts, and they bring to bear all the best practices in innovating and brainstorming, including using a room full of whiteboards, LEGO®, sticky notes, overhead projectors, and room to walk around.

A tension exists between the need to have experts in the room to make decisions about science objectives and which instruments can make measurements to answer them, and the need to wait until these are well-developed before selecting the specific instruments and inviting their leads to be Co-Investigators. Those very instrument leads are the top experts on what instruments can do and how to make the science interpretations, but once they are on the team, the decision is generally made that their instrument is going to fly. We purposefully kept our team as lean as was possible and attempted to avoid inviting any instrument providers before we knew we wanted their instrument.

The first payload concept for the mission was put together in January 2012, and the science team consisted of Elkins-Tanton, Asphaug, Weiss, Richard Binzel, Bills, and Wenkert. Our work was greatly enabled by Foundry members and others from JPL who joined the planning meetings. Later in 2012, as we continued with Foundry and did a first Team X planning session (see below), Bill Bottke, Simone Marchi, and Insoo Jun joined the team.

Finally, in spring 2013, JPL committed to the Psyche project and David Oh, electric propulsion expert and systems engineer, joined us as our Capture Lead, with the job to manage the proposal process.

In 2013, Carol Polanskey joined the team, along with Maria Zuber and Tom Prettyman. In May, we brought on Jim Bell as Deputy Principal Investigator. We led a session at the Lunar and Planetary Science Conference and held a Planetary workshop at the Carnegie Institution for Science that resulted in an edited volume of papers, *Planetesimals* (Elkins-Tanton and Weiss 2017), from Cambridge University Press on the state of the art of planetary science.

At this point, we had fairly closely defined five driving science objectives and were well into creating a Science Traceability Matrix. We knew we needed a magnetometer and imager, and we debated about the need to fly an ultraviolet/visible (UV/VIS) spectrometer; they were expected by the community at this time because the active missions of the day were all investigating rocky bodies with silicate crystalline minerals, well-suited for that kind of spectrometry. UV/VIS spectrometers are not, however, discriminating for metal. After careful debate we decided to fly a gamma-ray and neutron spectrometer instead, as it would be able to measure the compositions of the metal we expect to find. We were also forced to reconsider instrumentation because of a NASA ruling that required a certain percentage of the payload to be American, which further firmed our conclusion that this mission had to fly the simplest, highest-heritage payload possible to enable the science we had defined. At that time, we invited David Lawrence and Patrick Peplowski to join the team and provide the

GRNS. We eventually settled on a paradigm we have kept ever since: we have no descopes in payload. We have selected exactly the needed instrument suite.

It was clear that accomplishing the science objectives would require entering orbit at Psyche and remaining there for many months. Under the anticipated Discovery cost cap, we thought our only viable mission architecture candidate model was the Dawn mission, which had used its electric propulsion system to travel to and enter orbit around both (4) Vesta and (1) Ceres. However, there were key changes to be made. By 2013, components of Dawn's electric propulsion system had become functionally obsolete. Commercial electric propulsion systems used near-Earth offered the necessary performance and the benefits of strong flight heritage and a strong industrial base but were not formally qualified for interplanetary applications. NASA's follow-on developmental initiative at Glenn Research Center, the NASA Evolutionary Xenon Thruster ion thruster system, remained incomplete. A viable alternative needed to be found for Psyche.

To address this need, JPL issued a Request for Information seeking partners to provide an interplanetary spacecraft that used solar electric propulsion (SEP). A variety of solutions were proposed using both NASA and commercial electric propulsion systems, but one option that stood out above the others came from Space Systems/Loral (SSL, now Maxar Space Systems).

SSL was already an experienced provider of electric propulsion spacecraft for commercial communications satellites. It had flown over a dozen mission-unique electric propulsion spacecraft with solar power systems that generated over 20 kW—the same power level needed for Psyche. SSL had production volume—they regularly launched from four to seven product line spacecraft per year, and based on their commercial experience, they were uniquely willing to work on a firm-fixed-price contract. The potential benefit of such an arrangement for a cost-capped Discovery mission was clear. However, because SSL's commercial satellites all operated in Earth orbit, SSL lacked experience working on interplanetary spacecraft. Psyche would be an opportunity for SSL to benefit from NASA's experience working in deep space, while NASA would gain the benefits of SSL's experience building commercial SEP spacecraft.

At JPL's request, SSL proposed to build a SEP chassis at their production line in Palo Alto and deliver it to JPL for the completion of integration and testing. The SEP chassis would comprise the spacecraft's main composite structure, solar power system, propulsion systems, and thermal subsystem while the balance of the spacecraft, including the main computer, flight software, and deep space telecommunications transponder would be provided by JPL. The proposed architecture leveraged the strengths of SSL's experience building high-power electric propulsion for Earth-orbiting satellites with JPL's experience building autonomous software and fault protection for deep space missions. However, a potential disadvantage was that it would require an unprecedented level of coordination between two organizations which had never before worked together at a spacecraft level. This would be SSL's first time building a spacecraft traveling beyond lunar orbit, and because SSL's design comprised only part of the spacecraft, a further collaborative design iteration with JPL would be needed to fully understand the technical, cost, and management implications for the whole spacecraft.

1.2.3 Team X

This task was ideally suited to JPL's "Team X" concurrent engineering environment. Team X is a long-established part of JPL's Innovation Foundry that provides processes and infrastructure to bring together a diverse team of design specialists to work in a single location, sharing data in real-time to rapidly create an integrated point design of a space mission

(Sherwood and McCleese 2013). In March 2014, we invited SSL to participate in a joint design session at JPL's Team X facility in Pasadena. SSL sent a team of ~20 engineers to participate in two three-hour design sessions, held over a two-day period during which they worked side by side with JPL's engineering experts to complete the spacecraft design. For most participants on both sides, this was the first time they had ever interacted with their colleagues from the other organization. The combined team rapidly identified areas of synergy and created a point design for Psyche that incorporated heritage design elements from both SSL and JPL. The results showed that the SEP chassis architecture was technically feasible and that the cost of the combined spacecraft was appropriate for a Discovery class mission. More importantly, the design sessions showcased the ability of the JPL and SSL teams to work effectively together, which built confidence that the unique partnership could work despite the unique arrangement and lack of prior work experience. The results of this collaborative Team X design study were critical to the final decision to partner with SSL for Psyche's SEP chassis, which was made in April 2014.

1.2.4 Step 1 Proposal

A draft of NASA's Announcement of Opportunity for Discovery selection 13 was released on July 2, 2014, and we went immediately to work. JPL created an Excel spreadsheet of all the sections that were needed, and the team began to populate the sections with writers, editors, reviewers, and documentarians. We reached over 60 drafts of our Science Traceability Matrix before it was complete, and it has changed very little since Step 1. We wrote the 218-page Step 1 proposal and handed it to NASA in January 2015.

1.2.5 NASA Downselect, Phase A Proposal, and Selection

We received notification that Psyche had been selected for Phase A on September 30, 2015, followed by a written and verbal debrief of the Step 1 evaluation by NASA's Science, Technical, Management, and Cost (STMC) review board a month later. The bulk of the work on the Concept Study occurred over a nine-month period between November 2015 and July 2016. We submitted the study report to NASA in August 2016 and then held a site visit with the STMC review board three months later. The debrief report from Step 1 served as the primary roadmap used to prioritize work during Phase A. The report's findings included a list of major and minor "strengths" and "weaknesses," and although the report listed only a handful of major weaknesses, because this was a first-time evaluation for the mission concept it also listed 59 minor weaknesses, each of which had the potential to turn into a major weakness at the end of Phase A. Our key challenge was to maintain the strengths while addressing these weaknesses in the mission concept within the limited time and resources available for the concept study.

The strategy used in Phase A was the same we used in Step 1: support excellent science with an eminently feasible mission concept formulated to be as simple as possible. In addition to addressing weaknesses identified by the STMC and by the team itself, we sought to highlight the strength of the team (particularly in systems engineering) and to develop a straightforward implementation plan for the science mission and payload.

Technical work began with a series of subsystem level heritage reviews applied to all elements of the payload and spacecraft. These on-site reviews lasted two to four hours per subsystem. Face-to-face meetings were a key element of team building throughout the study period. For example, over 25 face-to-face meetings were held between JPL and SSL personnel during Phase A. These meetings built strong relationships that would endure through the

balance of the project. Out of the heritage reviews came a checklist of all technical work to be done in Phase A to address known weaknesses and establish the final technical baseline for the Concept Study Report.

In addition to conducting design work, we also conducted a series of risk reduction tests to mature critical elements of the Psyche spacecraft design. These tests included an end-to-end laboratory demonstration of the modifications needed to SSL's power system to operate Hall thrusters in deep space and a testbed demonstration of the compatibility of JPL and SSL's avionics interfaces. In addition, risk reduction testing of a demonstration model pulse tube cryocooler provided confidence that a new cooler design could reliably meet mission requirements for the Gamma-Ray Spectrometer instrument, resulting in a switch from a rotary cryocooler design to the pulse tube cryocooler during Phase A.

After the baseline design was declared "frozen" in April, we turned to the completion of the management and cost plans and the writing of the Concept Study Report itself. This involved multiple drafts, strategic editing sessions to "handcraft" text, and committee reviews conducted by JPL's Team X and by an excellence review committee. The final Concept Study Report, over 1000 pages long, was submitted to NASA for evaluation in August 2016.

Three months later, the evaluation team's site visit was conducted at SSL's facility in Palo Alto. Preparations took place over many weeks and included a mock site visit conducted with a stand-in board of independent reviewers three weeks before the real site visit. The mock site visit was critical to our preparations, as a particular challenge for the Psyche project management team was that none of us had ever previously participated in a site visit. In addition, because SSL had never previously worked for NASA as a prime contractor, most members of the STMC were not familiar with the company or its capabilities. Building NASA's confidence in SSL capabilities was an important goal for the site visit.

The work began in earnest a week before the presentation date, when we received written questions from the STMC and began the process of creating written and oral responses. We used the mock site visit to develop a seven-day workflow that engaged experts from across the team working around the clock to write responses. Each response was generated by a subject matter expert and then reviewed by two independent technical content reviewers and a technical editor before it was cleared for release. The workflow proved to be robust and produced over 150 pages of high-quality written technical material in the seven days prior to the final site visit. The oral presentations were similarly developed and dry run before a group of independent reviewers two days before presentation day and then were edited and cleared for final presentation to the STMC.

In the end, the site visit proved to be a valuable opportunity to demonstrate the strength of the team and to introduce the STMC and the members of the Discovery program office to SSL's manufacturing capabilities. A tour of SSL's spacecraft manufacturing facilities proved to be a highlight of the visit, as the visitors were able to view some 15 spacecraft under various stages of construction on SSL's factory floor. In addition, in feedback received after the site visit, the STMC stated that they could perceive the strength of the team, see the strong cooperation between team members, and appreciate the open communication philosophy embraced by the management team.

The final step of Phase A was for a subset of the team to travel to Washington DC and present for one hour to Thomas Zurbuchen, the NASA Associate Administrator. As part of our strategy for showing how ideal SSL was for prime partnership, the presentation was given primarily by Steve Scott from SSL and Lindy Elkins-Tanton as Principal Investigator.

The announcement that Psyche had won was made in January 2017. Many of the design decisions and outcomes of the build of the spacecraft have been described in other papers

(Oh et al. 2019, 2022, 2025a); the major change that our plan underwent was a launch slip from August 2022 to October 2023, caused by a variety of issues including building the spacecraft entirely during the COVID-19 pandemic of 2020–2021. During 2022, Ben Weiss from the Massachusetts Institute of Technology (MIT) replaced Jim Bell as Deputy PI, and Prof. Bell has stayed on as a highly valued Co-Investigator, lead of the Imagers, and lead of the science data center at Arizona State University (ASU).

1.3 Psyche's Place Amongst Small Body Missions

The minor bodies in our Solar System are a fascinating set of worlds that only become more interesting and varied as we visit them and expand our inventory of explored places. We continually work to categorize them—by size, by color, by location, by comparing them to the meteorites in our collections—but every world we visit has surprised us and often defied our expectations. Psyche will undoubtedly be no different.

Although humankind has trained their telescopes on asteroids for hundreds of years, our first close encounter with them came courtesy of Galileo and its flybys of (951) Gaspra and (243) Ida in 1991 and 1993, respectively (Veverka et al. 1994; Belton et al. 1996). Those images showed us little, rocky worlds, covered in regolith, cratered and weathered. Ida surprised us with a moon, Dactyl, though we have now come to understand that satellites and binaries are not uncommon among asteroid populations, and searching for them is standard procedure for asteroid missions.

The Near Earth Asteroid Rendezvous (NEAR) mission to (433) Eros in 2000 was our first chance to orbit and thoroughly explore an asteroid. We found Eros to be similar in many ways to Gaspra and Ida, all three are S-type asteroids of roughly similar size. But there were differences too. Space weathering effects on Gaspra and Ida appear similar to lunar weathering, with changes to both albedo and color evident in impacts that expose fresh material. Weathering on Eros, in contrast, results in large changes of albedo but almost no color variation, the reason for which is still a mystery (Pieters and Noble 2016).

Five years after NEAR visited Eros, the Japanese Hayabusa mission visited tiny (25143) Itokawa, an asteroid similar in composition to Eros, but vastly different in size, providing an opportunity to explore some of the consequences that size and gravity have on asteroid evolution. Hayabusa also returned a small amount of sample from Itokawa. These samples, along with the detailed chemistry obtained by NEAR, confirmed the link between S-type asteroids and ordinary chondrites (Veverka et al. 2000; Binzel et al. 2001; Nakamura et al. 2011). Despite this and other progress, linking asteroids to their parent bodies remains a challenge. The Psyche mission certainly faces that challenge; we have a large array of metal-rich meteorites in our collections, but none seems to perfectly match our understanding of Psyche so far.

The Dawn mission showed us the other end of the size scale, orbiting the two largest asteroids in the asteroid belt, protoplanet Vesta and dwarf planet Ceres. These two little worlds took very different evolutionary paths. Water-rich Ceres cooled fast and stratified into a rocky, clay-rich mantle and an ice- and hydrate-rich outer shell (Russell et al. 2016). In contrast Vesta, with little or no water, cooled slowly, differentiating into a core, mantle, and crust (Russell et al. 2012). Psyche will be the third largest asteroid we have visited and should further help us understand the different pathways planetesimals and planets follow as they form and evolve.

Our catalog of compositional differences is also growing, with Origins, Spectral Interpretation, Resource Identification, and Security – Regolith Explorer (OSIRIS-REx) and the Japanese Hayabusa2 missions, which have now visited - and successfully sampled - a couple

Table 1 List of asteroids that have been (white) or will soon be (grey) visited by spacecraft

Asteroid	Mission	Mean diameter	Class	Asteroid	Mission	Mean diameter	Class
Ceres	Dawn	946 km	C	Ida	Galileo	16 km	S
Vesta	Dawn	525 km	V	Steins	Rosetta	4.6 km	E
Psyche	Psyche	226 km	M	Donaldjohanson	Lucy	4 km	C
Patroclus	Lucy	113 km	P	Annefrank	Stardust	4 km	S
Menoetius	Lucy	104 km	P	Toutatis	Chang'e 2	2.5 km	S
Lutetia	Rosetta	100 km	M	Braille	Deep Space 1	1.6 km	Q
Eurybates	Lucy	64 km	C	Ryugu	Hayabusa2	0.9 km	C
Matilde	NEAR	58 km	C	Didymos	DART	0.78 km	S
Orus	Lucy	51 km	D	Dinkinesh	Lucy	0.7 km	S
Leucus	Lucy	40 km	D	Bennu	OSIRIS-REx	0.5 km	B
Polymele	Lucy	21 km	P	Apophis	OSIRIS-APEX	0.34 km	S
Gaspra	Galileo	17 km	S	Itokawa	Hayabusa	0.3 km	S
Eros	NEAR	16 km	S	Dimorphos	DART	0.16 km	S

of carbon-rich (C-type) targets, Bennu and Ryugu, respectively. Those samples are already providing a wealth of information and will continue to reveal their secrets for generations to come.

The recent Double Asteroid Redirection Test (DART) mission further added to our collection of imagery of S-type surfaces, including some dramatic close-ups of the surface of tiny Dimorphos (160 m diameter) as the impactor approached. This impactor, alongside results from Hayabusa2 and OSIRIS-REx, has shown that small rubble piles apparently have almost zero cohesive strength (Ferrari and Tanga 2022; Walsh et al. 2022).

Finally, as we look forward, our “sister” mission, Lucy (because it was also selected for flight in January 2017), will help us build up the diversity of our catalog listed in Table 1, conducting flybys of several Jupiter Trojans, a significantly different population from the main belt, and exploring an array of P-, D-, and C-types. Together, Lucy and Psyche, the first mission to orbit an M-class asteroid, will allow us over the next decades to make large strides in understanding the diversity of asteroids across the solar system.

1.4 History and Characteristics of (16) Psyche

Psyche was discovered at the Astronomical Observatory of Naples by Italian astronomer Annibale de Gasparis (1819–1892) on 17 March 1852 (de Gasparis 1852). Gasparis named the asteroid Psyche after the Greek mythological figure, the mortal wife of Cupid, and the personification of the Greek word for “the soul.” Psyche is the 16th asteroid spotted by astronomers since the discovery of Ceres some 50 years earlier and is one of eight found by Gasparis (Badolati 2007). It was one of the last minor planets to be assigned a planet-like iconic symbol, which is meant to resemble a butterfly wing under a star-like asterisk $\star$ (Paper Elemental Blog 2021).

Besides its orbital elements, little was known about the asteroid for more than a century. Starting in the 1960s and 1970s, a combination of advancements in telescopes and detector technologies, and renewed interest by space scientists in asteroids and their potential connections to meteorites, fueled new discoveries about Psyche. For example, optical and infrared radiometric, polarimetric, and photometric observations of Psyche revealed quickly

that it is a relatively rapidly rotating (spin period just over 4 hours), relatively large object (with an average diameter > 200 km) corresponding to about 1% of the mass of the Main Asteroid Belt (Gehrels 1970; Morrison 1974; Chapman et al. 1975). Early broadband multi-spectral observations revealed the asteroid to have featureless reddish visible-wavelength color properties that immediately suggested similarities with the colors of iron-nickel meteorites, earning it a prominent status among the M-class asteroids (e.g., Chapman and Salisbury 1973; McCord and Gaffey 1974; Zellner and Gradie 1976; Chapman 1976; Dollfus and Mandeville 1977; Tholen 1984; Bell et al. 1989).

Radar observations of Psyche also date back to the 1980s (e.g., Ostro et al. 1985; Magri et al. 2007; Shepard et al. 2008), with Psyche yielding one of the highest radar albedos among all measured main-belt and near-Earth asteroids with a value of 0.37 ± 0.09 (Shepard et al. 2017). The high radar albedo is arguably the strongest evidence suggesting that Psyche's surface and regolith is dominated by iron-nickel metal. Radar and ground-based adaptive optics and stellar occultation imaging observations have also helped to significantly refine the asteroid's size and shape, which is now known to be best approximated by Shepard et al. (2021) with a triaxial ellipsoid of approximate dimensions $280 \times 240 \times 170$ km (effective spherical diameter ≈ 222 km). However, significant concavities, likely related to the presence of large impact craters or basins, are also evident in the most current shape models (Shepard et al. 2021; Viikinkoski et al. 2018). Size estimates combined with mass estimates derived from orbital perturbations of the asteroid by other solar system bodies have enabled estimates of the density of Psyche, which is likely between 3400 and 4100 kg/m³ (see review by Elkins-Tanton et al. 2020), with the most recent estimate at 4172 ± 145 kg/m³ (Farnocchia et al. 2024), values that could be consistent with a mix of metals, silicates, and porosity rather than just an entirely metallic composition.

Additional evidence for a more complex composition, structure, and origin for Psyche comes from telescopic evidence for the presence of what are likely relatively small abundances of iron-bearing silicates (olivine and pyroxene) as well as hydrated mineral phases on the asteroid's surface, with progress in spatial and spectral resolution revealing surface heterogeneities (e.g., Binzel et al. 1995; Hardersen et al. 2005; Ockert-Bell et al. 2010; Sanchez et al. 2017; Takir et al. 2017).

There are significant remaining uncertainties in the physical properties, composition, and potential origin of Psyche, even though the most powerful modern instruments in ground- and space-based optical, infrared, and radar astronomy have been brought to bear on its study. The Psyche mission can now address these uncertainties with an up-close spacecraft investigation of this important but enigmatic small world.

1.5 Psyche Mission Goals and Science Objectives

The science requirements levied by NASA on the Psyche mission were refined from those developed from the proposal. To summarize at a high level, the mission objectives are to probe the history of Psyche and address the fundamental question whether Psyche is the remnant of a planetary core or has some other origin. Section 4 of this paper describes how the science data acquisition plan provides the means to satisfy these objectives within mission constraints.

The Psyche mission addresses two of the NASA Science Mission Directorate's Planetary Science Division goals and ties into the two most recent National Academies of Science and Engineering Planetary Science Decadal Surveys (National Academies of Sciences, Engineering, and Medicine 2011, 2022) with an investigation that has three broad goals:

Table 2 Relationship between the four science orbits and their science and operational goals. The goals of the Magnetometry Investigation span all orbits and orbit transfers but are called out only in Orbit C here

Orbit	Science and operational goals	Science measurements	Instrument
A	Characterize Psyche's gravity field, shape, and spin axis to enable navigation to lower science orbits Determine the relative ages of regions of Psyche's surface by counting craters with diameters larger than 1 km	Doppler and ranging Global multispectral imaging Off-nadir imaging Magnetic field monitoring	HGA and LGA Imager Magnetometer
B	Characterize Psyche's topography and morphology Identify silicate regions on Psyche's surface Search for oldhamite on Psyche's surface Develop a geologic map of Psyche's surface	Global nadir and off-nadir imaging Global multispectral imaging Magnetic field monitoring	Imager Magnetometer
C	Determine Psyche's gravity field Characterize Psyche's surface density variations Refine Psyche's shape model Determine whether Psyche has a remanent magnetic field (applies to all science orbits)	Doppler and ranging Global nadir and off-nadir imaging Magnetic field monitoring	HGA and LGA Imager Magnetometer
D	Discriminate between achondritic and chondritic silicates on Psyche's surface by measuring Fe, Si, K, Ca, and Al Determine the global average weight percent of nickel over Psyche's surface Determine the metal-silicate ratio of Psyche's surface	Gamma-ray and neutron measurements over 70% of Psyche's surface Magnetic field monitoring	GRNS Magnetometer

1. Understand a previously unexplored building block of planet formation: iron cores.
2. Look inside the terrestrial planets, including Earth, by directly examining the interior of a differentiated body, which otherwise could not be seen.
3. Explore a new type of world. For the first time, examine a world not made of rock or ice, but of metal.

The goals were further refined into the five science objectives for the mission:

1. Determine whether Psyche is a core, or if it is primordial unmelted material
2. Determine the relative ages of regions of Psyche's surface
3. Determine whether small metal bodies incorporate the same light elements into the metal phase as are expected in the Earth's high pressure core
4. Determine whether Psyche was formed under conditions more oxidizing or more reducing than Earth's core
5. Characterize Psyche's morphology.

The science objectives became the basis of the high-level science requirements of the mission, which were then transformed into a set of science measurement requirements. These measurement requirements drove the design of the instruments described in Sect. 2 and the orbital operations architecture of the mission found in Sect. 4. Based on the measurements and the instrument suite, the mission science was divided into four orbital phases. Table 2 provides a summary of how the science requirements map to the science orbit altitudes and measurements. Each science objective is shepherded by a specific working group within the Psyche team, and each is explored in depth in the accompanying papers by Elkins-Tanton

Table 3 List of science instruments and basic attributes

Science instrument	Magnetometer	Multispectral Imagers	Gamma-Ray and Neutron Spectrometer
Provider	Technical University of Denmark (DTU)	Malin Space Sciences Systems, Inc. for Arizona State University (ASU)	Johns Hopkins/Applied Physics Laboratory (APL)
Heritage	European Space Agency (ESA) Swarm	Mars Science Laboratory Mastcam; Mars 2020 Mastcam-Z; Mars Reconnaissance Orbiter MARCI	MESSENGER Lunar Prospector
Components	2 sensors 2 electronics boxes	2 camera heads 1 electronics box	2 instruments 2 electronics boxes
Mounting	On a dedicated 2-m boom	—X direction	On a dedicated 2-m boom
Mass	3.635 kg	5.12 kg	21.42 kg
Power	4.4 W	5.6 W standby and 8.7 W imaging, per camera	69.1 W (peak)
Specifications	Sampling frequency: 50 Hz Measurement uncertainty requirement at system level: < 1.5 nT/axis Range: $\pm 80,000$ nT Noise: 25 pT _{RMS} Resolution: 11 pT Thermal stability: <0.25 nT in operating temperature range [−125 °C, +60 °C]	1 broadband polychromatic filter and 7 narrowband color filters Field of view (FOV): $4.6^\circ \times 3.4^\circ$ Instantaneous field of view (IFOV): 50 μ rad pixel scale Focal ratio: $f/2.9$ Focal length: 148 mm	Gamma-ray energy range: 60 keV to 9 MeV Energy resolution: 1.92 keV @ 1333 keV (ground); 2.09 keV @ 1333 keV (in space)

et al. (2022), Marchi et al. (2022), Prettyman et al. (2025), McCoy et al. (2022), and Jaumann et al. (2022).

2 Psyche Instrumentation

As described in Sect. 1.2, the minimum set of three instruments, the Magnetometer, the Multispectral Imager, and the GRNS, plus X-band gravity science, were selected to meet mission science requirements. A duplicate Multispectral Imager was added for redundancy as that instrument is also needed for optical navigation. Details on each of the science instruments are provided in subsequent articles of this collection (Weiss et al. 2023; Bell et al. 2025; Lawrence et al. 2025; and Zuber et al. 2022), but a brief description is provided below as context for the flight operations plans. Table 3 summarizes their characteristics and heritage while Fig. 3 shows the layout of the instruments on the Psyche spacecraft.

While our goal was to fly the simplest, highest-heritage payload possible to complete the science portion of the mission, the NASA Discovery 2014 Announcement of Opportunity offered an incentive for adding a technology demonstration opportunity to the payload. DSOC was chosen from the options provided because the Psyche interplanetary trajectory enabled demonstration of laser communications where NASA is intending to use it: in the vicinity of Earth, at the distance of Mars, and beyond Mars orbit. DSOC is not required or intended for use during the Psyche baseline orbital mission; however, the spacecraft flight software supports transmission of both engineering and science downlink telemetry

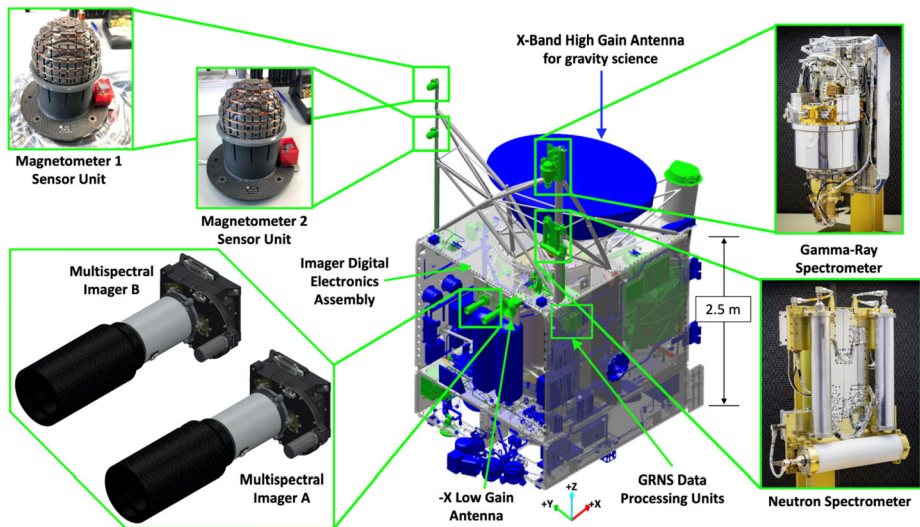


Fig. 3 Placement of science instruments on the spacecraft. The two Magnetometer sensor units are located on a 2-m boom with their electronics units (not shown) in a gradiometer configuration. The Gamma-Ray Spectrometer and Neutron Spectrometer are located on a separate 2-m boom with the Gamma-Ray Spectrometer at the 2 m point and the Neutron Spectrometer at the 1 m point of the boom. Each has its own electronics box mounted inside the spacecraft. The two Multispectral Imagers are mounted on the central cylinder of the spacecraft with views out of the $-X$ direction of the spacecraft. Their electronics are packaged into a single box mounted inside the spacecraft. Gravity science is performed with the X-band HGA as well as three LGAs mounted on the $+X$, $-X$, and $-Z$ decks of the spacecraft (not all are visible here). Blue components on the spacecraft drawing are provided by Maxar. Green components are from JPL or other partner institutions

via DSOC. Transmission of spacecraft engineering telemetry by DSOC was successfully demonstrated during cruise.

2.1 The Magnetometry Investigation

The Magnetometry Investigation (Weiss et al. 2023) is motivated by the mission science objective of determining whether asteroid Psyche is a differentiated body containing a metallic core or is a primordial unmelted aggregate (Elkins-Tanton et al. 2020). It addresses this objective by seeking to constrain the remanent magnetization of Psyche. Sufficiently strong magnetization would provide evidence that Psyche cooled and/or was aqueously altered in the presence of an ancient magnetic field. Remanent magnetization will be constrained by measuring the ambient magnetic field and in particular searching for an intrinsic field from remanent magnetization.

The Magnetometer consists of two identical sensor units positioned at 1.45 m and 2.15 m away from the spacecraft along a boom. Each sensor unit is connected by a harness to its own electronics located on the spacecraft bus. The two sensor units each measure the three components of the magnetic field. In combination, these enable a measurement of the gradient of the field, which will be used to characterize and suppress the contributions from the spacecraft magnetic field (Ream et al. 2022). Each sensor unit acquires measurements over 0–50 Hz, has a dynamic range of $\pm 80,000$ nT, and a required $3\text{-}\sigma$ instrument uncertainty of 0.22 nT per axis. The Magnetometry Investigation is led by the Massachusetts Institute of Technology, and the Magnetometer is provided by the Technical University of Denmark.

Table 4 Psyche Imaging science measurement requirements driving orbit design

Imager physical parameter measured	Pixel scale	Surface coverage	Target orbit	Target pixel scale
Mapping craters > 1 km diameter (1 Imager filter)	200 m	50%	A	35 m/pix
Metal-silicate ratio map (4 Imager filters ≥ 700 nm)	500 m	80%	B	15 m/pix
Mapping oldhamite on the surface (3 Imager filters)	200 m	80%	B	15 m/pix
Mapping topography to support gravity determination (1 Imager filter, ≤ 70 m height scale)	70 m	80%	B	15 m/pix
Mapping for topography (1 filter, ≤ 50 m height scale)	20 m	50%	B	15 m/pix
Geologic mapping (1 Imager filter)	20 m	80%	B	15 m/pix

The Magnetometer derives high heritage from the ESA Swarm Vector Field Magnetometer (Merayo et al. 2008).

The Magnetometer operates using the fluxgate principle in which a high-permeability toroidal core is cycled in and out of magnetic saturation by the application of a time-variable current applied to windings around the core (Primdahl 1979). An ambient magnetic field can be detected because the core will spend a larger fraction of each magnetization cycle saturated in one polarity compared to the other.

2.2 The Multispectral Imager Investigation

The Psyche Multispectral Imager (“Imager”; Bell et al. 2025) includes two redundant camera heads, each paired with their own electronics; the Imager is used to characterize the geology and topography of Psyche’s surface and to provide some constraints on the asteroid’s surface mineralogy. The cameras protrude from the –X panel of the spacecraft with a toe-out angle of 3.7° between them.

Major goals (Table 4) of the Imager Investigation include:

- (1) Characterizing the geology and topography of a metal-rich world to reveal potentially unique impact, erosional, weathering, and possibly even tectonic and/or volcanic processes that could be characteristic of this class of small body
- (2) Characterizing the relative ages of Psyche’s surface regions by mapping statistically significant numbers of impact craters across the asteroid’s surface at diagnostic spatial scales
- (3) Deriving topographic maps of the asteroid’s surface from digital terrain models to assist with crater counting and shape modeling, and to provide insights into internal structure and homogeneity via comparisons with lower spatial resolution gravity field, magnetic, and neutron/gamma-ray compositional measurements
- (4) Assessing the relative spatial abundances of iron-bearing silicates and metals on the surface of the asteroid, to help constrain the evolutionary history of the surface and to provide potential connections to specific meteorite classes
- (5) Searching for the presence of specific diagnostic sulfide minerals like oldhamite [(Ca, Mg, Fe)S] and/or troilite (FeS) that could provide information on oxidizing or reducing conditions predicted by differentiated versus primitive hypotheses for the asteroid’s origin.

These are first-order exploration goals and questions, and the Imager will directly address them by providing both polychromatic (clear) filter imaging and narrowband multi-spectral filter imaging (see Bell et al. 2025 and Winhold et al. 2025) at smaller spatial and vertical (topographic) resolutions as the orbital altitudes decrease during the mission. Geologic and topographic imaging goals will primarily be met using the polychromatic filter (540 ± 125 nm), which can acquire high spatial resolution (Table 4), high signal-to-noise ratio ($\text{SNR} > 100$) images at short enough exposure times to avoid target motion smear. Imaging goals related to mapping of iron-bearing silicates and the silicate/metal ratio will be achieved using a combination of short-wave near-infrared filters that provide continuum (722 nm) and in-band (840, 942, and 1014 nm) imaging at $\text{SNR} > 50$ designed to help detect and constrain the abundances of iron-bearing phases like pyroxene and olivine via mapping of the so-called “1- μm band.” Imaging goals related to detecting and potentially discriminating between sulfide phases will be achieved using imaging also at $\text{SNR} > 50$ with short-wave visible filters (442, 494, and 548 nm) positioned to provide maximum sensitivity to known sulfide absorptions in that wavelength region.

The Imager camera heads are powered and controlled by a separate Digital Electronics Assembly (DEA) that can compress the images via lossless or lossy compression algorithms and that can store more than 4000 images per camera in its own internal memory. The DEA is powered by and interfaces with the spacecraft’s main computer to command the camera heads and to control the transfer of locally-stored images to the spacecraft for eventual downlink.

2.3 The Gamma-Ray and Neutron Spectrometer Investigation

The Psyche GRNS instrument (Lawrence et al. 2025) includes two sensor subsystems—a Gamma-Ray Spectrometer (GRS) and a Neutron Spectrometer (NS)—and measures elemental abundances of Ni, S, K, Si, Fe, Ca, and Al, as well as Psyche’s metal-to-silicate fraction. Since the Psyche mission science objectives require measurement of Ni abundances, this measurement requires use of a high-purity Ge (HPGe) sensor due to its excellent energy resolution and detection sensitivity. In particular, the primary Ni gamma-ray line at 1.454 MeV is very close in energy to a ubiquitous K gamma-ray line at 1.46 MeV and therefore needs the energy resolution provided by HPGe technology. The excellent performance of HPGe technology also enables the measurement of other elements such as Al, Ca, and S.

The Psyche GRS is an evolution of the MErcury Surface, Space ENvironment, GEOchemistry, and Ranging (MESSENGER) GRS (Goldsten et al. 2007) that incorporates various lessons learned from MESSENGER experience (Burks et al. 2020). The GRS sensor is a 5-cm-long by 5-cm-diameter cylindrical HPGe crystal. To properly operate, the Ge crystal needs to be cooled to cryogenic temperatures. To facilitate this cooling by thermally isolating the sensor, the Ge sensor is enclosed in a cryostat designed and built by Lawrence Livermore National Laboratory. The sensor is cooled to its operating temperature of 85 K using a long-life, pulsed-tube cryocooler built by Lockheed Martin Corporation. As with prior GRS systems, the Ge sensor is surrounded by a boron-loaded anticoincidence shield. The anticoincidence shield serves three functions. First, the anticoincidence shield provides active background rejection for non-gamma-ray signals (e.g., charged particles) that occur in the Ge sensor. Second, the boron loading in the scintillator enables the measurement of fast neutrons, which is a key measurement for characterizing Psyche’s metal-to-silicate fraction. Finally, energy deposition measurements in the anticoincidence shield provide a direct monitor of the time varying galactic cosmic-ray flux, which is a normalization factor needed for deriving accurate elemental abundances.

The NS consists of three ^3He gas proportional sensors that use different material coverings to discriminate three separate neutron energy ranges. Thermal (neutron energy $E_n < 0.4$ eV) and low-energy ($0.4 < E_n < \sim 10$ keV) epithermal neutrons are measured using bare and Cd-covered sensors. High-energy ($E_n < 100$ keV) epithermal neutrons are measured with a polyethylene-covered ^3He sensor. This energy band is used to more effectively characterize the expected metal-rich content of Psyche (Yokley et al. 2019; Lawrence et al. 2020, 2021, 2025).

To reduce background from spacecraft-originating gamma rays and neutrons, both sets of sensors are mounted on a boom to physically separate the sensors from other spacecraft materials.

Finally, each sensor subsystem is operated by separate data processing units, which provide capabilities for low and high voltages, command processing, autonomous fault protection, signal processing, and data packet generation. The data processing units are located in the spacecraft body and are connected to the sensor subsystems via harnesses approximately 1–2 m in length.

2.4 The Gravity Science Investigation

The Gravity Science Investigation aims to map the gravitational variation of asteroid Psyche to provide constraints on its interior structure (Zuber et al. 2022). The gravity field will be measured via the spacecraft's X-band telecommunication system, which is capable of two-way radio link (7.2 GHz uplink and 8.4 GHz downlink) between the Deep Space Network (DSN) on Earth and the spacecraft, allowing high-precision range and Doppler measurements. The spacecraft trajectory is sensitive to gravitational perturbations, thus, accurately tracking the velocity of the spacecraft allows determination of Psyche's gravity field. The performance of the Doppler data is expected to be better than 0.1 mm/s at 60-s integration time and the performance of the range data is expected to be 1–2 m (Konopliv et al. 2014; Park et al. 2016). The Psyche spacecraft is also equipped with a camera that can be used to determine the shape and landmarks via stereophotoclinometry (Park et al. 2019). The Doppler and imaging data are primary for recovering the gravity field and the spacecraft orbit relative to the asteroid while the range data are primary for determining the heliocentric orbit of Psyche (Park et al. 2016, 2020).

The Psyche mapping plan has four orbital phases (see Sects. 3 and 4). For gravity science, Orbit C is the prime phase with the orbit altitude of ~ 190 km. During the Orbit C phase, radio tracking data will be acquired for the equivalent of at least 35 days, which is expected to be sufficient for a global sampling of Psyche's gravity field (Zuber et al. 2022). Gravity data will also be acquired in the lower-altitude Orbit D phase, during which time measurements of surface composition are prioritized.

The radio and imaging observables will be processed using JPL's MONTE (Mission Analysis, Operations, and Navigation Toolkit Environment) orbit determination software, which employs a least-squares approach to estimate gravity field (Evans et al. 2018). MONTE has been utilized and demonstrated in a similar capacity in numerous other planetary gravity investigations and is capable of processing all of Psyche's gravity observations (Park et al. 2016, 2020).

A detailed gravity simulation based on realistic modeling, measurement scheduling, and expected data performance shows that a degree and order 10 gravity field can be recovered (Zuber et al. 2022), which is equivalent to spatial block size, or half-wavelength, average resolution of ~ 36 km. Combining the gravity data with the shape, magnetometry, and compositional data will allow probing the interior structure of Psyche. Specifically, techniques

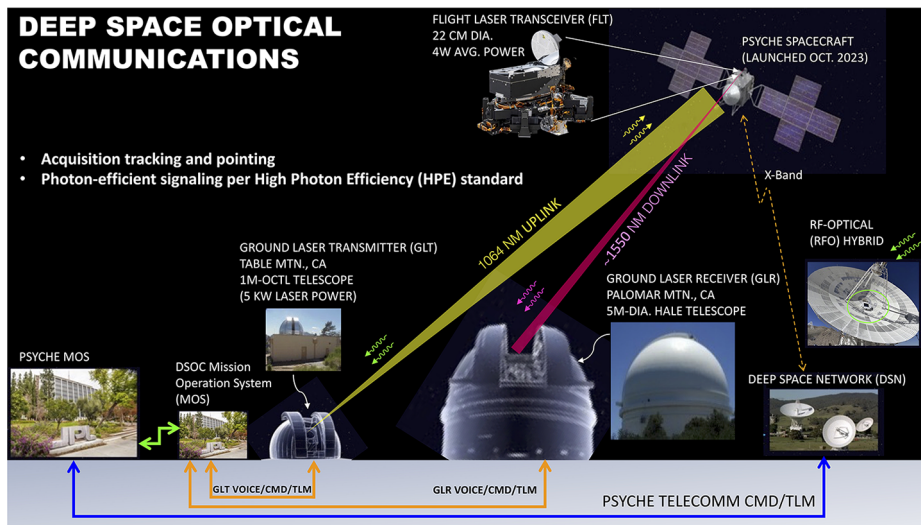


Fig. 4 DSOC technology demonstration architecture showing the FLT on board the Psyche spacecraft, the ground laser transmitter and ground laser receiver on the ground with the DSOC mission operations system for coordinating operations. Coordination between the flight and ground assets in addition to the Psyche project was key to DSOC project success

(Goossens and Smith 2023) developed to analyze gravity data acquired in the NASA GRAIL mission (Zuber et al. 2013) will allow recovery of lateral density variations with depth in spatial areas (cf. Goossens et al. 2025). The mass estimate from gravity data and the shape estimate from imaging data will provide the bulk density of the asteroid. The degree-2 gravity field will provide the gravitational flattening and spin axis precession, related to the radial distribution of mass. Correlation of the gravity field with the surface data (e.g., topography, magnetic field, and composition map) will provide mass, and potentially, structural variations (Park et al. 2020).

2.5 Technology Demonstration: Deep Space Optical Communications

The DSOC project provides the first technology demonstration of free-space optical communications from planetary distances (Biswas et al. 2024). Validating the free-space optical communications technology will enable an order of magnitude increase in data rates from future NASA science and human exploration missions (Deutsch 2014).

Figure 4 shows the operational architecture for the DSOC technology demonstration. The three key elements include: the flight laser transceiver (FLT) accommodated on the Psyche spacecraft; the ground laser transmitter, integrated to the Optical Communications Telescope Laboratory located at JPL's Table Mountain Facility near Wrightwood, CA; and the ground laser receiver integrated to the 5-m diameter Hale telescope at the Palomar Observatory, Palomar Mountain, CA. An optical link is initiated when the ground laser transmitter points a 1064 nm beacon laser at the FLT while the spacecraft "coarsely" points to Earth. The FLT searches out the laser beacon signal by scanning over the spacecraft-pointing uncertainty region. Upon acquisition the FLT actively tracks the beacon signal establishing a spatial reference used to point back the 1550 nm downlink laser, to the ground laser receiver. The DSOC mission operations system shown in Fig. 4 coordinates activities between the DSOC space and ground assets and Psyche mission planning.

The Psyche spacecraft trajectory is shown in Fig. 1 with blue dots to indicate the DSOC contacts during the first two years of cruise. As the Palomar observatory cannot operate in the daytime, DSOC operation must be suspended when the Psyche spacecraft's line of sight to ground laser receiver occurs during that period. Optical link operations occurred in the pre-conjunction phase (November 2023–July 2024) and resumed in the post-conjunction phase in early December 2024.

Pre-scripted DSOC operations sequences are uploaded to the spacecraft in advance. Sequence execution assisted by spacecraft avionics and FLT electronics interaction occurs on a pre-defined schedule. Ground activities at the ground laser transmitter, ground laser receiver, and the mission operations system are coordinated to the schedule unless sequence execution is aborted in advance due to predicted weather outages. On the few occasions when weather outages occur without advance prediction, the FLT continuously searches for the beacon signal until the end of the sequence.

Downlink data-rates to the ground laser receiver vary from 267 Mbps to 8 Mbps, from 0.175 AU to 2.68 AU using 2 W of average 1550 nm laser power. Low-rate uplink communications at 1.8 kbps was achieved out to 3.3 AU with the upper bound of ground uplink 1064 nm laser power being 3 kW. In the spirit of a true technology demonstration, comprehensive system telemetry is gathered in space and on the ground to learn how to operate future planetary optical communications services from deep space.

A significant departure of DSOC from previous free-space optical communications demonstrations is the substantial increase in distance. For example, NASA's Lunar Laser Communication Demonstration Project established 622 Mbps downlink in 2013 (Borson et al. 2014), from lunar range (0.00267 AU). Defining link difficulty as the product of data-rate in Mbps $\times$ squared-distance, AU², indicates more than a 1000-fold increase, or 60 dB increase, in difficulty for achieving 1 Mbps data rate. The DSOC project implemented several new technologies that enabled photon-efficient signaling combined with microradian accuracy laser beam pointing so that link difficulties up to 113 have been achieved to date.

Pulse position modulation and serially concatenated pulse position modulation error correction codes (Moision and Hamkins 2003; CCSDS Blue Book 2019) are used by DSOC for photon-efficient communications. A high peak to average power space laser transmitter (Dailey et al. 2019) coupled to 22 cm diameter silicon carbide optics (Driscoll et al. 2020) is used to transmit the downlink data-modulated laser beam. Flight electronics with the JPL legacy universal space transponder (Pugh et al. 2017) serve as a modem for generating the laser modulation signal and host the flight software and firmware for operating the FLT.

On the ground, a superconducting nanowire single photon detector array (Wollman et al. 2024) at the focus of the Hale telescope is used to detect the laser pulses. The electrical output from the array is time-stamped with a time-to-digital converter front-end signal processing assembly (Rogalin et al. 2021; Srinivasan et al. 2017). Subsequent signal processing of time stamps achieves temporal synchronization, parameter estimation, de-interleaving, and decoding to extract the information codewords.

The multi-beam uplink laser assembly has 8–10 lasers transmitted through sub-apertures of the 1 m Optical Communications Telescope Laboratory telescope. It is blind pointed to the spacecraft using spacecraft ephemeris provided by the Psyche team. Multi-beaming partially mitigates atmospheric turbulence-induced fades through incoherent averaging of the atmospheric turbulence-induced power fluctuations. The mean irradiance delivered at the FLT aperture is typically about 10 pW/m². This irradiance suffices for acquisition tracking and uplink communications.

The optical transceiver assembly (OTA) is mounted on struts equipped with magnetic actuators and sensors. When powered, the struts “float” or isolate the OTA from base disturbance. This allows 6-degrees-of-freedom control to search and acquire the beacon signal on

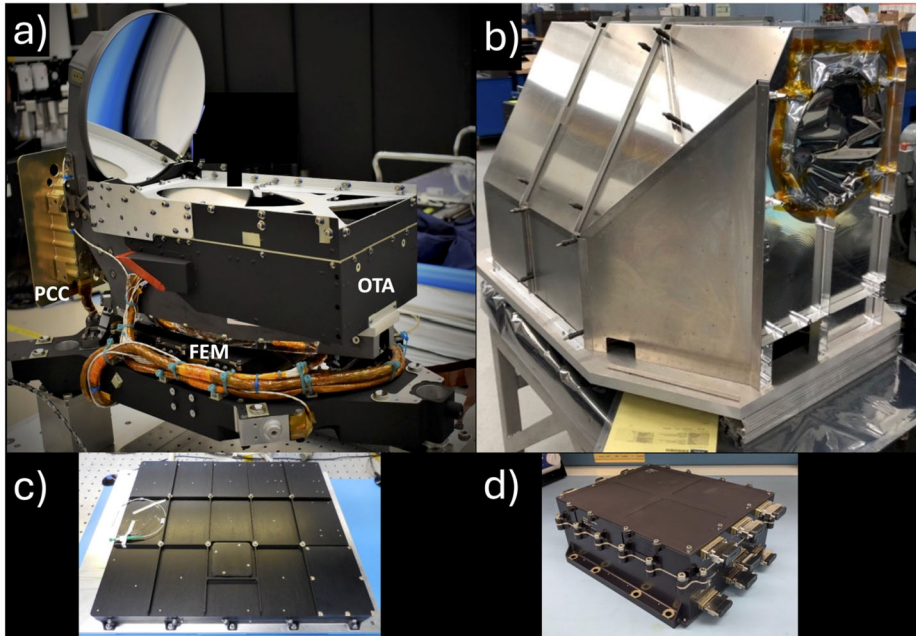


Fig. 5 (a) Optical Transceiver Assembly (OTA) with photon-counting camera and floating electronics module integrated; (b) DSOC accommodation kit for accommodating OTA; (c) Laser transmitter assembly mounted separately on spacecraft panel and fiber coupled to OTA; (d) Stationary electronics module interfaces to spacecraft electronics and houses flight software and firmware

the single-photon-sensitive photon-counting camera (Buck et al. 2020). Upon acquisition of signal, a low bandwidth control loop maintains spatial and temporal “lock” onto the beacon laser spot. The pointing reference established by the “lock” allows the onboard-computed point-ahead angle of the downlink beam to be implemented. An electronically controlled piezo-electric point-ahead mirror in the OTA transmit path implements the point-ahead angle. Figure 5 shows a photograph of the DSOC FLT.

The FLT operating from space is pointed to Earth with relatively small (tens of degrees) angular separation from the sun. The resulting stressful thermal environment is partially mitigated by a DSOC accommodation kit that is comprised of an interface plate that mates to the spacecraft and an enclosure for thermal management of the OTA. One-time deployable launch locks are integrated to the interface plate of the DSOC accommodation kit. A latching mechanism is used to stow the OTA when it is unpowered.

3 Planning, Constraints, and Considerations for Science Orbital Operations

After selecting a planetary target and a suite of instruments, the next step was to design an architecture for the science observations. The observation strategy for the Psyche mission was inherited from the Dawn mission to Vesta and Ceres. Dawn had demonstrated the use of SEP to efficiently observe a massive body with unknown size, gravity, and other physical characteristics by systematically mapping the body’s surface without regard to individual

surface features or characteristics (Polanskey et al. 2011). Having successfully developed and executed a science observing campaign for Vesta and Ceres (Polanskey et al. 2012, 2014, 2016), the same observation strategy was applied to Psyche, despite the project having a different spacecraft provider, SEP technology, and instrument suite (Polanskey et al. 2018).

Psyche is among the ten most massive objects in the main asteroid belt, so uncertainty as to Psyche's gravity plays a key role in planning. Without data from previous missions or even flybys of Psyche, the project must observe size, shape, and gravity during approach and feed that information into the design of the spacecraft trajectory without disrupting the pre-defined science observation plans. This is accomplished by maintaining the orbit resonance (the number of Psyche rotations over the number of spacecraft orbits) with emphasis on preserving the orbit period when redesigning the spacecraft orbit for a new asteroid shape model. In addition, hours or days of margin are added to the start time of each science orbit campaign to accommodate uncertainties in the time needed to thrust from one orbit altitude to the next (see Sect. 5.3). Therefore, the observing plan must also be responsive to changes in orbit timing, particularly since the files of sequenced commands that execute the plan on the spacecraft are designed in advance, well before the spacecraft arrives at Psyche. Anchoring each orbit's data acquisition to the geometric epoch when the spacecraft crosses from the dark (night) side to the lit (day) side of the body provides a straightforward mechanism to update observing plans to changing conditions. The strategy outlined in this section is proactive, rather than reactive, to the challenges posed by a low thrust mission orbiting a massive asteroid for the first time, one that may have significant mass density contrasts in its interior.

The observing community on Earth is using the latest techniques to improve our knowledge of the asteroid before the spacecraft's arrival. While the measured dimensions from each study vary, all models agree that Psyche is irregularly shaped. The irregular shape can impact observation strategies and orbit stability. The Psyche flight team has already gained experience adapting to new estimates of Psyche's size and shape over the course of the mission. For the Discovery proposal, the Kaasalainen et al. (2002) shape was used as the baseline model for trajectory planning, but by the time the mission was selected, a new model had been published by Shepard et al. (2017). The Shepard model was adopted as the baseline for Phase B and new science orbits were redesigned to the same resonances, requiring no significant changes to transition the observation plans. This exercise provided an early validation of the team's ability to maintain the original operations plan while adapting to the latest shape model. By 2020, there were additional shape models published by Hanuš et al. (2017) and Viikinkoski et al. (2018), so the project performed a sensitivity study to gravity, shape, and spin axis orientation. Through these bounding cases, the project demonstrated that updated orbit parameters could be derived that met all requirements and preserved the observation plans. The baseline shape model was changed to Shepard et al. (2021) when the mission replanning began for the 2023 launch date. Table 5 lists the current baseline physical characteristics and Fig. 6 shows the selection of recent Psyche shape models evaluated.

The following subsections describe the key aspects that influence the development of the architecture of the Psyche science mission.

3.1 Functional Redundancy and Heritage Approach to Observing

If there is anything given in spacecraft operations, it is that plans will change, and data will be lost. Data can be unavailable either because it was never acquired, transmission to Earth failed, or it was obtained under conditions that make it unsuitable for scientific analysis. To ensure that the science objectives of the mission are met, the Psyche science plan over-samples the dataset needed for each investigation by collecting data that are functionally

Table 5 Psyche baseline characteristics used for mission and science planning

Characteristic	Value (Shepard et al. 2021)
Effective diameter	222 (−1/+4 km)
Dimensions a, b, c	278 (−4/+8 km), 238 (−4/+6 km), 171 km (−1/+5 km)
Pole (was EMO2000)	right ascension, declination (36°, 6°) ± 2° EME2000
Optical albedo	0.16 ± 0.01
GM	1.601 ± 0.017 km ³ s ^{−2} ; bulk density 4172 ± 145 kg m ^{−3} (Farnocchia et al. 2024)

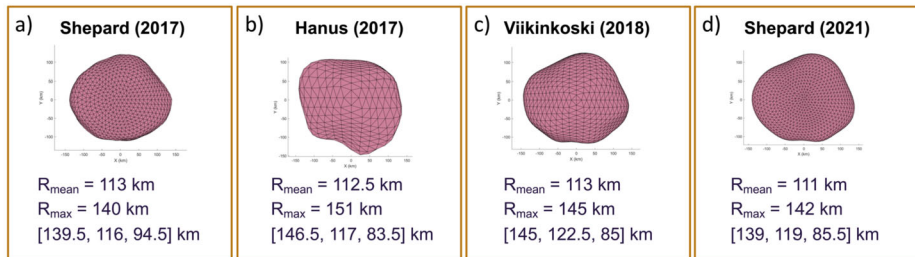


Fig. 6 Four recent Psyche shape models are shown for comparison. (a) Shepard et al. (2017) was the planning baseline for the 2022 launch opportunity. (b) Hanuš et al. (2017) was used to evaluate our plans in the “bounding cases” study. (c) Viikinkoski et al. (2018) was a second model used in the “bounding cases” study. (d) Shepard et al. (2021) is the current planning baseline for the 2023 launch mission. Our mapping campaign was shown to be successful with each of the shape models, demonstrating the resilience of our strategy for designing science orbits

redundant. Functional redundancy in this context means that each dataset designed to satisfy mission science requirements will be acquired more than once, although the circumstances of acquisition can be varied to allow the combination of those two datasets to yield better results than either dataset alone (Polanskey et al. 2011). Even if significant portions of either dataset are lost, there will be sufficient data to meet requirements without altering future plans.

Other strategies are used to ensure we meet and exceed the science requirements. For example, orbit altitudes are designed to provide better imaging resolution than required to allow for minor adjustments to orbit altitude in response to Psyche’s physical characteristics. For measurements that need to accumulate signal over time, the accumulation time is extended to provide margin against lost data or periods of reduced data quality. Data volume margin is held to protect against uncertainty in scheduling sufficient DSN tracking to return the data or to accommodate the need to retransmit data. Unlike spacecraft mass and power margins in the development phase that are intended to be used whenever needed, science margin is built into the plan to reduce the workload and time needed to replan the mission timeline if additional resources are needed. The margin built into the science plan is encumbered for the execution phase of the mission and is not available to the team to make use of during the development or design phases.

A series of circular science orbit altitudes with dedicated science requirements for each altitude enables sufficient redundancy and margin-keeping, as demonstrated by Dawn (Polanskey et al. 2016). Figure 2 illustrates the four science orbit altitudes chosen specifically to meet Psyche’s requirements. Psyche is a mapping mission that methodically observes the surface and in-situ environment to obtain global measurements. The specific

Table 6 Optical topography requirements for image data set (Polanskey et al. 2018)

Stereo technique	Optimal conditions	Useable conditions
Stereophotogrammetry	$5^{\circ} < \text{Phase Angle} < 90^{\circ}$	$5^{\circ} < \text{Phase Angle} < 90^{\circ}$
	$10^{\circ} < \text{Incidence Angle} < 60^{\circ}$	$5^{\circ} < \text{Incidence Angle} < 70^{\circ}$
	Emission Angle $< 55^{\circ}$	Emission Angle $< 70^{\circ}$
	Illumination Variation $< 10^{\circ}$	Illumination Variation $< 15^{\circ}$
	Stereo Separation $15^{\circ}, 15^{\circ}\text{--}65^{\circ}$	Stereo Separation $10^{\circ}, 15^{\circ}\text{--}65^{\circ}$
Stereophotoclinometry	$0^{\circ} < \text{Phase Angle} < 90^{\circ}$	$0^{\circ} < \text{Phase Angle} < 90^{\circ}$
	$10^{\circ} < \text{Incidence Angle} < 75^{\circ}$	$10^{\circ} < \text{Incidence Angle} < 85^{\circ}$
	$20^{\circ} < \text{Emission Angle} < 50^{\circ}$	$0^{\circ} < \text{Emission Angle} < 70^{\circ}$
	$20^{\circ} < \text{Illumination Variation} < 70^{\circ}$	$10^{\circ} < \text{Illumination Variation} < 85^{\circ}$
	Stereo Separation $0^{\circ}, 20^{\circ}\text{--}90^{\circ}$	Stereo Separation $0^{\circ}, 15^{\circ}\text{--}90^{\circ}$

observation technique with the highest inheritance from Dawn is the optical topography campaign. The Psyche spacecraft does not carry a laser altimeter, so topography must be derived from stereo imaging. While Psyche does carry two Imagers that are offset by 3.7° in their mounting, only one Imager is needed for the primary topographic data set. The second Imager is reserved as a backup. There can be many approaches to designing stereo observations, but Psyche will use the approach developed by the Dawn mission for both Vesta and Ceres. Psyche will image a series of complete maps of the surface with a different body-relative fixed attitude for each map. Modeling and experience have shown that a minimum of four maps of the surface (one acquired with a nadir attitude and three with different off-nadir attitudes) are necessary to derive a high-fidelity shape model for an asteroid (Raymond et al. 2011; Jaumann et al. 2022). The topographic solution improves with each additional off-nadir view. In addition to the number of different viewing angles, the quality of the image dataset is critical as well. Table 6 lists the criteria necessary to acquire an imaging set that will maximize the quality of the topography solution for each of the two stereo techniques (stereophotogrammetry and stereophotoclinometry) that the science team plans to use. The beta angle, the angle between the spacecraft orbit plane and the direction to the sun, is an additional constraint used to achieve optimal observing conditions for topography.

3.2 Science Orbit Requirements

The science orbits are designed to characterize Psyche both to enable navigation from one orbit to the next and to efficiently accomplish the science objectives and requirements of the mission. Fortunately, the observing requirements of the Psyche instrument suite are synergistic, so it was straightforward to select a series of orbit altitudes that met both navigation and science requirements. Since Psyche's rotation axis is tipped nearly into its orbital plane, the surface is only fully illuminated during the spring and autumn equinox every 2.5 years. Based on the baseline spin axis model, peak lighting of the surface is achieved in June 2029 as well as July 2031.

Creating the orbital operations timeline was like assembling a puzzle in which all the differently shaped pieces need to fit together to complete the picture. The first step was to select a series of orbit altitudes that meet the science and navigation requirements. The mission designers determined the maximum change between orbit altitudes that could be supported with the gravity knowledge available from the current orbit(s). For example, a

Table 7 Science orbit requirements are listed with the current baseline estimates noted in italics below each requirement

Orbit attribute	Orbit A	Orbit B	Orbit C	Orbit D
Science and Operational Objectives	Characterize gravity and shape, multispectral mapping	Multispectral mapping and topography	Gravity field mapping, improved topography, and magnetometry	Elemental mapping, improved gravity, and magnetometry
Averaged Altitude (<i>using shape model</i>)	600–2000 km <i>709 km</i>	< 400 km <i>304 km</i>	< 1.7 R _{Psyche} <i>193 km</i>	< 1 R _{Psyche} <i>75 km</i>
Inclination (deg)	90 ± 5 <i>90</i>	90 ± 5 <i>90–90.7</i>	90 ± 5 <i>89.7</i>	70% coverage <i>160</i>
% Surface Illumination (<70° incidence angle)	> 80% <i>81–75%</i>	> 80% <i>B1=68% B2=98%</i>	No requirement	No requirement
Beta angle (deg)	10–30 <i>[29–30]</i>	30–40 <i>[31–34], [36–38]</i>	≤ 45 <i>[35–38]</i>	No requirement <i>[55–85]</i>
Duration	≥ 40 days <i>56 days</i>	≥ 6 cycles <i>21 cycles</i>	≥ 100 days <i>100 days</i>	≥ 100 days <i>100 days + 63 days margin</i>

transition directly from approach to the altitude needed for the required imaging resolution was not feasible. Therefore, the original mission timeline that supported launches in 2021 and 2022, both arriving at Psyche in early 2026, included a four-orbit set:

- Orbit A was insensitive to Psyche's gravity while providing the first opportunity to characterize Psyche's gravity and confirm its spin axis orientation.
- Orbit B provided sufficient imaging resolution to satisfy mapping requirements, utilizing the best surface illumination conditions available.
- Orbit C was low enough to meet gravity science and magnetometry requirements but high enough to support a stable polar orbit with global ground track coverage.
- Orbit D provided close-up sensing of elemental composition.

This orbit plan provided a methodical platform for meeting science and navigation objectives. Evolution of this plan to the 2023 launch opportunity is discussed in Sect. 3.4.

Next, orbit resonances were selected, providing a ground track that would efficiently accumulate the surface coverage required at each altitude. The selected orbit resonance specifies the length of a global mapping cycle (the time for the ground track to begin repeat coverage), which is a critical building block of the science plan. The unit of a ground track repeat cycle is used to plan repeatable templates for imaging and other spacecraft operations. The science objectives dictated the number of mapping cycles needed at each orbit, which in turn defined the length of each orbit phase. Next, the orbit phases were placed on the timeline to assure that the surface coverage requirements could be met when paired with illumination conditions and that sufficient time was allocated to transfer between each orbit altitude. A summary of the orbit requirements is found in Table 7.

All orbits must also meet a stability requirement that the spacecraft does not crash or escape within 28 days, the maximum time allocated to recover from a significant anomaly and perform orbit maintenance maneuvers. This requirement drove the selection of a highly inclined orbit for Orbit D.

3.3 Data Budget and Margins

One of the more challenging issues on many flight projects is the allocation of data volume (the amount of data that can be sent back to Earth) between engineering activities and the various science instruments or scientific objectives. The Psyche mission data volume is also constrained by virtue of a fixed HGA that can only be pointed towards Earth when the spacecraft is not oriented for science observations. An articulating HGA was out of scope, which drove two important decisions during mission concept development. The spacecraft data storage capacity was increased to allow for a large accumulation of images during periods of high-frequency acquisition relative to the downlink intervals. In addition, the range of telemetry rates implemented in spacecraft flight software included the highest rates achievable at a DSN station during orbital operations rather than the average rate. This allows the flight team to command the downlink rates to rise and fall during each communications pass to take maximum advantage of the DSN antenna's capability as the spacecraft rises and sets during the antenna's viewing period.

Rather than have each instrument team competing for the data to support their own observations, the project allocates data volume in a structured manner. The first element of the data budget is the routine engineering housekeeping data that must be returned to operate the spacecraft safely, diagnose anomalies, and monitor trends. The total volume of spacecraft housekeeping data can be estimated for a given science orbit phase and subtracted from the total available data that can be transmitted to Earth per that science orbit's observation schema. Early in project development, the undefined telemetry needs of the spacecraft make it difficult to estimate the housekeeping data volume, but this can be measured and adjusted after launch during the interplanetary cruise phase. Next, the constant rate of science instrument housekeeping data is computed and allocated.

The Magnetometer and GRNS produce a constant stream of science data products, although the GRNS can customize that rate by command. The fixed rate science data volume is estimated and allocated, leaving the imaging data volume as the only discrete and variable dataset. Since Psyche is a mapping mission, our planning tools can model the number of image footprints to cover the surface and compute total imaging data requirements by multiplying the number of image stations by the required number of filters for that objective, taking into account the expected data compression. This approach was used during the proposal phase to specify the required average data rate of the spacecraft and correspondingly the size of the HGA.

The total data volume available to downlink mission data is computed from the time allocated to HGA tracking per the observing strategy of each science orbit phase adjusted for expected liens. These liens are accounted for by encumbering margin on the overall downlink capability. It is well known that the DSN resources are oversubscribed, and the Psyche project can only take advantage of DSN tracks that occur during specific times when the instruments are not collecting data so the antenna can point to Earth. We subtract a margin of 27% from the theoretical data return and encumber it as DSN-allocation margin for orbital operations. Remaining margin, if any, is unencumbered and can be used to address new issues or opportunities at Psyche. Each science orbit has a different estimated margin, but the overall margin available to the mission is roughly 30%. The data margin must be managed within each orbit phase because unused margin in Orbit A is not available to be used later in Orbit B. The Psyche project does not plan to use DSOC to transmit any of the required orbital science or engineering data for the mission.

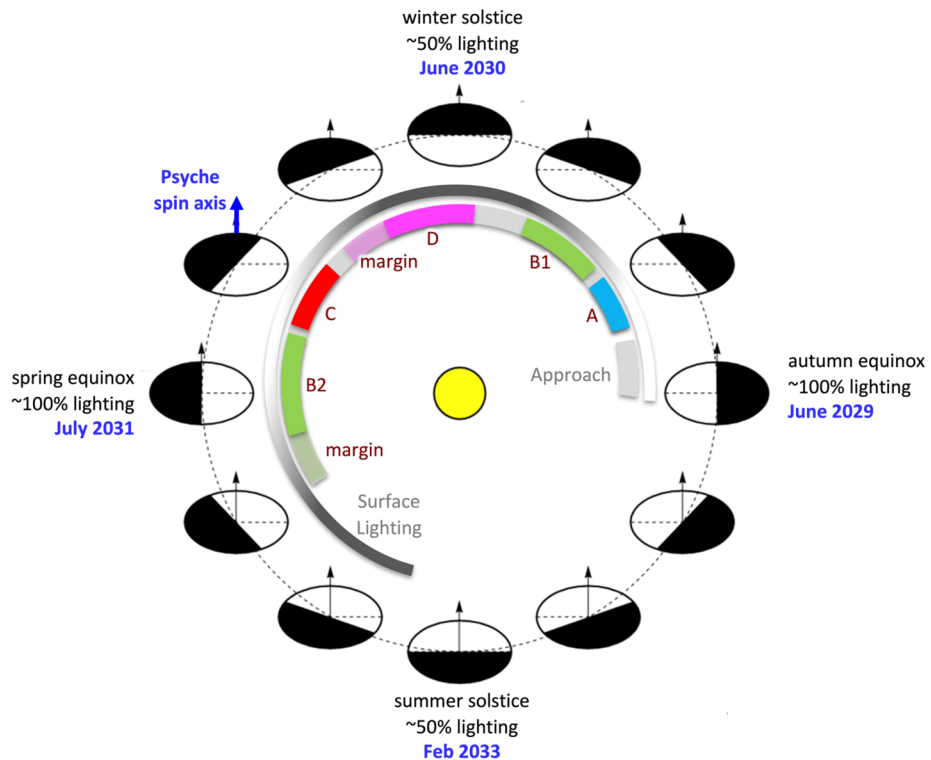


Fig. 7 Simplified graphic of Psyche’s surface illumination as viewed from the north ecliptic pole. Psyche spin axis is the “positive pole” and that hemisphere is the basis of the seasonal references. The science orbit phases are shown on the inside circle. Orbits A through C are polar orbits and Orbit D is a near equatorial orbit. The best surface illumination is available during the two equinoxes. Orbit B2 is the only orbital phase that occurs during optimal illumination conditions

3.4 Restructuring the Mission Timeline for the 2023 Launch Date

Following the launch slip from 2022 to October 2023, the mission design team conducted a study of different interplanetary trajectories to deliver the spacecraft into a Psyche orbit. The preferred trajectory begins orbital operations at Psyche in August 2029, after the peak in surface illumination as shown in Fig. 7. The next time Psyche will be fully illuminated is in July 2031. The most demanding imaging requirement of obtaining greater than 80% surface coverage at less than 20 m/pixel dictates that Orbit B should be scheduled during periods of optimal illumination, but Orbit B imaging is also necessary for the updated gravity and shape knowledge required to transition to lower orbits. The only solution to these constraints is to conduct two mapping (Orbit B) campaigns. Orbit B1 follows Orbit A with insufficient surface illumination to meet science requirements but sufficient shape model updates to support the transfer to lower orbit altitudes. Orbit B2 occurs in mid 2031 to complete the imaging needed to satisfy the science requirements.

Orbit C does not have surface illumination requirements, but the gravity science investigation requires a Sun-Earth-Probe angle greater than 30° for at least 70 days, although greater than 40° is preferred. In the scenario where Orbit C followed Orbit B1, the Sun-Earth-Probe angle would be less than 40° for most of Orbit C. In addition, Orbit C would

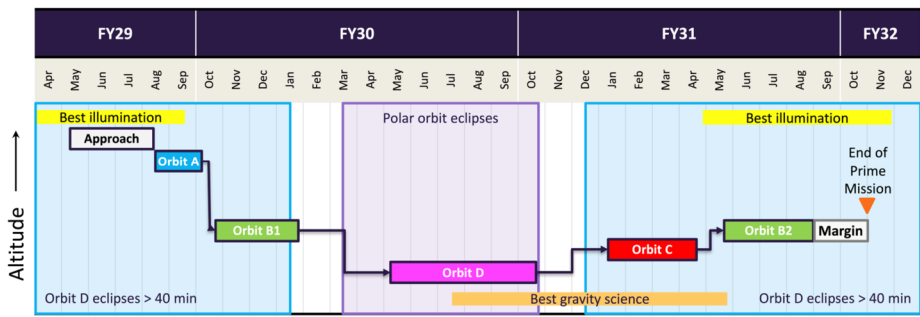


Fig. 8 The Psyche Approach phase, science orbits, and transfer orbits are shown on the mission timeline for the 2023 launch planning baseline. Illumination conditions are based on a model using a triaxial ellipsoid. The eclipse season for equatorial orbits is shown in pale blue and for polar orbits in pale violet. The period for optimal gravity science when the Sun-Earth-Probe angle is $> 40^\circ$ is shown in orange. Avoiding eclipse seasons while satisfying surface illumination and other observation requirements drove the design of the mission profile

occur during a period when Psyche would be blocking sunlight from reaching the spacecraft. While the spacecraft is qualified to go into eclipse, it is constrained to avoid eclipses longer than 65 minutes occurring as often as once every 6.6 hours. The Orbit C eclipse season would begin in March 2030 and continue through September 2030, and then by the time the spacecraft could complete the orbit transfer to begin Orbit D, the period when equatorial orbits experience eclipses would have begun. This disadvantageous set of geometric relationships drove the project to change the order of the science orbits.

An alternate consideration was to put the spacecraft in a high parking orbit and wait for the eclipse season for polar orbits to end, but that would add many months to the orbital operations timeline. Fortunately, Psyche is closer to the sun in its orbit during the 2029 scenario, providing more power to the solar arrays and enabling more efficient thrusting during the transfer periods between science orbits. This enabled the design of a transfer trajectory between Orbit B1 and Orbit D without having to stop at Orbit C first. This novel solution avoids eclipses in both Orbit D and C and has other advantages. In the previous mission timeline, the elemental composition measurements would be acquired at the end of the mission and could take advantage of any remaining operations margin; this is no longer possible under the new timeline. Therefore, margin was added to the duration of Orbit D to mitigate data loss from any solar energetic particle events that would mask Psyche-relevant data collection and to avoid starting Orbit C too early before the end of its eclipse season. Orbit C was then placed following Orbit D in a period where the Sun-Earth-Probe angle is $> 40^\circ$ for the entire duration, allowing the final transfer to Orbit B2 to complete at an opportune time to begin imaging the surface in the northern hemisphere as it becomes illuminated.

The result illustrated in Fig. 8 is a mission duration of 26.5 months that meets all science observing requirements, avoids eclipses, maximizes conditions for gravity science, and still has timeline margin to handle unexpected interruptions. The adaptability of the mission profile to a completely new set of constraints speaks to the advantages of having science orbit phases that can be treated as mostly independent building blocks in the timeline. If there are no delays in the interplanetary trajectory that impact the arrival date at Psyche, then the lower resolution multispectral imaging objectives could be satisfied in Orbit A, leaving only the higher resolution geologic mapping and optical topography objectives to Orbit B2.

3.5 Lighting Variability

As the mission timeline could need adjustment after arrival at Psyche if certain attributes (such as Psyche's spin axis orientation) are significantly outside the current error estimates, the mission planning team analyzed the impact of spin pole uncertainties on lighting conditions. Psyche was modeled as an oblate spheroid based on the Shepard et al. (2021) best fit triaxial ellipsoid, but with the semi-major axis of the spheroid approximated by the average of the two largest radii in the best fit ellipsoid. Assuming a pole orientation [right ascension (RA) and declination (dec)], maximum acceptable solar incidence angle for surface imaging, and epoch (which translates to relative position of the sun with respect to Psyche), the model determines the geodetic latitude at which the limiting incidence angle occurs. The lighting percentage is calculated as the percentage of surface below that latitude compared against the total surface area of the spheroid. Both hemispheres are considered separately, because the areas around both poles can simultaneously experience unacceptable lighting near the equinox for incidence angle limits of less than 90° . This simple model, intended for use in quick what-if studies, was compared against a detailed time-varying simulation incorporating more complex shape modeling (Shepard et al. 2021 shape model), body rotation, and self-shadowing, and was found to estimate the lit surface percentage to within 2%. Figure 9 shows the percentage of the surface that is illuminated during the nominal timeline of the Psyche mission. A 3-standard deviation worst case pole error shows the impact of variation in optimal lighting conditions and how the strategic placement of Orbits B1 and B2 provide the conditions to meet science surface coverage objectives under any possible illumination in this study. While our plans are designed to be robust to the expected uncertainty in Psyche's physical characteristics, we also acknowledge that Psyche's real shape and spin pole orientation may require adjustments to our plans.

4 Science Implementation by Orbit Phase

The following subsections describe, in mostly chronological order, the details of each science orbit and orbit transfer phase that includes science data collection. A summary of the science goals that informed each orbit can be found in Table 2. The Psyche science data in each phase builds on data collected earlier in the mission. All efforts were made to modularize and standardize the spacecraft and instrument activities to reduce complexity of operations while simultaneously designing in the ability to be flexible to changes in the timeline and other conditions.

Low-thrust missions allow timing of each phase to fluctuate with routine re-optimization of the trajectory. The date of capture into orbit around Psyche is allowed to drift in response to the effectivity of cruise thrusting and any unplanned loss of thrust. The time between science orbit phases may also change once the characteristics of Psyche are known and realistic spacecraft transfer sequences are developed. The Psyche mission operations concept embraces these uncertainties, and the flight team operational processes are designed to respond nimbly to expected timing changes. However, long-range planning tasks can benefit from a stable planning baseline. Table 8 contains the dates currently assumed for the planning baseline, knowing that the actual dates are guaranteed to change. The following subsections describe the layout of the science observing plans independent of the actual start and end dates of each science orbit phase, so it is expected that certain geometric traits of the asteroid (such as surface illumination) will vary based on actual execution times as well as actual tilt of the asteroid spin axis as discussed above.

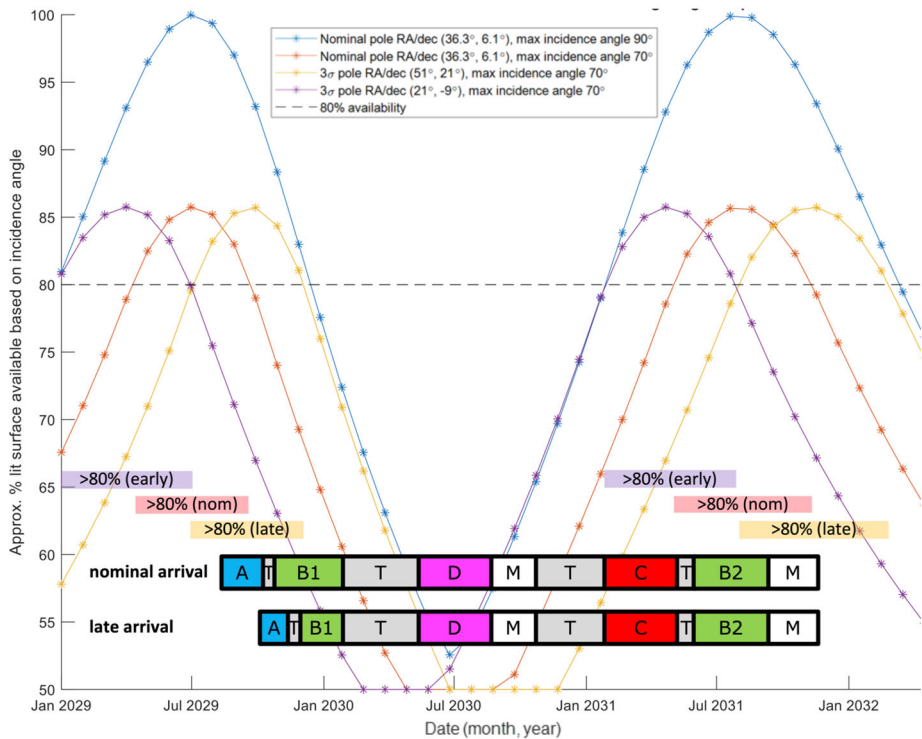


Fig. 9 Variability in illumination conditions for Psyche assuming different spin axis orientations. The blue curve is for Psyche's nominal pole RA/dec (36.3° , 6.1°) and maximum incidence angle of 90° . The salmon curve is for the nominal pole with maximum incidence angle of 70° . The yellow curve shows a worst case 3-sigma variation in Psyche's pole RA/dec (51° , 21°) and the purple curve is the opposite worst case 3-sigma pole RA/dec (21° , -9°), both with maximum incidence angle of 70° . The science orbit phases are shown with respect to the lighting curves for both the nominal date for injection into Orbit A and a worst-case injection date. Imaging in Orbit B2 will satisfy the requirement to provide surface coverage of greater than 80% even with a worst-case injection date

Table 8 Orbital operations phases

Operation phase	Duration (days)	Start date (baseline for planning)
Approach	100	May 9, 2029
Orbit A	56	August 17, 2029
Transfer to Orbit B1	17	October 12, 2029
Orbit B1	92	October 29, 2029
Transfer to Orbit D	98	January 29, 2030
Orbit D	163	May 7, 2030
Transfer to Orbit C	90	October 17, 2030
Orbit C	100	January 15, 2031
Transfer to Orbit B2	23	April 25, 2031
Orbit B2	100	May 18, 2031
Operations Margin	67	August 26, 2031
End of Mission	n/a	November 1, 2031

The spacecraft attitude modeling is performed with the Science Opportunity Analyzer science planning tool (Ortega et al. 2021). This is the same tool that was used to create the mission architecture in the proposal phase. To uniquely define spacecraft attitude, the primary pointing vector (i.e. Imager pointed to Psyche nadir) and secondary pointing vector (sometimes referred to as twist) must be specified. For missions with solar arrays, the twist is often controlled to keep the solar arrays facing the sun although the arrays may be slightly off-pointed to control angular momentum of the spacecraft.

4.1 Observations and Navigation on Approach to Psyche

Once the spacecraft reaches a distance where imaging Psyche will provide improvements on ground-based observations, the project will embark on an imaging campaign to support optical navigation (OpNav). Since the navigation team relies on the same Multispectral Imager used for the science observations to perform this mission critical navigation function, redundant imagers were built and mounted to the $-X$ axis of the spacecraft to protect the mission from a single point failure. While there is a 3.7° offset in the mounting orientation of the imager boresights, the mission requires only one camera to complete all engineering and scientific goals, and a single Imager will be operated throughout the completion of imaging requirements. Both Imagers are calibrated during the approach phase, so the OpNavs alternate between Imager A and Imager B ensuring that Imager B is ready if needed during orbital operations. As a project policy, OpNav and science imaging are shared between both teams. The labeling of the activity as OpNav or science is based on which team requires and designs the observation.

The primary objective of the approach imaging is to refine estimates of Psyche's spin axis orientation and shape to enable the design of and delivery to Orbit A. These attributes of Psyche are also important science requirements that feed into the shape derivation and gravity modelling. This objective requires images taken with the polychromatic (clear) filter. Additional observations using both polychromatic and/or all narrowband filters are planned to calibrate the image exposure times to be used during the orbital operations phase and to search for evidence of any secondary objects in orbit around Psyche. Analysis has demonstrated that the existence of orbiting secondaries would not pose a threat to the Psyche spacecraft, so the search for satellites is purely of scientific interest.

The approach phase is a transition from cruise operations that are focused primarily on thrusting to orbital operations with long coasts during the science orbits and thrusting between those orbits. Thrusting is still a critical part of this phase, and the imaging periods need to either be reserved in advance or scheduled during periods when it is optimal to coast. During this period there will be a change from a Sun-centered ephemeris to a Psyche-centered ephemeris. Science operation processes specific to the orbital operations phase will be integrated with the processes still being used from the cruise phase in a way that is unique to this period.

4.1.1 Optical Navigation Plan

On approach to Psyche, the key objectives for OpNav are to provide optical observations for the day-to-day orbit determination activities and to update key Psyche physical properties, such as the spin axis, spin rate, and shape model. The goal of these objectives is an accurate insertion into the reference Orbit A science orbit. During the first seven weeks of approach, with Psyche less than 50 pixels diameter, OpNav is based on body center-finding techniques, such as cross-correlation and limb scans. The only geometric requirement for

these techniques is a phase angle less than 100° – 110° , to allow part of the body to be illuminated. Once Psyche exceeds ~ 50 pixels diameter, the main OpNav observables are landmarks, constructed with stereophotoclinometry. Landmarks are body-centered vectors determined via small digital terrain models, or landmark maps. The construction and maintenance of such a network of landmarks allows the estimation of the rotational properties of Psyche and the large and small-scale shape. These landmark maps also enable terrain-relative navigation when Psyche exceeds the field of view (FOV) of the Imager.

Stereophotoclinometry places several geometric constraints on the imaging and illumination geometry (i.e., on the emission and incidence angles). Ideally the emission angle will vary both in elevation and azimuth, with elevation angles ranging from nadir to 60° off-nadir and azimuth from 0° – 360° . In practice, on approach to Psyche such an emission range is accomplished by pointing the Imager in the nadir and off-nadir Psyche directions and by imaging for a full rotation period over a range of sub-spacecraft latitudes, enabling east-west and north-south azimuth camera directions. Ideally the incidence angle will have an elevation range of 20° – 70° and a 0° – 360° azimuth. Such a variation in the incidence angle is difficult to accomplish. In practice we rely on the beta angle being less than 15° to avoid very low incidence angle observations. The rotation of the body allows for a variation in the incidence angle, but the solar azimuth is restricted to morning and evening observations rather than a wider range. The range of solar elevation also imposes a similar constraint in the phase angle in the range of 20° – 70° . In addition, to maximize the surface coverage and to allow more favorable observing conditions for the rotational frame estimation, there is a need to image, for at least a limited time, at sub-spacecraft latitudes as close to Psyche's equator as possible.

The approach low-thrust trajectory design optimizes xenon fuel usage and arrival time and tends to produce trajectories that reach Orbit A from a high phase angle while keeping the sub-spacecraft latitude mostly near the north pole of Psyche, which is largely not illuminated. As a result, several state constraints had to be added to allow, in a statistical sense, the trajectory to meet the OpNav imaging geometry needs. These constraints are discussed below.

At two different fixed epochs of 48.015 days and 12.358 days prior to Orbit A insertion, or approximate range of 93,000 km and 3800 km respectively, we impose a phase angle constraint in the 0° – 90° range, which keeps the phase angle from growing too large during the period where landmarks are the key OpNav observations. In the current reference trajectory these constraints keep the phase angles in the range of 35° – 69° .

At a range of 10,000 km, or approximate epoch of 16.5 days to Orbit A, a constraint of a sub-spacecraft latitude between 0° – 30° North latitude has been added. This constraint forces the trajectory to move closer to Psyche's equator, while staying always at north sub-spacecraft latitudes. In the current reference this constraint keeps the trajectory between 30° and 17° North latitudes during key OpNav sessions. Figure 10 provides examples of several simulated OpNav images during the approach phase, showing the increasing resolution and differing illumination conditions and including a simulated surface texture for context.

Operational considerations for thrust cycles create another constraint. The mission design divides the approach phase into seven thrust design cycles. Design cycles 5–7 target insertion into Orbit A based on the Psyche team's knowledge of physical parameters. To improve the spin axis knowledge in design cycles 5 and 6, a constraint has been added to slow the spacecraft down relative to Psyche, starting at a range of 22,000 km, or an epoch of 28.504 days relative to Orbit A insertion, so as to increase the number of OpNav imaging sessions and ground processing time, during which landmark observables are being constructed.

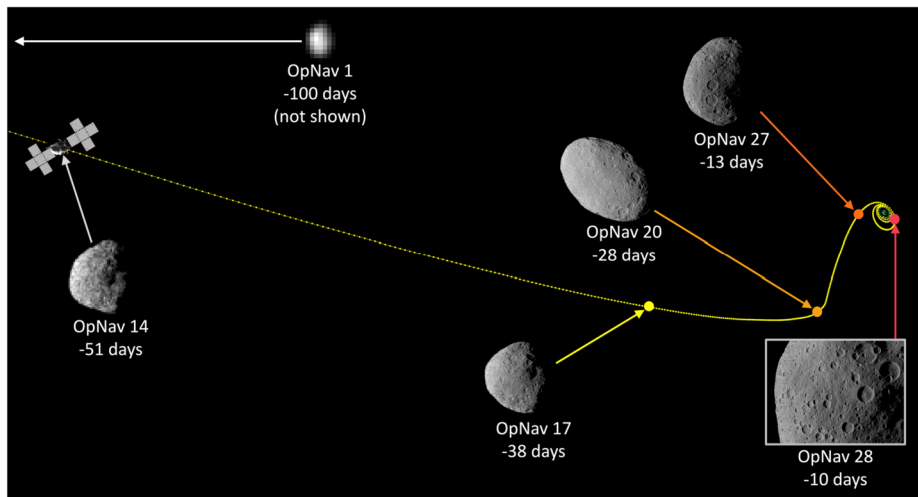


Fig. 10 Six simulated OpNav images shown on approach to Psyche. Psyche's diameter changes from 6.7 pixels to 2900 pixels between OpNav 1 and OpNav 28. For this approach trajectory, the phase angles range between 35° to 72° and all but OpNav 28 are taken from a northern sub-spacecraft latitude. Psyche surface texture is simulated

4.1.2 Multispectral Imager Calibrations

Ten of the OpNav opportunities will include additional science imaging with all seven narrowband filters (five opportunities for each Imager) to allow the Imager team to refine the image exposures planned for orbital operations. The first pair are acquired early during the first week of the approach phase, and the remaining four pairs are during the last five weeks of approach when Psyche fills a larger portion of the camera FOV. Two straylight tests for each Imager are also planned towards the end of the approach phase.

4.1.3 Psyche Satellite Search Observations

Two opportunities are planned during the approach phase to search for natural satellites by imaging the star field within Psyche's Hill sphere, where any bound satellite orbits are expected to reside. To date, no moons have been detected within Psyche's Hill sphere by ground-based observations. Viikinkoski et al. (2018) established an upper limit of undetected satellites of 1.5 km diameter within 150 km of Psyche and 800 m diameter within 2000 km of Psyche. The primary challenge of these observations is to design image exposures that will provide sufficient signal to detect small moving objects without excessive smear from spacecraft jitter. The details for these observations will be planned during the cruise phase. While McFadden et al. (2015, 2018) conclude that large, intact protoplanets do not have satellites, the fact that Psyche's spin pole is approximately in its orbit plane argues that the asteroid was disrupted early in its history and thus could have been subject to different or unique circumstances regarding formation and retention of satellites, compared to "normal" large asteroids. Therefore, if satellites are observed around Psyche, they could contribute additional information to discriminate between models of Psyche's origin.

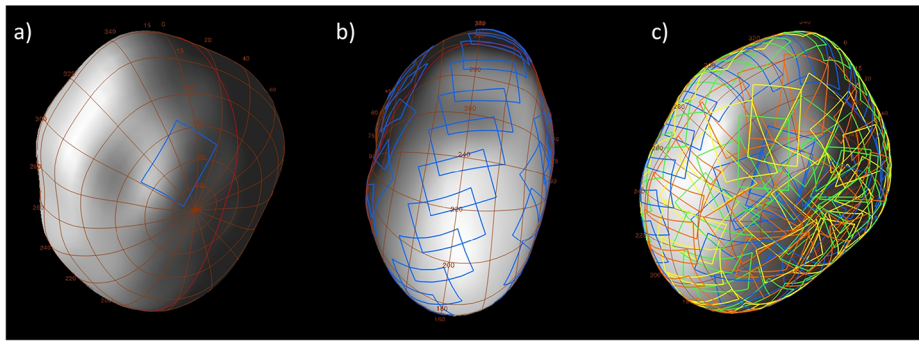


Fig. 11 Imaging in Orbit A taken from the perspective of the spacecraft as modeled by the Science Opportunity Analyzer software. (a) One Imager FOV is shown positioned near the dark-to-lit terminator in Orbit A. (b) One orbit of nadir imaging. The spiral shape of the orbit ground track and imaging swath results from the spacecraft orbit period being greater than the asteroid rotation period. (c) Four orbits of imaging, less than half of the 9-orbit cycle, provide nearly global surface coverage of the illuminated surface

4.2 Orbit A: Characterizing Psyche

Insertion of the spacecraft into Orbit A marks the beginning of the orbital operations phase of the mission. Unlike spacecraft with chemical propulsion, SEP missions do not utilize discrete orbit insertion maneuvers. Instead, all thrusting from the beginning of the cruise phase through the approach phase comprise an extended maneuver designed to gradually enter Orbit A. The interplanetary thrusting changes the spacecraft's heliocentric trajectory to slowly become more and more like (16) Psyche's. Thrusting does not target gravitational capture by the target body. Rather capture occurs simply as part of normal electric propulsion thrusting during the approach phase to Orbit A.

4.2.1 Objectives

While some imaging science requirements may be achieved in Orbit A, the primary motivation for this orbit is to characterize Psyche by obtaining gravity measurements and deriving a shape model sufficient to design the transfer trajectory to Orbit B1. Imaging with all filters in this cycle will contribute to the global mapping of silicates and metals, if not intimately mixed within the rock, and the search for oldhamite. Coverage requirements for these objectives are difficult to satisfy with the predicted illumination conditions. However, all imaging objectives can be satisfied in Orbit B2, so the data acquired in Orbit A simply provides an opportunity to begin mapping early and enables fine tuning the observation parameters and techniques for subsequent orbits. Mapping of craters with diameters greater than 1 km over 50% of the surface can be fully accomplished during Orbit A. Figure 11 illustrates the scale of the Imager FOV on Psyche's surface in Orbit A and how imaging coverage builds up over the course of a nine-orbit mapping cycle.

To determine Psyche's gravity sufficiently to design and build the spacecraft sequences that command thrusting to the Orbit B altitude, the navigation and science team will use X-Band Doppler data from the HGA and LGA to create the first global gravity model of Psyche. One of the most important scientific results to come from the gravity investigation during approach and/or Orbit A is a precise measurement of Psyche's mass, which will finally give us an accurate density and tell us whether Psyche really is metal rich.

Table 9 Orbit A planned characteristics (*values expected to change after arrival). $R_{\text{Psyche}} = 111 \text{ km}$

Orbit A start date*	8/17/2029	Average orbit period	32.6 hours
Orbit A end date*	10/12/2029	Orbit resonance	70:9
Duration	56 days	Orbits per repeat cycle	9
Average radius*	811 km (7.3 R_{Psyche})	Ground track repeat cycle duration	12.2 days
Average altitude*	709 km	Number of ground track repeat cycles	4.5
Beta angle range*	[29–30]°	Number of orbits in phase	41
Maximum ground speed*	61 m/s	Average Imager pixel scale*	35 m

4.2.2 Implementation

The orbit altitude was constrained to be greater than 600 km by the need to limit perturbations from Psyche's gravity to less than 1% and lower than 2000 km to ensure that the images acquired satisfy the mineralogy resolution requirement of 200 m/pixel with margin. The current baseline altitude of 709 km provides an image pixel scale of 35 m/pixel, substantially better than the required 500 and 200 m/pixel for the metal to silicate ratio determination and oldhamite mapping objectives.

The 56-day duration of Orbit A was driven by operational considerations of the timeline for planning and building the thrust sequences to Orbit B1, so it is longer than needed to map the surface with all Imager filters with 100% redundancy. Table 9 lists the attributes of the current Orbit A planning baseline.

We use global image maps created with polychromatic-filter imaging to improve the Psyche shape model. To efficiently accommodate both imaging and gravity objectives, the spacecraft will be commanded to point the prime Imager towards Psyche when the spacecraft is traversing the lit portion of the orbit, then turn to point the HGA towards Earth to return accumulated data and collect X-band Doppler data for gravity science while the spacecraft is traversing the dark side of the orbit. Figure 12 is a cartoon of this operations schema including the expected spin axis (+Z) orientation of Psyche.

After 12.2 days and nine orbits, the spacecraft will have completed its first ground track repeat cycle, and the following orbit will begin to cover terrain that has previously been mapped. This repeated coverage provides redundancy and opportunities to observe the surface differently. The allocation of time between the spacecraft pointing to acquire imaging and pointing the HGA to Earth for communications is illustrated in Fig. 13.

Over the first 12-day mapping cycle, imaging will be acquired at a strictly nadir attitude. The orbit resonance (70:9) was selected to accommodate measurements of Psyche's potential magnetic signature, although the field would need to be strong to be detected at this altitude. The ground track repeat pattern is designed so that each point on Psyche's surface lies within 10° of arc of the ground track while providing overlapping swaths of image footprints to obtain complete coverage of the illuminated surface. The ground tracks are spaced closely enough that duplicate imaging coverage can be acquired in one mapping cycle.

The second mapping cycle will be a repeat of the first, offering an opportunity for the Imager team to update image exposures based on results from the first cycle and redundant coverage for any terrain that was missed.

The third mapping cycle is allocated to off-nadir imaging with the polychromatic filter to improve the Psyche shape model for optical navigation and science planning. This also allows more time for the imaging operations team to evaluate and update parameters for their next full-color imaging campaign in the fourth cycle. Improving the shape model with stereo imaging early in the orbit phase allows time to feed those results into the thrust planning for

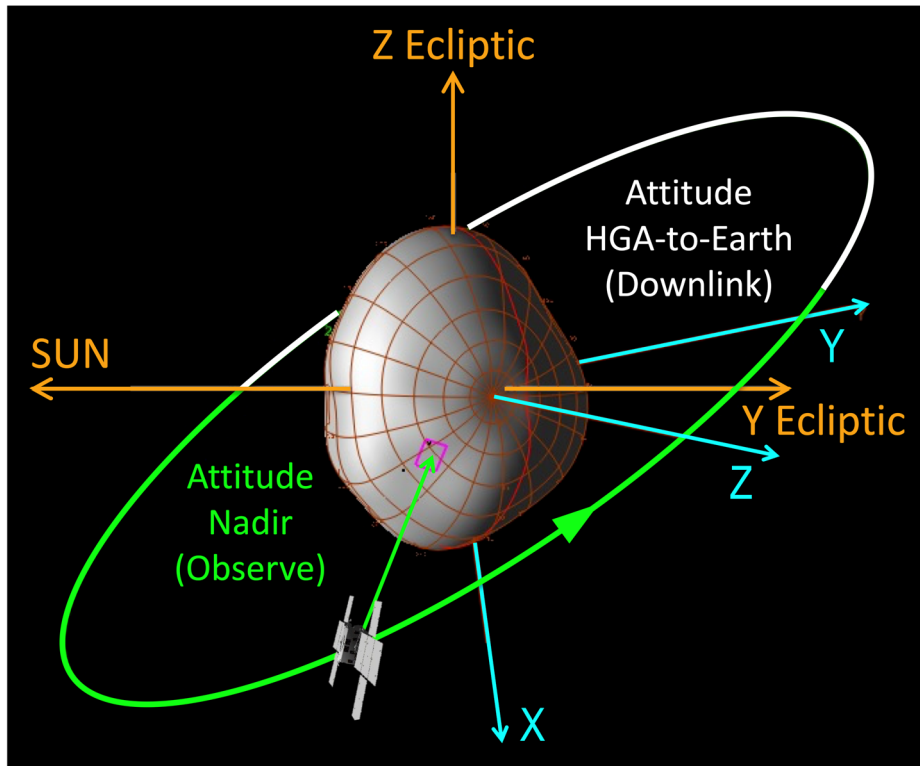


Fig. 12 The spacecraft transitions between data downlink as it passes over the dark side of the asteroid during its orbit and science observing while passing over the lit side of the asteroid. The spacecraft enters the hit hemisphere in the south and travels to the north. Activities are planned relative to the terminator crossings of each orbit. This scheme allows the imaging sequences to be developed as a repeating template that can be adjusting if the timing of the terminator crossing changes. The magenta rectangle is a projection of the Imager field of view on the surface. The Psyche body-fixed axes are turquoise. The spacecraft is not to scale

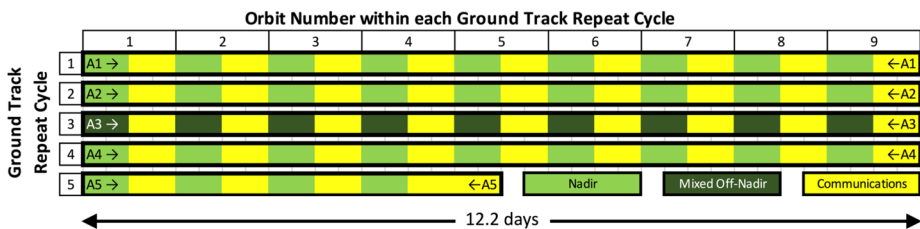


Fig. 13 The operations time in Orbit A naturally divides into 4.5 ground track repeat cycles. Each 9-orbit cycle provides the opportunity to completely map the surface with the Imager and map gravity using the telecommunications subsystem. The time spent pointing the Imager nadir or off nadir is colored green and the communications periods are yellow. Cycle 3 is dedicated to off-nadir OpNav imaging

the transfer to the next orbit and has little impact on the final multispectral coverage. Off-nadir pointing in Orbit A attempts to control the camera azimuth and elevation, relative

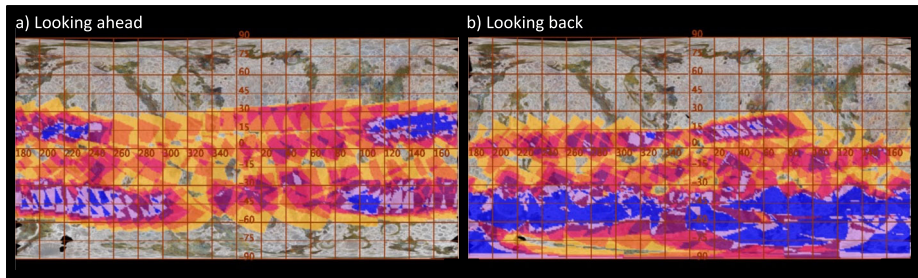


Fig. 14 Examples of different off-nadir imaging coverage for one orbit of Orbit A plotted on a latitude/longitude map. The background texture map is an artist rendering. (a) The spacecraft is pointing the Imager ahead of the orbital velocity vector. (b) The spacecraft is pointing the Imager behind the orbital velocity vector. The lightest orange indicates coverage with a single image and duplicate coverage is shown with increasingly dark colors until blue, which indicates coverage with six images or more covering the same area

to the local surface, by using a specific imaging strategy in each of the nine orbits of the mapping cycle.

For each orbit the camera follows a specific azimuth direction relative to the spacecraft, scanning either left, right, ahead, or behind the spacecraft. While in each of these four azimuth directions, we execute a 1-dimensional scan pattern to map as much of the surface as possible. In the first four of these orbits, the turn angle is “large” to image the surface at local emission angles in the range of 40° – 60° and in next four of these orbits the turn angle is “small” to image the local surface at emission angles in the range of 15° – 40° . The key assumption with this plan is that at the limit of complete surface coverage, each patch of surface will have been imaged in four distinct and widely separated directions in camera azimuth and two distinct directions in emission angle. The ninth orbit aims at imaging the illuminated limb of Psyche by positioning the Imager boresight to $\sim 20\%$ off the body.

Figure 14 demonstrates two examples of off-nadir attitude mapping. Looking ahead and looking back shows some overlap in coverage where the same patch of surface is imaged in these two azimuth directions, but also areas where, due to the irregular body shape, there is no overlap.

The fourth and partial fifth cycles are a repeat of the first two with optimized imaging exposures. These cycles are primarily intended to improve on the images taken previously and fill in gaps created by ground track deviations from the plan that result from delivery errors and small forces perturbing the spacecraft. The final cycle is not long enough to complete all nine orbits but provides one additional coverage map. Multispectral imaging continues for the first four orbits of the cycle, after which it is time to begin thrusting to Orbit B1.

4.2.3 Illumination Constraints and Mapping Coverage

Peak illumination conditions occur during the approach to Psyche, so orbital science operations begin with the north polar region in shadow down to about 60° north latitude. While 81% of the surface is theoretically illuminated with incidence angles below 70° at the start of Orbit A, northern latitudes will become less illuminated throughout this phase. These conditions may be sufficient to satisfy multispectral coverage requirements in the first cycle, but it is likely that parts of the northern hemisphere will need to be imaged again towards the end of the mission. However, these early maps in combination with lower resolution

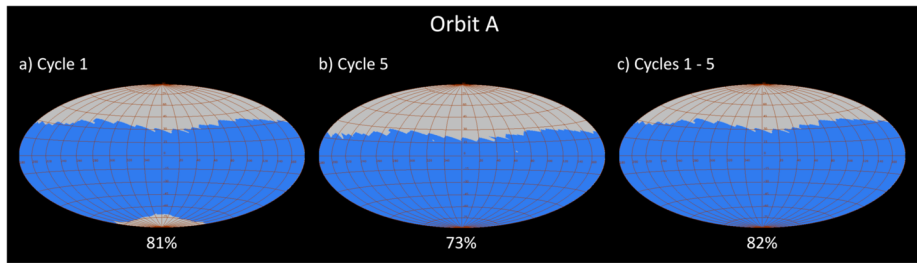


Fig. 15 (a) The first cycle of Orbit A provides 81% surface coverage for incidence angles $< 70^\circ$. (b) As the sun moves towards the southern hemisphere, overall coverage decreases by the end of the phase although additional terrain in the south becomes visible with good illumination. (c) The cumulative coverage is slightly above the required 80% for multispectral imaging, but this could change with small variations in spin axis or arrival date. Coverage is shown using a Mollweide projection

approach imaging will provide the multispectral baseline for future planning and help the project better prepare to complete the multispectral mapping much later in the mission when illumination improves in the northern hemisphere. Figure 15 illustrates the surface coverage that can be achieved in Cycle 1 and Cycle 5 of Orbit A. The combination of the two cycles provides a minor increase in overall mapping coverage.

Following completion of Orbit A activities, 17 days of thrusting are allocated to transfer the spacecraft to the Orbit B altitude to begin Orbit B1. A short trajectory correction maneuver with the electric propulsion system may be required several days after nominal thrusting completes to fine-tune the orbital parameters to achieve the required reference orbit.

4.3 Orbit B: Geologic and Topographic Mapping of Psyche's Surface

4.3.1 Objectives

Many of the driving requirements for imaging will be completed at the altitude defined for Orbit B. The most challenging requirement is to obtain the images needed to create a geologic map of at least 80% of the surface at a resolution ≤ 20 m/pixel. Global multispectral imaging is also required to search for evidence of the sulfide-bearing mineral oldhamite that would indicate that Psyche was formed under reducing conditions, and evidence of large regions of achondritic or cumulate silicates on Psyche's surface that could indicate that Psyche is a stripped core. Data sets sufficient to derive a global shape model using stereo imaging techniques are also planned for this orbit phase and comprise the majority of the timeline. As noted earlier, the surface illumination during the Orbit B1 is insufficient to meet imaging coverage requirements; however, the spacecraft will return to the same altitude to conduct Orbit B2 towards the end of the mission. Most attributes of Orbits B1 and B2 are similar, so the two phases will be described together in this section.

4.3.2 Implementation

Designed to meet all imaging requirements for topography and multispectral imaging, Orbit B is required to have at least six mapping cycles in each phase, two at a nadir attitude and four with off-nadir attitudes. The baseline plan accommodates ten cycles in Orbit B1 and 11 cycles in Orbit B2, providing significant margin. The mean orbit altitude must be < 400 km

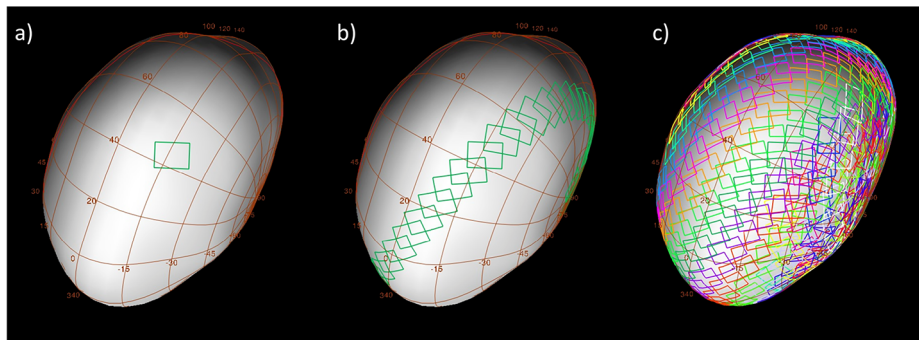


Fig. 16 Imaging in Orbit B1 taken from the perspective of the spacecraft. (a) One Imager FOV. (b) One orbit of nadir imaging. The spiral shape of the orbit ground track is evident. The twisting of the FOV is driven by the secondary pointing axis that strives to keep the solar arrays oriented towards the sun. (c) One complete cycle of 17 orbits of imaging provides global coverage of the illuminated portion of the surface

Table 10 Orbit B planned characteristics

Orbit B1 start date	10/29/2029	Average period	11.6 hours
Orbit B1 end date	1/29/2030	Orbit resonance	47:17
Orbit B1 duration	92 days	Orbits per ground track repeat cycle	17
Average radius	406 km (3.7 R_{Psyche})	Ground track repeat cycle duration	8.2 days
Average altitude	304 km	Operational repeat cycle duration	9.2 days
Average inclination	90.7°	Number of operational repeat cycles	10
Orbit beta angle range	[31–34]°	Total number of orbits	190
Maximum ground speed	66 m/s	Average Imager pixel scale	15 m
Orbit B2 start date	5/18/2031	Average period	11.6 hours
Orbit B2 end date	8/26/2031	Orbit resonance	47:17
Orbit B2 duration	100 days	Orbits per ground track repeat cycle	17
Average radius	406 km (3.7 R_{Psyche})	Ground track repeat cycle duration	8.2 days
Average altitude	304 km	Operational repeat cycle duration	9.2 days
Average inclination	90.0°	Number of operational repeat cycles	10
Orbit beta angle range	[36–38]°	Total Number of orbits	206
Maximum ground speed	65 m/s	Average Imager pixel scale	15 m

to satisfy the imaging requirements for the 20 m pixel scale, and the baseline altitude provides 25% margin on the requirement. Over 17 orbits (one ground track repeat cycle), the spacecraft ground track provides complete imaging coverage. Figure 16 illustrates the projection of the Imager's footprints from the vantage point of Orbit B1. Each FOV drawn on the surface is called an “image station” because it could correspond to a single image acquisition with one filter or a complete set of all eight filters.

Because the ground track coverage is the driving requirement for this orbit, the final altitude in operations will be selected to preserve the planned orbital period and resonance. This selection will be made once the gravity field and shape knowledge are improved with data from Orbit A. These parameters influence the design of the orbit transfer, so the science and mission design teams collaborate on the final orbit selection. Table 10 lists the attributes of the current baseline design for both Orbit B phases.

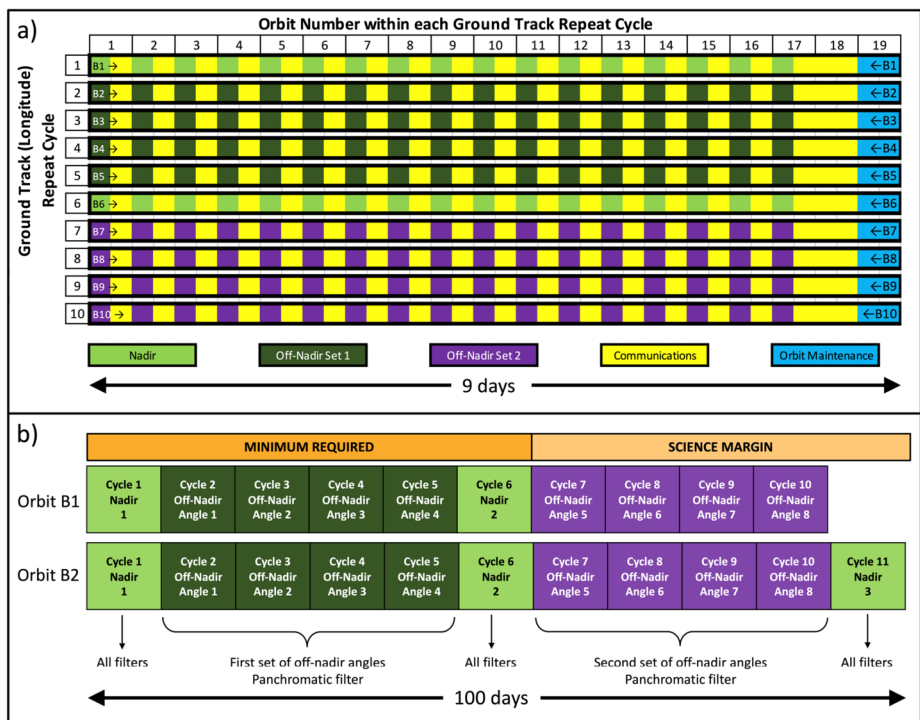


Fig. 17 (a) Graphical depiction of the 10 sequence cycles for Orbit B1. Each ground track repeat cycle is 17 orbits with additional orbits added for extra downlink and orbit maintenance maneuvers. Yellow indicates periods when the HGA is pointed to Earth for communications and shades of green and purple indicate when the Imager is pointed nadir or off nadir to observe. The illuminated surface is fully mapped 10 times in Orbit B1. (b) Summary timeline of all cycles in Orbits B1 and B2 demonstrating the ample science margin in the plan

Like Orbit A, the spacecraft will be pointed towards Psyche on the illuminated side of each orbit with LGA tracking for gravity science, and the HGA will be pointed towards Earth to return the imaging data and collect two-way Doppler tracking data on the dark side of the orbit. However, with an 11.6-hour orbit period, the 40-minute duration of the spacecraft turns consumes a significant portion of the dark-side pass.

Images acquired with all filters during nadir cycles have a high enough data volume to preclude returning all images with this downlink strategy alone. An extra orbit devoted only to data return will occur at the end of each 17-orbit ground track repeat cycle. The current timelines include multiple opportunities to acquire complete nadir maps with all eight camera filters in Orbit B (two in Orbit B1 and three in Orbit B2) as shown in Fig. 17. Despite the extra downlink-only orbit, the number of images acquired in the multispectral cycles cannot be returned to Earth with only the downlink available during that cycle. Therefore, the nadir cycles are separated in time to allow some of those images to carry over to the next polychromatic-filter cycle and trickle down after the data from those orbits are returned. This separation of nadir cycles also provides an opportunity for the illuminated regions to change, increasing coverage, and for the Imager team to evaluate the data quality and make any adjustments to exposures or other parameters prior to the subsequent multispectral imaging campaign.

In addition, studies by the mission design team indicate that regular orbit maintenance maneuvers may be needed to keep the ground track consistent with the plan by compensating for the small forces imparted to the spacecraft by the cold gas thruster firings needed to desaturate the momentum wheels. Therefore, another orbit has been reserved for potential thrusting at the end of each cycle. The resulting duration of the operational cycle in Orbit B is 19 orbits, equivalent to 9.2 days. Imaging of the surface continues for 10 cycles, lasting 92 days (that last cycle does not include an extra playback orbit or orbit maintenance orbit). The end of Orbit B1 observing is timed to complete before the polar eclipse season starts at this altitude. While the spacecraft could withstand short eclipses, operations are simplified by transitioning before the start of eclipse season. Figure 17 illustrates the layout of the ten cycles of observations in Orbit B1.

4.3.3 Illumination Constraints and Mapping Coverage

As described earlier, the placement of Orbit B1 on the timeline is not ideal for illumination conditions. It is understood that at most 67% of the surface will be illuminated with incidence angles $< 70^\circ$ at the start of Orbit B1. However, Orbit B1 provides an excellent opportunity to map the south polar regions. During the 10th cycle, the entire southern hemisphere is fully observable with incidence angles $< 70^\circ$.

The northern hemisphere becomes illuminated in Orbit B2. Starting with viewing similar to the beginning of Orbit B1, the sun moves towards the north throughout Orbit B2, continually revealing new terrain. The timeline accommodates 11 cycles of Orbit B2, and by the end of the phase, the surface is well illuminated up to 45° North latitude. The final cycle of Orbit B2 will be imaged at a nadir attitude to allow for multispectral coverage to the highest extent towards the north pole. All imaging requirements could be met in Orbit B2 alone with a cumulative coverage of 91% of the surface, but when combined with the south polar views, the coverage increases to 92%. The operations margin of the mission is allocated to the end of Orbit B2. If any operations margin is consumed during the earlier orbit phases, then Orbit B2 will be delayed and execute during periods with better lighting without risk of introducing a gap between B1 and B2 coverage. If the operations margin is not consumed, then Orbit B2 observing can extend to the nominal end of prime mission and provide coverage up to 75° North latitude. There is a solar conjunction towards the end of the nominal observing period when the spacecraft cannot communicate with the DSN due to interference from the sun, but there is nothing to preclude the spacecraft from continuing to image the surface during conjunction. The images would be stored onboard until conjunction has ended, and the data transmission can resume. Figure 18 illustrates the illuminated surface available for observing in both Orbits B1 and B2.

4.3.4 Accommodation of Optical Topography Imaging Requirements

The imaging science requirements for Orbit B can be operationally categorized into two separate but compatible campaigns: multispectral nadir mineralogy mapping described above, and single-filter off-nadir topography. To acquire an off-nadir imaging dataset that supports use of optical topography, the imaging plan must meet a set of lighting and stereo separation requirements as described previously in Table 6. The two numerical optical topography techniques used by the Psyche project, stereophotogrammetry and stereophotoclinometry, benefit from imaging datasets with slightly different attributes. For more details on these techniques see Jaumann et al. (2022). The goal of the observing campaign is to select off-nadir pointing angles that will satisfy both sets of stereo criteria. Selecting off-nadir angles

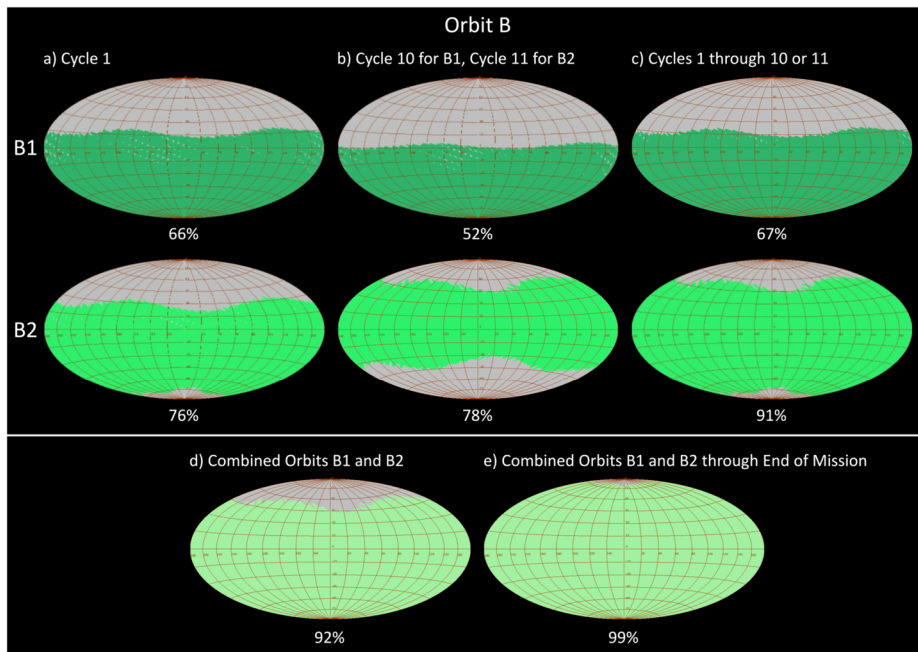


Fig. 18 The illuminated surface available with incidence angles $< 70^\circ$ at the beginning and end of Orbits B1 and B2 is shown here. (a) Cycle 1 coverage for both Orbits B1 and B2 are shown. The improved surface illumination in Orbit B2 is evident in the coverage results. (b) Coverage results for the final cycle for Orbits B1 (Cycle 10) and B2 (Cycle 11) are shown. (c) The accumulated coverage over each science orbit phase for Orbits B1 and B2. (d) Coverage results from combining Orbits B1 and B2. (e) The addition of bonus mapping in the operations margin at the end of Orbit B2 brings the total mapping coverage to 99% for the mission

in groups of four that form a rectangle centered on nadir when plotted in terms of look direction provides a more robust stereo dataset that accommodates the requirements for both stereophotoclinometry and stereophotogrammetry.

The current baseline that satisfies our requirements is the set of eight fixed off-nadir attitudes shown in Fig. 19 (a) where “Ahead” refers to looking ahead of or behind the spacecraft velocity vector and “Cross” is looking perpendicular to the velocity vector. On the Psyche spacecraft we command these attitudes in terms of the “local vertical” and “local horizontal” directions. With eight off-nadir cycles, we were able to construct two different sets of complementary off-nadir attitudes. The additional cycle at the end of Orbit B2 returns to the nadir attitude. The imaging plan was modeled using these attitudes and then scored against the topography criteria in Table 6 with the CKVIEW multi-mission planning tool (Matz and Roatsch 2018). CKVIEW evaluates the attributes of all images of each surface element against the criteria and provides a score of the number of contributing views, enabling science planners to predict how well the observation plan will support stereo image processing. Four or more views of each surface element are desired for robust topography extraction although three views can be sufficient. The results shown in Fig. 19 (c) and (d) for the “useable” criteria demonstrate that greater than 80% of the surface satisfies the topography criteria for both techniques using the complete datasets of Orbits B1 and B2. The combination of the extensive south polar mapping in Orbit B1 with the optimal equatorial surface illumination in Orbit B2 creates a robust imaging dataset for topography. Stereophotogrammetry

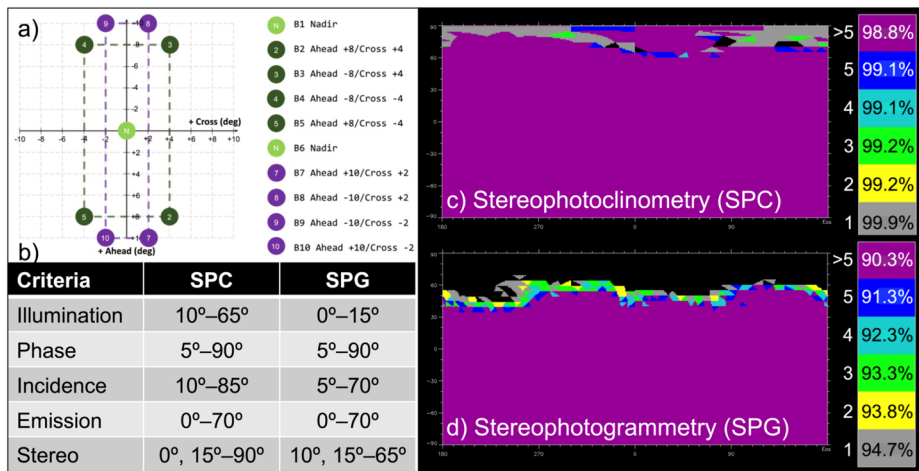


Fig. 19 Summary of Orbit B topography results. (a) Representation of off-nadir angles planned for Orbit B. The dashed green and purple rectangles emphasize the two sets of four off-nadir angles chosen. (b) Definition of “useable” stereo criteria used to evaluate expected performance in reconstructing topography from the planned imaging campaign for both stereophotoclinometry and stereophotogrammetry. (c) Stereo coverage on a latitude/longitude body map using the stereophotoclinometry criteria. Colors defined in legend show how many stereo views are available for each element of the surface. Coverage results are cumulative with the requirement being > 80% for four or more views. Results of 99.1% satisfy requirements with margin. (d) Stereo coverage using the stereophotogrammetry criteria. Results of 92.3% satisfy requirements with margin

toclinometry techniques can incorporate images with incidence angles above 70° resulting in higher coverage results than shown in Fig. 18.

Orbit B1 provides topography for only the southern hemisphere, but that is sufficient for us to design and implement the thrust sequences for the complex orbit transfer to Orbit D.

4.4 Transferring to and from Orbit D

4.4.1 Objectives

Between each science orbit, the spacecraft must thrust with the electric propulsion subsystem to achieve the next set of orbit requirements. The objectives of these orbit transfers are strictly driven by navigation and engineering to safely transport the spacecraft to the next science orbit. The transfers from Orbit B1 to D and from Orbit D to C are unique because they include a plane change between polar and equatorial orbits plus additional objectives for the GRNS and Magnetometry Investigations.

The GRNS Investigation benefits from this fortuitous opportunity to bookend their primary science phase, Orbit D, with two high-altitude calibrations where the spacecraft is far from gamma rays and neutrons originating from Psyche’s surface. GRNS requires 40-days of background calibration at an altitude greater than 3.5 body radii after capture into Psyche orbit and in the vicinity of Orbit D. This requirement is satisfied during these two large orbit transfers as long as the GRS is configured in its full science mode, cooled to cryogenic temperatures.

The Magnetometry Investigation is searching for evidence of Psyche’s remanent magnetic field by sampling the region around Psyche at different geometries with respect to the solar wind and with respect to the body. The additional viewing points provided by transfer

orbits, which allow for continuous magnetic field observations over a large range of distances, will augment science opportunities for the Magnetometry Investigation to capture structures like a magnetopause or magnetic tail, whose actual distance from the body is not known. Data collected during the transfer orbits will aid in characterizing the shape of the interaction region between the asteroid and the solar wind, which is critical for extracting the fields caused by any remanent magnetization of the asteroid.

4.4.2 Transferring Between Polar and Equatorial Orbits

The orbit plane change required to transition from a polar orbit to a nearly equatorial orbit is most efficiently accomplished at a higher altitude than Orbit B1. Orbital velocity decreases with altitude and rotating a smaller velocity vector requires less change in spacecraft velocity, or delta-V (ΔV), than a larger one. Consequently, it can be more efficient to temporarily raise and lower the orbit to perform an inclination change than to execute the maneuver at the original altitude. This is especially true for a low-thrust spacecraft that cannot quickly perform maneuvers at optimal locations. Therefore, before spiraling down to Orbit D, the spacecraft will spiral up to altitudes significantly higher than Orbit A (see Fig. 20). These orbit transfers are significantly longer than the other transfers, at 98 days for transfer into Orbit D and 90 days to transfer into Orbit C. These duration estimates currently contain 50% margin, and the transfer to Orbit D also must accommodate a period of solar conjunction when the sun is between the spacecraft and Earth and blocks our ability to communicate with the spacecraft. Finally, the orbit transfer from Orbit B1 to Orbit D needs to achieve the high-altitude plane change before eclipses are predicted to occur for polar orbits but arrive in Orbit D after the period when eclipses are expected in equatorial orbits.

The specific implementation details of these orbits transfers are still being designed as they depend on resolution of other trades that the spacecraft team is working during the cruise phase.

4.4.3 Instrument Activities During These Orbit Transfers

While all aspects of the orbit transfer design are driven by the need to navigate the spacecraft between the polar orbits Orbit B1 and C on either side of the equatorial Orbit D, activities for GRNS and the Magnetometer can be accommodated without impacting the engineering objectives. Aside from these orbit transfers, Orbit D is the only phase where GRS will be operating its HPGe detector at cryo-temperatures. Prior to configuring for Orbit D, the HPGe crystal will be annealed near the end of Orbit B1 operations. The detector will be cooled and the instrument configured for its high-rate science collections. This allows GRS to collect calibration data during the transfer up to high altitudes before and after collecting its primary dataset in Orbit D. The requirement to accumulate 40 days of calibration data can be achieved by combining the data from both orbit transfers.

No additional commanding is required for the Magnetometer during the orbit transfers; however, the quality of the magnetic field measurements is degraded while the Hall thrusters are operating. There are frequent coast periods for communication with the spacecraft when the thrusters are off, and data collected during these coast periods is an important contribution to this study. It is possible that the thruster signatures can be calibrated out of the data if it would appear to be of high value. However, even after calibration, the noise floor would still be about twice as high during thrusting intervals than during coasting intervals, so the quality of the fully calibrated data would be degraded for these intervals.

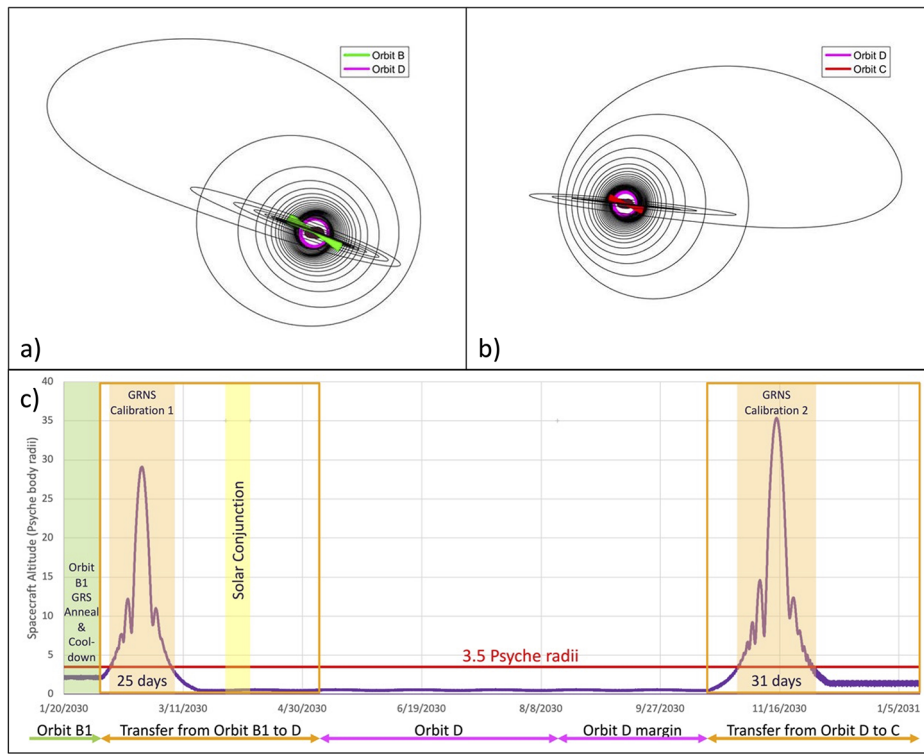


Fig. 20 Illustrations of an example (a) orbit transfer from Orbit B1 to Orbit D and (b) orbit transfer from Orbit D to Orbit C showing the plane change at high altitude. Note that these transfer orbits reach altitudes of 29 and 35 Psyche radii, respectively, far exceeding the highest science orbit altitude. (c) This graphical timeline demonstrates the placement of the GRNS calibrations during the high-altitude periods of the orbit transfers before and after Orbit D. The purple curve is the spacecraft altitude in units of Psyche body radii. Thicker portions of the line are due to altitude fluctuations. The horizontal red line at 3.5 body radii is the altitude above which the GRNS can obtain useful background calibration data. The 40-day calibration requirement is satisfied by combining the 25 days from the first transfer with the 31 days from the second transfer. The orbit transfer durations are designed to have 50% margin as they are expected to increase in duration once all constraints are accommodated. The transfer to Orbit D includes a 10-day solar conjunction period

4.5 Orbit D: Determining Psyche's Elemental Composition

4.5.1 Objectives

Orbit D differs significantly from the other science orbits in the mission. Its design is driven by the requirement to orbit less than one Psyche body radius from the surface to provide gamma-ray and neutron data collection with sufficient signal from Psyche's surface compared to the background counts from galactic cosmic rays. Data from GRNS will provide global abundances of key elements that could answer the question of whether Psyche is a core. Identifying the major elements in silicate compositions may help discriminate between achondrites, cumulates, and chondritic compositions. A global average nickel abundance measurement of less than 4 wt% indicates Psyche formed under reducing conditions, whereas abundances greater than 12 wt% indicate highly oxidizing conditions. Lastly, mea-

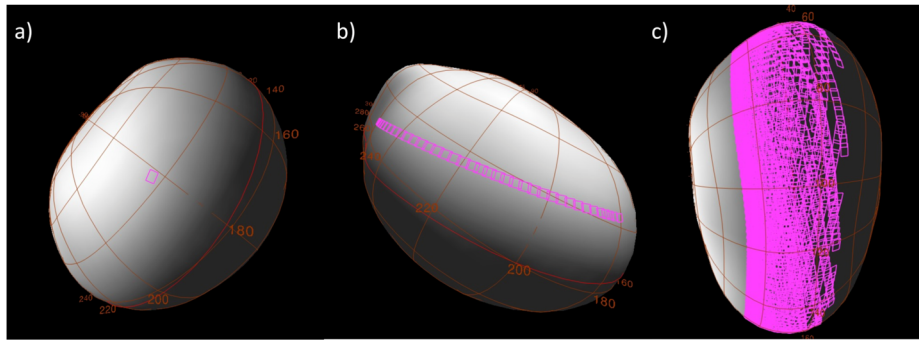


Fig. 21 Imaging in Orbit D taken from the perspective of the spacecraft. (a) One Imager FOV. (b) One nadir swath of 37 images covering about 25% of the lit-side pass. The trajectory hugs a band around the terminator for the entire phase. (c) The equivalent of 25 days (171 orbits) of imaging with 37 images/orbit. Placement of the images over features of interest viewable from a nadir attitude will be planned during Orbit B1 execution and the transfer to Orbit D

surements of the average abundance of light elements that appear to be in the metal phase can be compared to models of their abundances in Earth's core.

Gravity science requires the equivalent of 35 days of DSN tracking with the Sun-Earth-Probe angle greater than 40° . While Orbit D begins with very low Sun-Earth-Probe angles and a solar conjunction, there are 118 calendar days where the Sun-Earth-Probe angle meets requirements corresponding to 59 cumulative days after accounting for a more conservative 50% DSN duty cycle. With high quality gravity tracking at low altitude, we expect to recover the gravity field to degree and order 12 for Psyche's equatorial band, the only region sampled from this high inclination orbit (Zuber et al. 2022). The global gravity dataset is acquired later in Orbit C.

OpNav requires a minimum of 24 images per lit side pass to support orbit determination; however, any science imaging acquired in Orbit D can be used to meet the OpNav requirement as long as it is distributed across the observing period. The data budget limits acquisition to 37 images per orbit which is about a quarter of the images needed to cover each lit-side swath; therefore global mapping is not possible at this altitude.

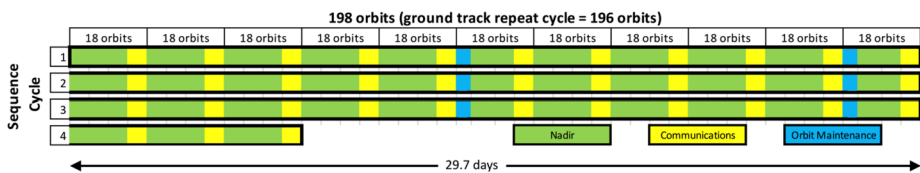
Opportunistic imaging is planned to capture higher resolution imaging of features of interest identified during Orbits A and B1 that are visible along the spacecraft ground track in Orbit D. The high ground speed will limit useful imaging to the polychromatic filter to reduce exposure time and avoid smear. The equatorial orbit constrains the terrain accessible for imaging as shown in Fig. 21. Off-nadir pointing is restricted to 10° around nadir. This allowance is intended to provide relief from accumulated pointing errors but can also be used operationally if needed.

4.5.2 Implementation

As described in the previous section, an almost equatorial orbit (160° inclination) is required to achieve the required orbit stability for the current Psyche shape model. While the baseline orbit is sufficient to provide good coverage for the GRNS instrument, the final orbit inclination will be re-evaluated once the spacecraft is in Orbit A and Psyche's gravity field is mapped. If possible, the inclination will be reduced to provide better coverage for most science investigations. The equatorial inclination benefits the Magnetometry Investigation

Table 11 Orbit D planned characteristics

Orbit D start date	5/7/2030	Average orbit period	3.6 hours
Orbit D end date	10/17/2030	Orbit resonance	153:196
Duration	163 days	Orbits per ground track repeat cycle	196
Average radius	198 km ($1.8 R_{\text{Psyche}}$)	Ground track repeat cycle duration	29.12 days
Average altitude	75 km ($0.8 R_{\text{Psyche}}$)	Operational repeat cycle duration	29.7 days
Average inclination	160°	Number of operational repeat cycles	5.6
Orbit beta angle range	[55–85] deg	Number of orbits	1086
Maximum ground speed	125 m/s	Imager pixel scale	4 m/pixel

**Fig. 22** Graphical depiction of the four cycles for Orbit D. Each ground repeat cycle is 196 orbits divided into roughly 75% nadir observing (green) for GRNS and 25% with HGA pointed to Earth for communications (yellow). The 18-orbit template was selected to provide the opportunity to evaluate spacecraft health every two days

by allowing sampling of new orientations of Psyche with respect to the solar wind. Details of Orbit D are listed in Table 11.

An advantage of the new mission profile is that eclipses are no longer expected in Orbit D. The spacecraft has been qualified to withstand eclipses up to 65 minutes in duration if circumstances change due to updates in Psyche's physical parameters or a shift in operational timeline, as originally included in the plan for the 2022 mission.

Unlike Orbits A and B1, the short 3.6-hour orbit period precludes turning the HGA to the DSN every orbit, so the spacecraft is pointed towards Psyche for 13.5 orbits to collect science data and then the HGA is pointed towards Earth for 4.5 orbits of telecommunication and Doppler data for gravity science. The turns can occur on the lit or dark side of the orbit because imaging is very limited in this phase. There is a 3-orbit period allocated to orbit maintenance at the beginning of the sixth and eleventh science periods. One ground track repeat cycle in Orbit D is 196 orbits. The operational repeat cycle shown in Fig. 22 is 11 sets of 18 orbits for a total of 198 orbits.

The GRNS will already be configured in its full science collection mode coming into Orbit D and is expected to require little commanding during this phase. GRNS does not need the surface to be illuminated to collect gamma rays and neutrons, but the spacecraft is required to point within $\pm 10^\circ$ of nadir during the data collection periods with a reconstructed Psyche-relative pointing knowledge uncertainty of 1° per spacecraft axis.

Modeling of GRNS instrument performance indicates that 52 days of data collection while pointing nadir is sufficient to accumulate enough signal to detect calcium (Ca) (Lawrence et al. 2025). The Ca requirement was used because it is the longest accumulation time of any of the required elements. Nickel, a key element in the science requirements, requires only 36 days of accumulation time and iron can be detected in less than a day. The 52 days of nadir accumulation will take 72 calendar days to acquire in Orbit D assuming a 75% nadir duty cycle and an orbit maintenance period. This provides 91 calendar days of margin in Orbit D. These values are based on an assumed orbit altitude of $0.8 R_{\text{Psyche}}$. There

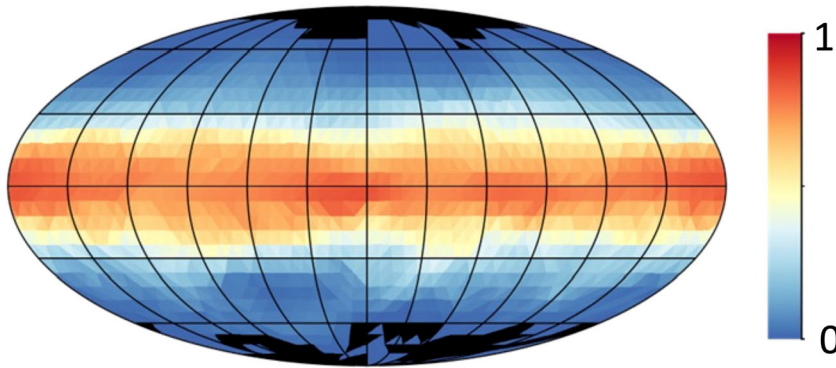


Fig. 23 Surface sampling. The map shows the relative contribution of surface regions to the response of the GRNS while in the Orbit D with the Shepard et al. (2017) shape model. Since the orbit has high inclination, the equatorial band between $\pm 20^\circ$ latitude contributes more than the mid-to-high latitude regions. This is primarily due to the solid angle of surface parcels at higher latitudes being lower than those at the equatorial sub-spacecraft points. When averaged over all orbital measurements, 76.5% of the surface contributes $>5\%$ of the maximum signal to the instrument response. This threshold criterion is used to evaluate coverage for different orbits. The sampling map is from an Orbit D “bounding cases” study for which several permutations of Psyche’s shape and corresponding orbits were considered. The map was normalized so that the maximum sampling intensity across all permutations considered was 1. The baseline orbit provides ample sampling of the equatorial region

may be additional time at other attitudes where GRNS would have an unobstructed view of Psyche by virtue of the boom. GRNS can sense Psyche even when not nadir pointed if the spacecraft is not blocking its view of the asteroid, but this time is considered bonus data and not part of the formal requirement. Events that would consume Orbit D margin are strong solar energetic particle events that can dominate the Psyche-originating gamma rays and neutrons for days to weeks, or a spacecraft safe mode entry that would ramp down the GRS high voltage. To avoid warming the HPGe crystal during safing events, which would then require a lengthy anneal of the crystal to recover, spacecraft fault protection does not always power off the GRS cryocooler during transition to spacecraft safe mode (see Sect. 5.3.8 for details).

4.5.3 GRNS Surface Coverage

Due to the large field of view of GRNS, pointing the instrument nadir in Orbit D provides greater surface coverage than what would be expected from inspecting the spacecraft ground track alone. The instrument response was modeled for the Orbit D scenario to validate the requirement that greater than 70% of the surface is sampled. The results of this analysis are shown in Fig. 23 for the Shepard et al. (2017) shape model and 2022 launch trajectory. The current Orbit D trajectory is similar, and the timeline now includes an additional 63 days of margin at this altitude.

To investigate the signal intensity and spatial resolution of GRNS in Orbit D, Fig. 24 provides a histogram of solid-angle-equivalent altitudes (h_{eq}) for a series of orbital locations sampled uniformly in time, analogous to GRNS data acquisition. The equivalent altitude has units of body radii and is the altitude above a sphere of unit radius that gives the same solid angle as that of the Psyche shape model (Prettyman et al. 2019). As such, h_{eq} has an inverse relationship with signal intensity. For comparison, low altitude elemental mapping orbits by Dawn at Vesta and Ceres had mean altitudes of about 0.8 body radii (Prettyman

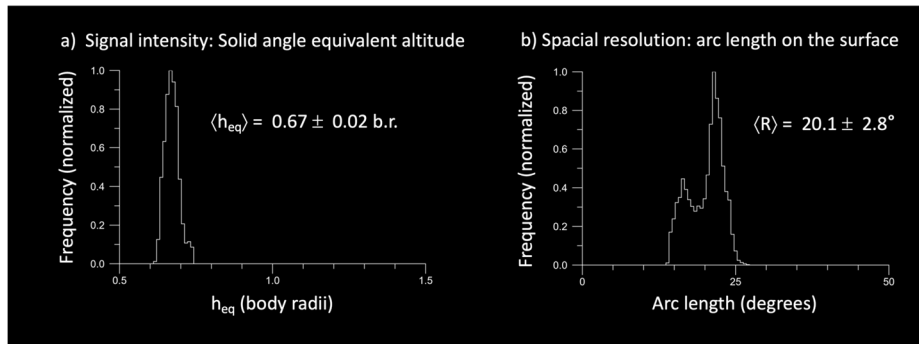


Fig. 24 Signal intensity and spatial resolution for the Orbit D baseline. (a) A histogram of solid-angle-equivalent altitudes (h_{eq}) for a series of orbital locations sampled uniformly in time. (b) A histogram of the width of the GRNS instrument response (1-sigma arc length on the surface) determined for the same time series of locations

et al. 2012, 2017). The measurement geometry of the Psyche's baseline Orbit D, with a mean h_{eq} of 0.67 body radii, would have higher signal intensity than achieved by Dawn. The variability of h_{eq} indicated by the histogram reflects variations in orbital altitude and the shape of Psyche. Also shown is a histogram of the width of the GRNS instrument response (1-sigma arc length on the surface) determined for the same time series of locations. Surface regions with different compositions can be resolved by the instrument if they are separated by about 2-sigma (Prettyman et al. 2019). The average 1-sigma width is 20° , which implies a full width half maximum resolution of about 40 degrees arc length on the surface. This would enable elemental analyses of roughly 10 separate regions in the equatorial band.

4.6 Orbit C: Gravity Science, Improved Topography, and Magnetometry

4.6.1 Objectives

Orbit C is designed to provide the conditions necessary to meet the gravity science requirements that require 35 days of tracking with the DSN at a Spacecraft-Earth-Probe angle greater than 40° in a polar orbit with global ground track coverage at an altitude less than 1.7 Psyche radii. The Orbit C baseline altitude of 174 km is approximately 1.7 Psyche radii, satisfying the gravity science requirement. To support gravity modeling, additional imaging from nadir and off-nadir attitudes provides a higher resolution shape model for the illuminated southern hemisphere to supplement the model produced with data from Orbit B. The smaller projected FOV on the surface increases the number of image stations needed for coverage (Fig. 25) and precludes imaging with all narrowband filters within the available data budget, although some filters may be added once in orbit if the data budget allows. There are no imaging surface coverage requirements for Orbit C.

The altitude is also sufficiently low to meet the magnetometry requirement to determine whether Psyche has a remanent magnetic field, although that requirement is expected to have been satisfied in Orbit D. Magnetometer requirements are compatible with and less driving than gravity science and imaging so specify no special attributes of Orbit C.

4.6.2 Implementation

A polar orbit is required to provide the necessary ground track coverage to meet the gravity science requirements. The 100-day duration of this phase is driven by the gravity science

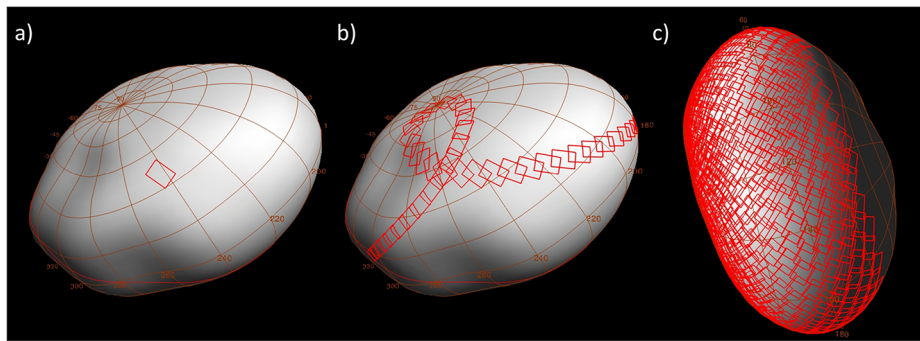


Fig. 25 Imaging in Orbit C taken from the perspective of the spacecraft. (a) One Imager FOV projected on the surface in the southern hemisphere; (b) One orbit of nadir imaging; (c) One cycle of nadir imaging coverage if every orbit was able to be imaged. The operations plan skips multiple orbits for communications passes as shown in a later figure. The northern hemisphere is not well illuminated

Table 12 Orbit C planned characteristics

Orbit C start date	1/15/2031	Average orbit period	7.2 hours
Orbit C end date	4/25/2031	Orbit resonance	77:45
Duration	100 days	Orbits per ground track repeat cycle	45
Average radius	295 km (2.7 R_{Psyche})	Ground track repeat cycle duration	13.46 days
Average altitude	193 km (1.7 R_{Psyche})	Operational repeat cycle duration	15.6 days
Average inclination	89.7°	Number of operational repeat cycles	6.4
Orbit beta angle range	[35–38] deg	Number of orbits	333
Maximum ground speed	71 m/s	Imager pixel scale	9 m

objective to acquire the equivalent of 35 days of tracking with the DSN at a Spacecraft-Earth-Probe angle greater than 40° . Conveniently, Orbit C begins with the Sun-Earth-Probe angle of 136° and ends at 56° , clearly exceeding that requirement. This provides high confidence that the requirement to resolve Psyche's gravity field to degree and order 10 can be satisfied by the operational plans (for details see Zuber et al. 2022). Details of Orbit C are listed in Table 12.

The architecture of Orbit C is a blend of the approach used for Orbits A and B with that of Orbit D. The short 6.7-hour orbit period motivates grouping multiple orbits when the spacecraft is pointed for imaging followed by grouping orbits pointing HGA-to-Earth for improved gravity science and communications. We allocate 75% of each cycle to imaging with LGA tracking and 25% to HGA communications, neglecting the small amount of time lost to spacecraft turns between those attitudes. Twice per cycle, two orbits are reserved for orbit maintenance maneuvers. As a result, some lit side passes each ground track repeat cycle must be sacrificed while the Imager is turned away from Psyche. The 52-orbit operational cycle shown in Fig. 26 was developed for the original 49-orbit Orbit C resonance and has been adapted to the current 45-orbit resonance. The resulting operational cycle is seven orbits longer than the spacecraft ground track repeat cycle, which intentionally moves the downlink passes to occur over different longitudes each cycle, avoiding systematic gaps in coverage. This strategy also benefits gravity science by providing evenly distributed HGA Doppler tracking over all longitudes. Scanning the chart vertically, each longitude band (as represented by an orbit number in the cycle), has at least two nadir observing passes.

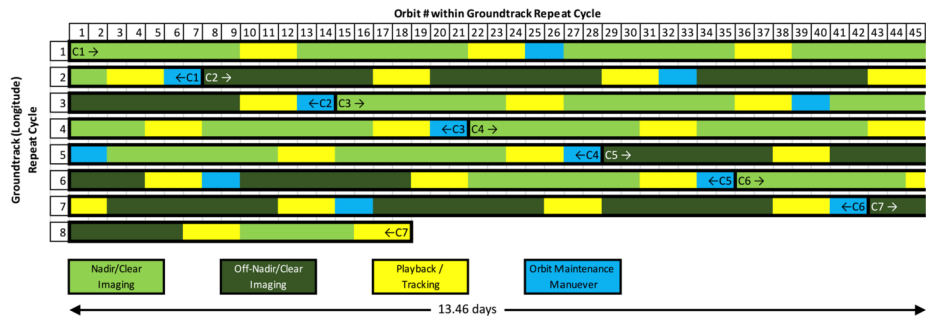


Fig. 26 Orbit C architecture showing observing periods (green), HGA communications (yellow), and orbit maintenance maneuver windows (blue). Each ground track repeat cycle is 45 orbits in duration and each operational repeat cycle is offset from the previous ground track repeat cycle by seven orbits to shift the HGA tracks

Most have at least one HGA pass as well. Future optimization of the template could yield increased off-nadir opportunities for some cycles.

To improve upon the Psyche shape model, four observation cycles are performed on a best-effort basis with the spacecraft at off-nadir angles. The lower altitude drives the selection of larger off-nadir attitudes (dark green periods in Fig. 26) than used in Orbit B. Most portions of the surface will only have two off-nadir angles in addition to nadir.

4.6.3 Illumination Constraints and Mapping Coverage

Orbit C begins with 56% of the surface illuminated, primarily the southern hemisphere, but more of the northern surface is illuminated as the cycle continues. Figure 27 shows how the imaging swaths of the illuminated surface are combined throughout the orbit phase.

The magnetic field mapping coverage provided to the Magnetometry Investigation is computed differently than for the Imager. To retrieve the magnetic field and address the Magnetometry Investigation objectives, the spacecraft trajectory must provide good coverage not only of the surface but also of the three-dimensional magnetic topology around the body. For a given magnetized feature within the body, the magnetic topology extending into space changes continuously due to its interaction with the highly time-variable solar wind. To mitigate this variability, the data are binned for different solar wind conditions and in two reference frames: the Psyche-body-fixed frame, which rotates with the body, and the Psyche-solar-orbit frame, which is tied to the solar wind-body interaction (see Weiss et al. 2023 for details).

To assess the quality of the science orbits for magnetic field retrieval, the body's surface is divided into $10^\circ \times 10^\circ$ bins in longitude and latitude in the Psyche-body-fixed frame. First, we count how many total measurements would be taken in each bin, assuming the Magnetometer operates in its nominal science mode at a frequency of 50 Hz. This can yield up to 10^5 measurements per bin. However, the total number of measurements in a given Psyche-body-fixed frame bin is distributed over several Psyche-solar-orbit frame locations as the spacecraft orbits the body and the sun vector changes, resulting in different magnetic topologies extending above the bin. We estimate the number of different topologies by counting the number of unique Psyche-solar-orbit frame views of the same Psyche-body-fixed frame bin (where each Psyche-solar-orbit frame view is defined as the spacecraft being within a $10^\circ \times 10^\circ$ bin in longitude and latitude around a central value).

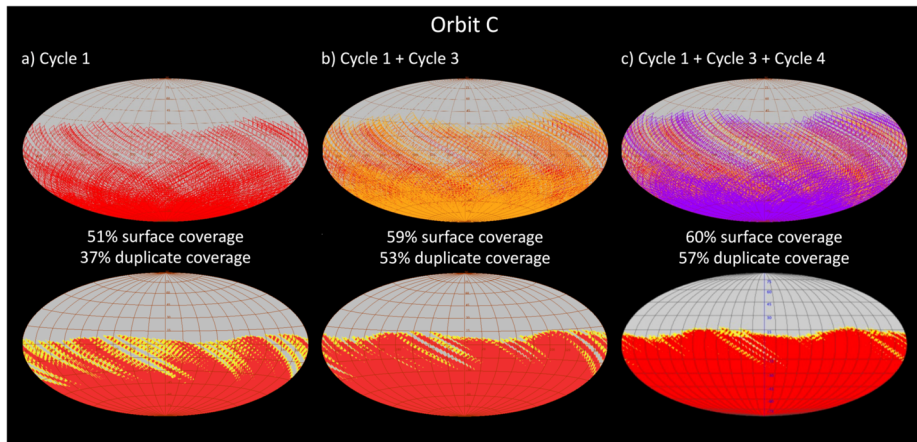


Fig. 27 Buildup of nadir coverage with incidence angle $< 70^\circ$ in Orbit C following the operations schema shown in Fig. 26. (a) Cycle 1 coverage: Top map shows individual Imager FOV and bottom map shows total coverage. Yellow indicates areas imaged only once and red indicates areas imaged more than once. (b) Combined coverage from Cycles 1 and 3. Top map shows individual FOV for Cycle 1 in red and Cycle 3 in orange. Bottom map shows total coverage. (c) Combined coverage from nadir Cycles 1, 3 and 5. Top map shows individual FOV with Cycle 5 in purple, and bottom map is total coverage. By Cycle 4, gaps from downlink passes and maneuvers in earlier cycles are filled in with robust redundant coverage. The northern hemisphere is not well illuminated during this period

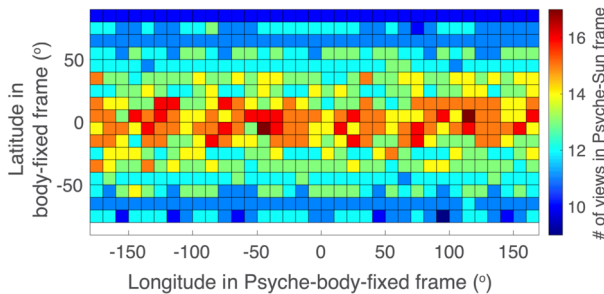


Fig. 28 Colored boxes show the number of unique views in the Psyche-solar-orbit frame from which a given area on the body would be observed by the spacecraft during Orbit C. Each view corresponds to a different interaction between the solar wind and the same region of Psyche's surface. Data are binned over longitude and latitude in the Psyche-body-fixed frame. Each bin is $10^\circ \times 10^\circ$, and the spacecraft position is sampled at 1-s intervals. The views in the Psyche-solar-orbit frame are also binned at $10^\circ \times 10^\circ$. Each view comprises tens of thousands of measurements taken during different segments of the orbit

The results for Orbit C are shown in Fig. 28. We find that each possible magnetic feature would be observed from more than 10 different topologies (and likely more, due to solar wind variability), with each configuration measured many thousands of times. This provides confidence that statistical analysis and related techniques will enable detection of the underlying signal.

Following completion of operations in Orbit C the spacecraft has a 23-day orbit transfer to begin Orbit B2. Activities in Orbit B2 were previously described in Sect. 4.3.

Table 13 The completion of each instrument science measurement requirement can be tracked across the science orbits. Yellow cells indicate significant progress is made towards that objective. Green cells indicate the measurement requirement is met or exceeded. Blue cells indicate additional measurements provide margin against requirements or science margin toward meeting that objective. Purple cells indicate enhanced resolution over limited areas. The “to be determined” (TBD) in Orbit C reflects the plan to update the topography campaign prior to arrival

Observation Type	Science Measurement Requirement	Capability in Baseline Mission				
		Orbit A	Orbit B1	Orbit D	Orbit C	Orbit B2
Imaging nadir (color filters)	Metal to silicate fraction: 80% of surface at pixel scale <500m.	82% @ <37m	67% @ <17m			~99% @ <17m
	Oldhamite mapping: 80% of surface at pixel scale <200m.	82% @ <37m	67% @ <17m			~99% @ <17m
Imaging nadir (clear filter)	Crater counting: 50% of surface at pixel scale <200m.	82% @ <37m	67% @ <17m	Limited @ <5m	60% @ 11m	~99% @ <17m
	Geologic mapping: 80% of surface at pixel scale <20m.		67% @ <17m	Limited @ <5m	60% @ 11m	~99% @ <17m
Imaging off-nadir (clear filter)	Spin axis determination: accuracy to 0.5° 1σ.	Continued OpNav imaging in Orbit A				
	Topography: 50% of surface at <50m vertical scale.		~56% @ <34m		TBD @ <26m	~99% @ <34m
	Topography: 80% of surface at <70m vertical scale.		~56% @ <34m		TBD @ <26m	~99% @ <34m
	Topography for gravity field: Provide a minimum of 4 off-nadir mapping cycles in Orbit C.				4 cycles Significant mapping overlap provides margin.	
Magnetometer observations	Magnetic dipole: Global dataset with ground track designed so that each point on Psyche's surface lies within 10° of arc of the ground track in Orbits A, B, C.	4 global datasets <3° arc	10 global datasets <4° arc	3 partial datasets between +/-20° latitude only <2° arc	7 global datasets <3° arc	9 global datasets <4° arc
Gravity science tracking	Gravity Field: HGA tracking (Doppler + ranging) in Orbits A & B for at least 40% of each orbit when Sun-Earth-Probe angle > 20°; 35 cumulative days of tracking in Orbits C/D with Sun-Earth-Probe angle > 40°.	~43% HGA tracking	~37% HGA tracking plus >10% LGA tracking	74 days between +/-20° latitude only	100 days global coverage	
GRNS observations	Element measurements (driving case - Calcium): 52 cumulative days at nadir in Orbit D.			71 days		

4.7 Completion of Science Observation Objectives by Science Orbit

As described in the previous sections, the science orbit design combined with the science observation strategy provide the measurements identified to address the mission science objectives. Table 13 connects the instrument measurement requirements to the corresponding science orbit and indicates progress towards completing those objectives.

4.8 Extended Mission Concept

The Psyche team is aware that the elemental mapping campaign would be enhanced by reducing the orbit radius to 1.4 Psyche radii (~44 km altitude) or lower. There will undoubtedly be regions of Psyche's surface of elevated scientific interest that could benefit from higher resolution imaging at a lower altitude. The primary mission will improve knowledge of Psyche's gravity and shape as well as understanding how the flight system performs in orbit and what consumables are remaining. This would enable planning for lower orbit operations; details of any specific orbit altitude and other science requirements would be developed during the primary mission and proposed to NASA for consideration.

5 Mission Operations

5.1 The Psyche Team

Psyche mission operations is a joint venture of the Psyche science team and flight operations team along with other partners such as the DSOC project, the DSN, the Small Bodies Node

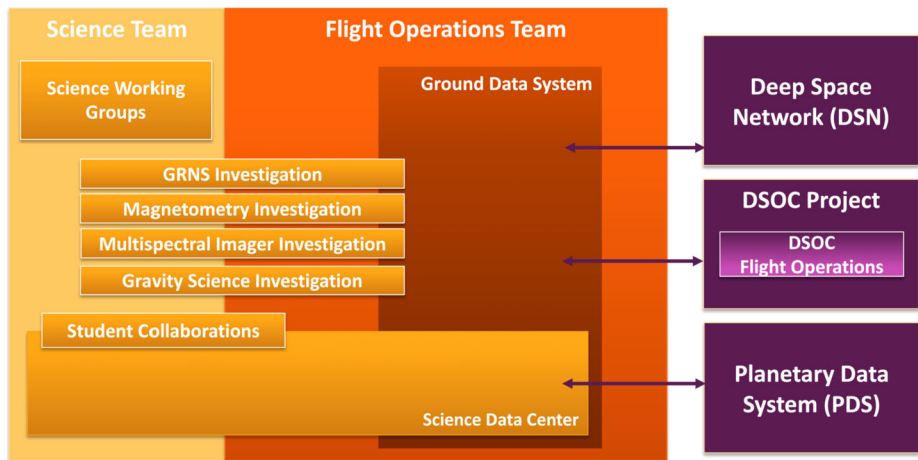


Fig. 29 Project decomposition and external interfaces. The instrument investigation teams and the Science Data Center have responsibilities to both the science team and the flight operations team. While all NASA projects interface to the DSN and PDS, the interface to the DSOC project is unique to Psyche

of the Planetary Data System, and sponsors at NASA. Interfaces between the Psyche project and other close partners are shown in Fig. 29. A vital component of the team not discussed elsewhere in this article is the Student Collaborations Program headquartered at ASU that engages hundreds of undergraduate students each year in capstone projects and in the Psyche Inspired program as described by Bowman et al. (2023). Beyond students, this group also develops the Psyche Innovation Toolkit online courses and leads the Psyche Outreach Interns and Docents as well as assists with other public engagement and media outreach.

Despite the delineations in Fig. 29, the project has never entertained a divide between the science and engineering components of the team. All members of the Psyche project and many external partners are invited to our twice-yearly project team meetings, which feature a mixture of speakers from all disciplines. Prior to orbital operations, these meetings are used to educate the entire project on each other's work.

Perhaps the most critical aspect of the team culture the Psyche leadership team attempts to foster is the idea that every voice needs to be heard. This is an explicit invitation for any person working on any part of the team to speak up with issues or concerns at any time. We have an unofficial motto that “the best news is bad news brought early,” allowing enough time for the issue to be resolved. Concerns need to be met with support and inquiry and their bearers embraced. This approach has allowed, again and again over the years, the most junior people on the team, those with their hands on the hardware or their fingers into the code, to raise issues in time to avoid bad outcomes. In fact, the times that problems surprised us were almost always times when someone had not been listened to attentively enough.

We have also led a team norms workshop to realign the team after initial checkout and the stress of making the launch was complete. The team came up with the following norms:

- Behaviors of Individuals: We work to build an environment of mutual trust.
- Ways we Interact: Each person's opinions and questions matter and are heard and valued.
- Actions of Leadership: Leaders recognize each team member's professional excellence and their importance on the team.

The team will continue to revisit and stress-test these norms statements by working through dilemmas and case studies in practice for orbital operations and its attendant higher stress.

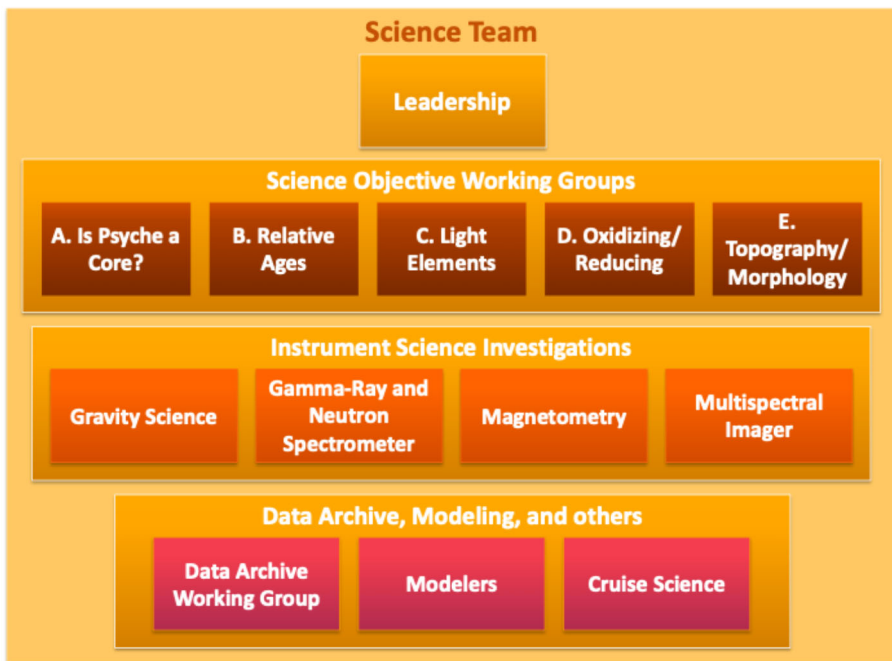


Fig. 30 The working group organization within the science team. In practice, there is significant overlap in the membership between science working groups so it is the responsibility of the working group leads to ensure that each science objective receives dedicated attention and analysis. Members of the instrument science investigations and other groups are also included in the science objective working groups

5.1.1 Science Team and Working Group Structure

The science team, led by the Principal Investigator, consists of the Co-Investigators approved by NASA, scientific Collaborators sponsored by the Co-Investigators, and Participating Scientists selected by NASA prior to Psyche arrival. All members of the Psyche team, both science and engineering, are bound by the Psyche Team Guidelines document and Psyche Communications Plan. The science team is organized into three tiers of working groups (Fig. 30). The top tier is the five Science Objective working groups, one for each of the science objectives. These groups synthesize the relevant data analysis from multiple instruments to address that specific science objective and facilitate the science by ensuring that all relevant data, people, and ideas are involved. By design, the Science Objective working groups are led by Co-Investigators who are not also instrument investigation leads. This encourages a multidisciplinary approach to exploring the key science questions of the mission. While one of the team's leading hypotheses for Psyche's origin is that it is a remnant of a differentiated planetesimal, all of the many possibilities (Elkins-Tanton et al. 2022) must be considered. The science team's organization and philosophy are to keep an open mind and avoid formulating strong conclusions until all mission science data has been collected, synthesized, and evaluated. This discipline is embedded in the ongoing discussions within the working groups and the team at large. Details of the science framework for each working group are described later in this volume (Elkins-Tanton et al. 2022; Marchi et al. 2022; Prettyman et al. 2025; McCoy et al. 2022; Jaumann et al. 2022).

The next tier contains the four instrument science investigations that assure calibrated, performant data for each of the three science instruments plus the gravity science investigation. These working groups also ensure that their data is interpreted properly by the other members of the team. The working groups in the third tier are not connected by a theme. The Data Archive working group provides the interface between the Psyche project and the Small Bodies Node of the Planetary Data System. This group works closely with the Science Data Center. There is another loosely defined group of Co-Investigators who are primarily modelers. They have no formal leadership but coordinate their modeling efforts. Finally, when topics expected to be of broad interest to the Psyche team arise, ad hoc working groups can be formed to meet specific needs.

One example of the last category is the Cruise Science working group that promotes and advocates for opportunistic science analysis activities during the interplanetary cruise phase using data collected for purposes of calibrating the science instruments (e.g., solar proton measurements by GRNS or interplanetary magnetic field data) or from engineering subsystems (e.g., single event upset data from memories used in the spacecraft avionics). Potential scientific topics include heliophysics, emphasizing space weather such as validating/improving the physics-based space weather forecasting tools for solar energetic particles and solar wind, understanding the propagation of coronal mass ejections and solar energetic particle events in interplanetary space using multi-point observations (e.g., Geostationary Operational Environmental Satellite, Europa Clipper, Jupiter Icy Moons Explorer, BepiColombo, etc.), and extrapolating the solar modulation index from the GRNS data.

5.1.2 Flight Operations Team

The spacecraft is operated at JPL in Pasadena, CA. The flight operations team is made up of the following teams: Spacecraft, Mission Planning and Sequencing, Mission Design and Navigation, Science Operations Support, Ground Data System, and Science Data Center at ASU. These teams are supported by project leadership/management, mission management, and mission operations assurance management. The Psyche project organization is shown in Fig. 31, with the flight operations team being the group of teams below mission management.

These teams work together to operate the mission—everything that needs to happen to get the spacecraft to Psyche and then acquire the science data. A brief description of each team's role follows. The functions of the Spacecraft team include the monitoring and analysis of spacecraft health and performance, planning of future spacecraft activities, and generation of spacecraft commands (both sequenced and real-time) to execute planned events. The Science Operations Support team is responsible for coordinating with the instrument teams on instrument health, as well as developing the instrument activities during cruise and at the asteroid. The Mission Planning and Sequencing team is primarily responsible for sequence/command generation, modeling, review and verification, and DSN scheduling. The Mission Design and Navigation team is responsible for tracking and predicting the spacecraft's position, designing the trajectory and maneuvers required to get to Psyche, as well as adjusting the orbits while there. The Ground Data Systems Operations team is responsible for the execution and monitoring of data processing of Psyche mission data during passes, monitoring and maintaining the Ground Data System software and interfaces used to perform operations, and for providing user support. In addition, outside of DSN passes the Ground Data System Operations team performs data accountability for downlink products received and processed by the telemetry processing system, archives and restores (when necessary) data produced by the telemetry processing system, and produces the data

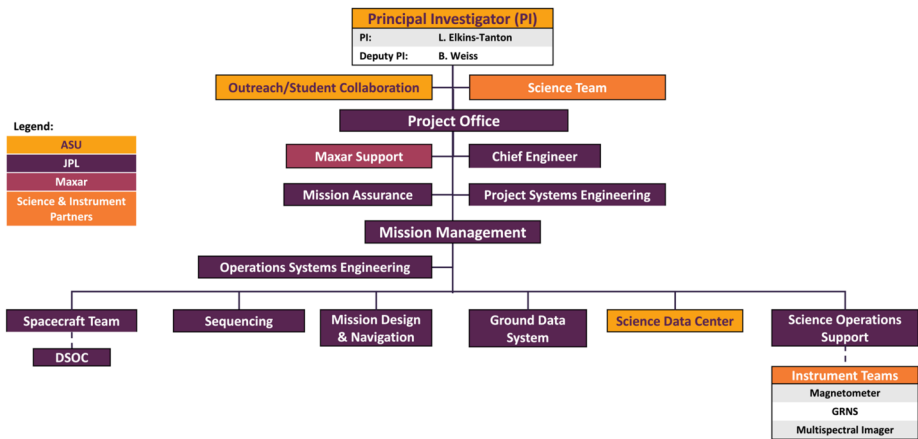


Fig. 31 Psyche Phase E project organization for operations. Many aspects of the Psyche organization are common to deep space flight projects. Unique to the Psyche project is the partnership with Maxar. While the spacecraft is operated from JPL, Maxar provides on-call support for anomaly investigation and other studies as needed

accountability report. Finally, the Science Data Center serves as the central clearing house of PDS products from the Psyche mission, generated by instrument teams and the science team. The Science Data Center is also tasked with being a source of instrument data and telemetry, used by the instrument teams to generate the RAW PDS4 data products that feed into downstream processing to generate CALIBRATED and DERIVED products. In addition to storing and disseminating project PDS4 data products, the Science Data Center team is responsible for the generation, validation, and delivery of mission PDS4 archives to the Small Bodies Node, which curates data relevant to comets, asteroids, and interplanetary dust (see Sect. 6).

5.2 Science and Instrument Operations

Science and instrument planning and sequencing are performed by the Science Operations Support team. This team has responsibility for science system engineering, science planning, science instrument sequencing, telemetry monitoring for instrument health and safety, instrument anomaly resolution, and tracking progress towards meeting the science observing (Level 2) requirements. Science operations support includes staff at JPL as well as the instrument operations teams at the home institutions of the instrument science investigations. Science data processing and archiving are the responsibility of the instrument science investigations that span both the science team and the flight operations team.

5.2.1 Science Planning and Sequencing

The Psyche science operations concept and processes were developed to support implementation of the mission's science goals and objectives that were defined early during the project's proposal phase. These science goals and objectives set the foundation for every aspect of the mission's design, planning, and execution (see Fig. 32). At the start, the initial proposal team developed the mission concept, which included specifying the target destination, selecting the science instruments, outlining the navigation timeline, and identifying

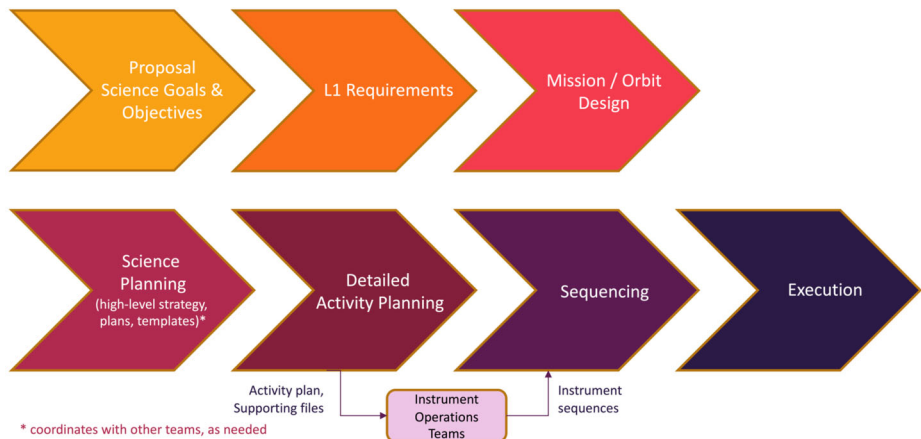


Fig. 32 Top-down approach drives science planning. Activity plans and sequences flow from top-level science goals rather than instrument-driven requests. This approach eliminates time-consuming negotiations between instrument teams for resources and ensures that the science data collected addresses the primary science objectives

the required engineering flight system capabilities. During the project's early development years, these high-level concepts were then translated into detailed requirements, which provided guidelines for spacecraft design, operations, and scientific data collection. As these requirements cascaded downward, they became progressively more detailed, extending to all subsystems and teams involved in the mission.

The mission's science objectives also led to requirements that informed the design of the science orbits, which aim to ensure that the spacecraft will be positioned in a way that allows optimal observations of Psyche. Orbit design needs to account for factors such as the spacecraft's distance from Psyche and the time available for scientific observations versus communication with Earth and downlink of data. All orbits need to be stable to allow consistent data collection from the Magnetometer and GRNS while also allowing imaging of the asteroid at various angles and lighting conditions.

Starting in the development phase and continuing throughout the mission, the mission planners who support operations use the requirements and resulting orbits to create high-level activity timelines, balancing the spacecraft's science objectives with its engineering and operational needs. These planners strategically lay out the timeline of when and how various science and engineering activities should be scheduled. They consider things such as periodic maintenance and calibration, Mars flyby activities, optical navigation and approach imaging, and time allocations for observing and downlinking data while orbiting the asteroid.

For the periods designated for science, a science planner then works to add the next level of detail to the plans by creating conflict-free observing timelines that align with the broader mission operations activity plan. These science planners must ensure that resources such as data volume, pointing attitude, and telemetry rates are carefully considered and allocated, so that the spacecraft can safely collect and transmit the necessary data within available capabilities. Depending on the science orbit phase, the science team may prioritize imaging of Psyche's surface or the collection of data for elemental mapping, magnetometry, or gravity science. With relatively infrequent commanding, the Magnetometer, NS, and anti-coincidence shield of the GRS will operate continuously throughout the mission, with Orbit

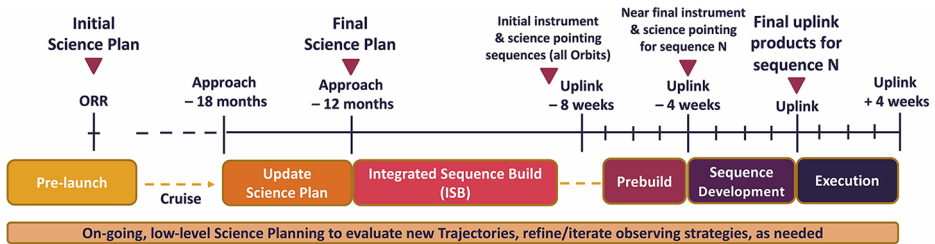


Fig. 33 Timeline for the uplink process from initial science plans through sequence development and execution. Progressive refinement of science activity plans leads to straightforward sequence implementation. Detailed science plans are developed pre-launch and then refined during cruise as additional information becomes available about the asteroid, the trajectory and arrival date, and spacecraft and instrument capabilities. The uplink planning process is designed to be strategic and iterative

D also dedicated to full gamma-ray science utilizing the instrument's high-purity germanium detector. The Imagers, which are crucial for detailed visual observations of Psyche's surface, require the most intensive observation planning. The Imager must be actively managed, and science planners must determine the optimal spacecraft pointing attitudes, ground track placements, and image footprint strategies. This includes decisions about the cadence at which individual images will be captured to achieve maximum surface coverage, along with collecting any ancillary engineering and instrument housekeeping data to support future data analysis.

Science planners work closely with instrument teams during the entire planning process to develop the detailed science observing and data collection strategies. These strategies are then captured in various ways, such as in imaging templates, and then integrated into official science activity plans. The science activity plans and supporting files then guide the instrument operations teams to add the final level of detail by crafting the specific command sequences that will be uplinked to the spacecraft from the DSN to execute the desired observations and other instrument activities. It is this top-down approach outlined in Fig. 32 that puts the Psyche mission's science goals and objectives at the heart of every decision made throughout the design and planning lifecycle and therefore ensures that the mission will provide the necessary valuable data about the target Psyche.

Throughout the mission, the high-level and detailed science operations planning processes will go through several cycles. The general timeline for process steps and refinements can be seen in Fig. 33. A couple years before approach to the asteroid, the entire science plan will be reviewed and updated as needed in response to improved information about the navigation timeline and for any changes in resources or spacecraft and instrument capabilities. At this stage in the mission, knowledge will have improved regarding arrival time, lighting conditions, timeline margins, and operational strategies for aspects like managing spacecraft momentum. The goal of this update period is to adjust and solidify the activity timeline and details enough to feed into an approximately year-long process called the Integrated Sequence Build.

During this early sequence implementation process, a full set of flight sequences will be developed for the orbital operations period, which will then serve as the initial command set to tweak and improve before actual execution once in orbit. The flight operations team will go through the process to plan, build, review, test, and document complete sequences based on the science and engineering activity plans and notional DSN tracking schedule. Besides providing a first build and validation of sequences during approach to the asteroid, science orbits, and orbit transfers between phases, this exercise also helps train and engage

all flight operations team members and identify tool limitations. The output of the process will be command sequences ready for flight. Ideally only small adjustments will be needed to align with the actual DSN schedule, arrival time to Psyche, and new knowledge about orbit characteristics from data gathered while closer to the asteroid. Liens are written to document corrections and lessons learned needed for the official sequence development.

The final steps before command files are uplinked to execute on the spacecraft occur in the Science Prebuild and Sequence Development processes. For each defined sequence execution period, the Prebuild process for science is a 3-week cycle during which the science operations team takes the previously-generated instrument activity plans and sequences, current DSN track allocations, latest ephemeris and trajectory files, and any known engineering activity inputs and creates final activity plans and science-related flight sequences. During this timeframe, cross-coordination with other teams is ensured by holding a group “kick-off” meeting to review activity plans, assess the sequence development schedule, assign responsibilities, and disposition liens and changes from the Integrated Sequence Build process. A design “tabletop” meeting is also held where all teams discuss science and engineering activities in more detail, review the intent of the planned activities and changes from previous plans, and outline details of instrument activities such as off-nadir angles and data production rates. This allows overall contextual awareness and sharing of information across the broader flight operations team. Science planners generate initial command files and interface with other teams as needed for input and validation while reviewing all sequences. The output of the process is a set of pre-validated science pointing and instrument sequences that are conflict-free and constraint-checked, which become inputs to the final 4-week Sequence Development process.

A key element of the Psyche uplink process that simplifies the transition from the Integrated Sequence Build process to the Prebuild process is scheduling most spacecraft activities relative to geometric epochs related to the spacecraft’s orbit around the asteroid. One common example is the epoch when the spacecraft crosses from the dark side to the lit side of each orbit. Psyche’s irregular shape results in each lit-side traverse having a slightly different duration and timing relative to the subsequent terminator crossings. Without the ability to easily update the timing, a sequence built based on an early plan would become misaligned with the illumination conditions if it started one or two orbits later than the original plan. Connecting commanding for each imaging swath to the dark-to-lit terminator crossings allows for the entire plan to shift and synchronize with the current conditions by simply updating a file of epoch times generated from the latest spacecraft ephemeris. Timing uncertainties are a standard characteristic of low thrust missions and extensive use of epoch-relative timing enables the team to manage those timing uncertainties with low risk to the uplink process. The epoch update process not only streamlines the transition from the Integrated Sequence Build process to the Prebuild process but also enables final adjustments during the later Sequence Development process. Timing adjustments arise from uncertainties in arrival date, orbit transfer durations, updates to science orbit characteristics, and/or interruption of the plan by spacecraft anomalies.

The main purpose of the Sequence Development process is to generate, integrate, verify, and validate commands that respond to or implement the activity plans generated in the Prebuild process. At this stage, all operations teams submit their complete set of command sequences to be modeled as an integrated set for checking constraints, timing conflicts, flight rules, and command syntax. Sequences may also be tested on testbeds that are running the latest flight software and are connected to instrument engineering models or simulations. The Sequence Development process includes two “passes” through the review and update cycles and provides opportunities for final time epoch, image exposure, and spacecraft ephemeris updates. The last step in the process once sequence reviews and approvals

are obtained from all teams is to hold a command approval meeting for all participants to give their official approval for the set of files to uplink and execute onboard the spacecraft.

5.2.2 Instrument Data Collection and Calibration During Cruise

The cruise phase provides a valuable opportunity to operate the science instrument suite and collect valuable calibration data. Early in development, the project recognized the importance of exercising both the instruments and the instrument operations teams during cruise and therefore planned for continuous operation of the Magnetometer and GRNS as a key preparation for orbital operations. Both instruments require minimal commanding unless powered off for special engineering activities or anomalies so do not add significant burden to the team during routine data collection. The Imagers are operated less frequently in cruise because they require dedicated spacecraft pointing to collect calibration data, but a robust calibration campaign was developed pre-launch.

Throughout cruise the Magnetometer monitors the interplanetary magnetic field embedded in the solar wind to provide a zero-level calibration of each sensor. The properties of the solar wind allow the zero levels of all three Magnetometer axes for each sensor unit to be determined, based on the dominant Alfvénic nature of the fluctuations in the solar wind, with fluctuations in direction but not in field strength (Leinweber et al. 2008). No special commanding is necessary although the data used for these calibrations are restricted to periods when the Hall thrusters are not operating. In addition to the interplanetary magnetic field, the Magnetometer is also sensitive to fields generated by the spacecraft subsystems (de Soria-Santacruz et al. 2020). Correlating changes in the measured field to activities on the spacecraft during cruise allows the team to remove these noise sources. For most of cruise the Magnetometer operates in its nominal 50 Hz data sampling mode although there are periods of low downlink capability where the instrument is commanded to its 10 Hz mode.

Portions of the GRNS are also powered on throughout cruise and will monitor energetic particle effects throughout the spacecraft's traverse from Earth to Psyche. Although the GRS is in reduced science mode and the HPGe sensor not cooled to cryogenic temperatures, the GRS anticoincidence shield provides energetic particle and fast neutron measurements (Lawrence et al. 2025). In addition to providing a long-term time-dependent baseline monitor of the galactic cosmic ray background, the anticoincidence shield and NS sensors observe solar energetic particle events, solar X-ray flare events, and gamma-ray burst events. Quick look data from the gamma-ray bursts are shared with the Gamma-Ray Burst Network and used along with other space assets to triangulate the location of these events in space (Hurley et al. 2010).

Every six months during cruise there are periods of higher instrument activity known as Periodic Maintenance and Calibration. The template for these instrument and engineering activities as executed onboard the spacecraft is shown in Fig. 34. This set of activities is designed to calibrate the instruments, maintain their overall health, test various behaviors, and improve the team's overall understanding of their capabilities. Each calibration period is typically four days in duration with daily two-way communications with the DSN. While most activities do not require real-time DSN coverage, these daily passes provide additional downlink opportunities and allow the calibrations to be monitored more frequently.

During a typical calibration period, both Imagers perform photometric, radiometric, and – occasionally – geometric calibrations. The first two calibrations allow the team to determine each Imager's photometric and radiometric accuracy and parameters. The third calibration characterizes their geometric distortion through the creation of a 4x3 mosaic centered on a large star cluster. While the celestial object used for each calibration is subject to

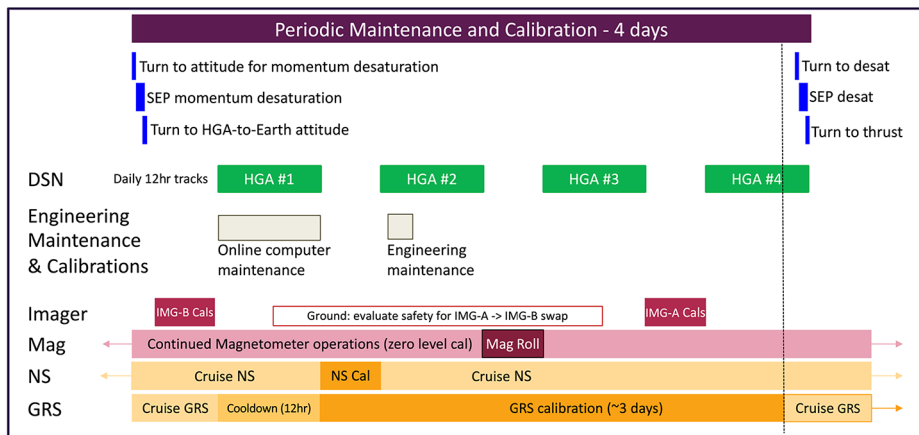


Fig. 34 Layout of the activities for a typical periodic maintenance and calibration windows. The need to thrust with the SEP system for most of cruise leaves limited time to calibrate the science instruments or perform maintenance on the spacecraft. All activities must be efficiently integrated within the 4-day timeline

change, the format of these calibrations is fairly homogeneous. First, multispectral images are collected for a bright, standard star using multiple filters and exposures. Next, the spacecraft is turned towards an object that can serve as a reflected solar light “standard” (such as Earth, Mars, Jupiter, or Saturn) and additional images are collected. If a geometric calibration is included, then the activity completes with a mosaic of a star cluster. Otherwise, a third star may be chosen for additional imaging.

The GRNS operates in its full science mode during cruise periodic maintenance and calibration windows. Early in the calibration window, the high-purity germanium (HPGe) crystal within the GRS is cooled to 85 K and a high voltage bias applied, initiating calibration data collection that lasts all four days. These measurements are designed to check the HPGe’s energy resolution and track its depletion voltage over time. Meanwhile, both spectrometers are configured to collect additional raw and diagnostic data to support trend analysis of all sensors. Other parameter tuning activities and experiments utilizing the GRNS may also be included as time permits.

Finally, the Magnetometer operates continuously throughout each calibration period. During this timeframe, the spacecraft is not thrusting so the magnetic field measured is free of thruster noise and invaluable for calibrating the sensors. In addition, the spacecraft bus is commanded to roll around the Y-axis eight or nine times while the solar arrays remain fixed on the sun to measure the magnetic field created by the solar arrays. By rotating the spacecraft through this full range, the magnetic signature of the arrays as function of the array angle may be determined. The impact of the arrays on the magnetic field may then be removed from future measurements through the ground data processing pipeline.

The most unique phase during cruise is the long optimal coasting period before the Mars gravity assist. Long duration coasting provides an opportunity to perform the equivalent of a periodic maintenance and calibration activity (Magnetometer roll, GRNS calibrations, Imager geometric/photometric calibrations) without the typical timing constraints intended to limit interruptions to mission thrusting. The Magnetometer has an extended quiet period in which to perform zero level and gradiometer calibrations, and additionally the Imager is operated over a period of a few weeks to characterize the instrument’s susceptibility to single event upsets caused by the space environment.

In addition, the Mars flyby provides an opportunity to rehearse aspects of science operations and perform instrument calibrations involving a target body in advance of arriving at Psyche. The GRS HPGe crystal is annealed to restore the performance (i.e. energy resolution) of the detector, after which it performs a calibration/characterization of at least 10 days including Mars closest approach. The Imager performs extended target calibrations with Mars ranging in apparent size from $> 25\%$ of to completely filling the fields of view of both cameras, and images of Mars at different exposures are taken to assess smear and characterize the in-field stray light. Imaging is performed with both Imagers (although not simultaneously, to limit risk to the hardware) to ensure they are fully calibrated and to determine alignment relative to the spacecraft and each other.

5.3 Spacecraft Operations

The following subsections will describe a few of the characteristic aspects of spacecraft operations of note for the Psyche mission. The spacecraft design is described in detail by Oh et al. (2025a). Spacecraft operations typically entail everything that needs to be done by the flight operations team on the ground as well as the functions that execute on the vehicle to accomplish the mission requirements. The ground activities include monitoring the health and performance of the system, trending the performance, responding to anomalies, planning, and building command sequences for future activities. Those ground activities have been designed in conjunction with the capabilities of the spacecraft. There are many functions that are necessary for mission success but aren't unique to the Psyche mission, such as thermal control of the spacecraft and battery charging/discharging. The subsections below are included as they are notable because the Psyche mission uses SEP to travel to a small body in deep space. NASA has flown two previous deep-space SEP missions, Deep Space 1 and Dawn. During these, JPL has developed the expertise needed to properly accommodate SEP's needs and to take best advantage of its capabilities in a deep-space mission, including systems engineering principles in development and operations, mission operations methods, trajectory design, and navigation (Rayman and Williams 2002; Rayman et al. 2007; Rayman 2020).

5.3.1 Solar Electric Propulsion in Operations

The Psyche mission utilizes SEP to provide the necessary thrust to maneuver the spacecraft after launch until arrival at Psyche and to maneuver in orbit around Psyche. The SEP system consists of several components: solar arrays, power electronics, power processing units, SPT-140 Hall effect thrusters, xenon storage and distribution, and dual axis positioning mechanisms (DAPMs). The solar arrays generate electricity from the sun's energy and are gimballed in one axis so that they can remain pointed towards the sun. At 1 AU from the sun, the arrays can produce ~ 21 kW, which goes down to ~ 2.7 kW at Psyche distances from the sun (~ 3.1 AU). The power electronics process electricity from the arrays to regulate the voltage on the power bus, charge the battery, or discharge the battery when the arrays are not producing enough electricity. The power processing units use electricity from the power bus to operate the Hall effect thrusters, which have an input power from about 1 kW to over 4.5 kW. This input power results in about 250 mN of thrust at full power. The SPT-140 thrusters use the electricity to ionize xenon gas and accelerate the ions with a magnetic field which generates the thrust. The xenon gas is stored in seven tanks for a total of 1085 kg at launch and distributed to the thrusters via tubing, valves, and pressure regulators. Finally, the thrusters are mounted to the spacecraft body via the DAPMs with two thrusters

per DAPM: one DAPM on the +X side of the spacecraft and one on the −X axis side. The DAPMs are gimbaled in two axes so that the thrusters can be positioned appropriately. One thruster is operated at a time on the spacecraft.

In order to propel the spacecraft, the thruster in use is positioned such that the thrust produced is aligned with the center of mass of the spacecraft and the flight software adjusts the gimbal position to minimize the torque generated by the thruster. A phenomenon that must be addressed for these thrusters is swirl torque: when the ions are accelerated, they produce a small, but continuous torque about the thruster exhaust direction (Oh et al. 2014, 2025a,b). This torque can't be countered by gimbaling the thruster while in use. The torque is absorbed by the reaction wheels by torquing the wheels in the opposite direction, accelerating the wheels. Since the wheels have a maximum speed, another source of torque must be applied periodically to reverse the torque buildup in the wheels. This is achieved on the Psyche spacecraft by pausing mission thrusting, turning the spacecraft approximately 35° to 40° around the Y-axis of the spacecraft to position a thruster on the opposite side to generate a torque to reverse the reaction wheel speed build up. The swirl torque magnitude is related to the thrust level – it is higher at full thrust and lower at lower thrust levels.

As described above, the thrust produced by the SPT-140 thruster is very small compared to chemical thrusters used for primary propulsion. This is countered by their much greater efficiency. Thruster efficiency, measured as specific impulse, varies with input power, which determines the thrust, and ranges from 1283 s at 58.2 mN (0.91 kW) to 1794 s at 279.3 mN (4.57 kW) (Snyder et al. 2020). Since the thrust is small, the thrusters are fired for thousands of hours (one at a time) to accomplish the interplanetary cruise to the asteroid and the maneuvers in orbit. The trajectory design accounts for all activities incompatible with optimal thrusting, primarily turning the spacecraft HGA to Earth once per week to transmit recorded engineering and instrument cruise data and to receive commands. There are also shorter, more frequent interruptions to unload the accumulated angular momentum in the reaction wheels (primarily from electric propulsion thruster swirl torque). These thrust interruptions result in a thrusting duty cycle of ~85%. This percentage varies depending on the thrust level (higher thrusting levels need more frequent swirl unloads) and other spacecraft activities. The cruise phase is made up of Cruise 1 which is prior to MGA and Cruise 2 which comes after MGA. Cruise 1 requires approximately 15 months of thrusting out of the 31-month period between launch and MGA. Cruise 2 requires approximately 33 months of thrusting out of the 40-month period between MGA and arrival at Psyche. These are approximate numbers and will change as performance models are updated during cruise.

The Mission Design and Navigation team utilizes performance models to design the trajectory—the path the spacecraft will take through the solar system to meet up with Psyche—as well as the required thrusting levels, durations, and direction to achieve that trajectory. The performance models include power available from the solar array, thrust and xenon mass flow rate as a function of Power Processing Unit input power, swirl torque as a function of thrust, other spacecraft electrical loads, and available duty cycle. All these models provide a prediction of the performance for the duration of cruise, either by solar distance or time. An interesting aspect of SEP, or low thrust mission design in general, is that the maneuver or thrusting done now or in the immediate future depends on what the predicted performance will be next week, next month, next year, all the way to the destination. Therefore, a key aspect of these models is ensuring that there is appropriate margin so that it is unlikely to fall short of predicted performance in the future. Additionally, if thrusting is planned to be performed for some portion of time and that does not occur, it is termed 'missed thrust.' The Mission Design and Navigation team designs the trajectory to accommodate a minimum amount of missed thrust and still arrive on time. For the Psyche

mission, the Cruise 1 period is very robust to missed thrust and the spacecraft could miss more than the minimum without impacting arrival. In fact, an anomaly on April 1, 2025, resulted in 77 days of missed thrust in Cruise 1 but had minimal impact on MGA timing and arrival at Psyche. However, Cruise 2 is generally more sensitive to missed thrust with a specific portion the most sensitive, but it is still meeting the requirements and objectives. The trajectory is being designed to build in more robustness, like planned arrival earlier than required, so that if there is another significant missed thrust event, it minimizes the impact to the science mission.

SEP missions are the most efficient, from a propulsion perspective, if they are thrusting at the ‘maximum sustainable thrust.’ Since the spacecraft loads vary as heaters cycle on and off, or the reaction wheels are torqued, the total load is variable. The spacecraft load on average, though, is relatively constant. To thrust at the maximum sustainable thrust level, the total average electrical load would be below, but near, the maximum power produced by the solar arrays such that when heaters cycle on, the increased load is carried by the battery, and when they cycle off, the excess power available is used to recharge the battery. Since there are inefficiencies in the system and battery charging/discharging, the average load needs to be below the maximum by some amount. If all the predictions were perfect, then each segment of thrusting would be designed at this point with no issue. However, there are uncertainties in the models and spacecraft performance (remember that each thrust segment design is unique in the spacecraft attitude relative to the sun, thrust level, and distance to the sun) that result in the thrust design not being at this exact maximum sustainable thrust. To mitigate the risk of discharging the battery too much by designing the thrust level to be closer to the peak power point intentionally, or inadvertently, there is a function in the spacecraft flight software that monitors the battery voltage so that if it goes below a threshold, the electric propulsion thrust level is autonomously reduced to allow more power to be available to recharge the battery. This functionality will continue to reduce the thrust level until the battery gets back above the threshold. Without this functionality, there is a risk that the battery would discharge to the point where fault protection would step in and terminate thrusting. While reducing thrust introduces an error from the designed trajectory, it is not nearly as impactful as missing thrust.

5.3.2 Mission Design Challenges

The Psyche mission requires an exceptionally high post-launch propulsive ΔV , 6.5 km/s, achieved entirely through a low-thrust SEP system. For comparison, NASA’s only other deep space SEP missions, Deep Space 1 and Dawn, accomplished 4.3 km/s and 11.5 km/s, respectively. Of the 6.5 km/s, 5.8 km/s is needed to get the spacecraft to Psyche capture (electric propulsion ΔV). Another 0.2 km/s is necessary for the orbital transfers bringing the mission electric propulsion deterministic ΔV requirement to 6.0 km/s. Additional ΔV of about 0.5 km/s is allocated to accommodate statistical maneuvers, angular momentum management, and missed thrust. The Mars gravity assist provides an additional 0.4 km/s to the spacecraft’s speed; however, the mission design is also using it for a heliocentric orbit plane change, which is very costly if done via a propulsive maneuver. The net effect of the Mars gravity assist is equivalent to a 2.0 km/s heliocentric velocity change.

While the Δv provides an interesting comparison with the far lower values for planetary missions that use chemical propulsion, it is generally considered to be a less significant metric for SEP missions. Optimizing the trajectory design is not only very complex but also very different from that for conventional missions. As indicated in Sect. 5.3.1, electric propulsion thrusting usually occurs for months or years at a time with only brief interruptions, so as to

achieve the proper changes in the trajectory, subject to constraints of the mission plan. Some periods during cruise are more valuable for controlling the trajectory and some are less. Indeed, there are periods when it is optimal not to thrust. The thrust and Isp depend on available electrical power, which depends on heliocentric range (see Sect. 5.3.1). As a result, changes in when thrusting can or does occur (whether because of nominal or off-nominal changes in the mission plan) can lead to significant changes in the ΔV for the same mission objectives.

Low-thrust trajectory design is inherently more complex than traditional chemical propulsion. For instance, chemical propulsion maneuvers typically rely on linear mapping using the State Transition Matrix. In contrast, Psyche's transfers between science orbits required a sequence of maneuvers to guide the spacecraft to the next orbit's entry point—spanning multiple revolutions around the asteroid and requiring time-varying thrust vectors. Designing these low-thrust trajectories demands different capabilities from both the software and operational processes. The Psyche flight team uses Mystic (Whiffen 2006) and Veil (Parcher and Whiffen 2011) for trajectory design and maneuver planning. The continuous operation of SEP adds to the computational complexity, requiring careful optimization throughout the process (Whiffen et al. 2016). Additionally, low-thrust trajectories are highly sensitive to electrical power modeling, as the ability to sustain thrust years into the future influences present-day thrust direction decisions. Accurate power modeling is crucial to ensuring sufficient control authority throughout the mission. Errors in power predictions can compromise the spacecraft's ability to achieve its objectives.

Traditional interplanetary missions using chemical propulsion measure design margins primarily in mass or ΔV . However, low-thrust missions introduce additional constraints, including power availability, and thrusting time. Over its five-year SEP operation, Psyche must carefully manage thrusting time margins. The thrust duty cycle—the ratio of SEP thrusting time to calendar time—is determined by mission planning and must account for forced coasts, including Earth-pointing maneuvers, data downlinks, instrument activities, reaction wheel unloads, and periodic maintenance. Because SEP thrusting must be prioritized whenever possible, low-thrust missions require more integrated planning than chemical propulsion missions. Coordination between mission planners and trajectory designers ensures a robust mission plan with adequate margins for reaching Psyche.

Managing missed-thrust events is critical to mission success, so trajectory design must also account for missed thrust. The Psyche mission design is required to tolerate up to 14 days per year of unplanned coast or a single 12-day interruption during the cruise and approach phases. Pre-launch trajectory validation included margins for power availability, thrust duty cycle, and thruster performance. Throughout the mission, these margins are carefully managed and incrementally released during monthly trajectory redesigns.

The Mission Design and Navigation team routinely analyzes missed-thrust sensitivity, identifying critical mission phases where heightened readiness is required. During these periods, additional ground tracking passes are scheduled to facilitate early anomaly detection.

The Psyche mission employs two types of Thrust Verification passes to monitor SEP performance:

1. **Nominal Thrust Verification Passes:** During these 4-hour sessions, the onboard transmitter communicates via an LGA while the SEP operates at reduced power to accommodate transmitter needs. These passes provide essential spacecraft health data and tracking information for Orbit Determination.
2. **No-Downlink Thrust Verification Passes:** Lasting approximately 2 hours or less, these passes operate on the principle that if the spacecraft is thrusting nominally, no downlink

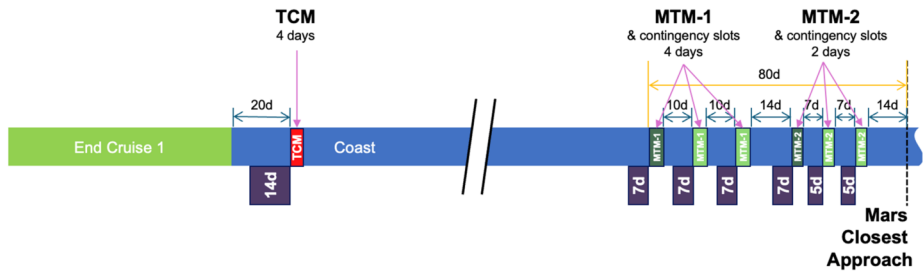


Fig. 35 Thrusting architecture for Mars gravity assist with the Trajectory Correction Maneuver (TCM) and windows for Mars Trim Maneuvers (MTMs). MTM-1 is deterministic and MTM-2 is a statistical correction. Both maneuvers include contingency cases shown in light green. This architecture was designed to achieve the most accurate delivery at Mars closest approach without overloading the flight team’s ability to generate the associated spacecraft command files

signal is received. If an anomaly forces the spacecraft into safe mode, it transmits diagnostic telemetry, which DSN operators report to the flight team. Rapid recovery from safe mode is essential to minimizing missed-thrust durations, a consideration reflected in the team’s flight operations procedures.

By continuously refining operational strategies and maintaining robust trajectory margins, the flight ensures resilience against anomalies while maximizing SEP efficiency to achieve mission success.

5.3.3 Mars Gravity Assist Navigation and Engineering

The Psyche mission requires a gravity assist from Mars to gain ΔV and adjust its heliocentric orbit plane. Monte Carlo simulations, run in the software package Veil, were conducted to develop a safe mission plan, incorporating both deterministic and statistical thrusting strategies for navigating the Mars flyby. In the baseline architecture (Fig. 35), two correction maneuvers follow the end of Cruise 1 to ensure a safe and optimal Mars gravity assist. A Trajectory Correction Maneuver is scheduled 20 days after Cruise 1 thrusting completes, while a Mars Trim Maneuver—part of the MGA architecture—is planned for 80 days before the spacecraft’s closest approach to Mars.

Thrusting during Cruise 1, followed by the Trajectory Correction Maneuver, will target a biased aim point away from Mars to meet NASA’s planetary protection requirements. The Mars Trim Maneuvers will then adjust the aim point to maximize the Mars gravity assist, optimizing the Cruise 2 trajectory for Psyche’s arrival.

5.3.4 Orbit Transfers

An exciting aspect of the Psyche mission is the opportunity to explore a previously unvisited solar system body. However, this also presents challenges, such as the lack of an accurate or detailed gravity model before arrival. To address this, the mission is designed so that knowledge gained in one phase informs the next during orbit operations.

Psyche’s rotation period and axis will be estimated through optical measurements during approach, while GM and low-order gravity field terms will be derived from radiometric tracking data. This information will be used to refine the Orbit A reference orbit and operations plan. Subsequently, updated gravity field and rotation data from Orbit A will guide the design and operations of Orbit B. This iterative process will continue throughout the orbit operations phase.

5.3.5 Time Correlation

The spacecraft avionics keep time by counting seconds, where zero seconds corresponds to 12:00 pm on January 1, 2000 (J2000). This time is referred to as SCLK (spacecraft clock) and is the reference for on board commanding and downlink data time tagging. The time is based on an oscillator in the electronics that may drift as result of temperature changes and age. Therefore, the team correlates the SCLK time to SCET (spacecraft event time, effectively UTC) by processing time correlation data packets from the spacecraft. The correlation activity produces a SCLKSCET file that is used to translate a given SCLK time to SCET. The SCLKSCET file contains one or more rows for each correlation, so it is a continually growing file for the life of the mission. The data in the file reflects a reconstructed time for times in the past, while the last entry is a prediction of the on board SCLK rate into the future.

Because the spacecraft uses SCLK to dispatch commands sequenced to execute at a specific time, the uplink process uses a SCLKSCET file to convert the sequence input in SCET to SCLK with the latest available correlation. The team uses the same SCLKSCET for all sequenced commanding in a given period of time so that conflicts aren't introduced by using different correlations for commands issued near the same SCET time.

Again, since the spacecraft operates in SCLK, all data is downlinked with SCLK time tags. The Ground Data System (GDS) automatically reads the latest SCLKSCET file to add the SCET time tag to the data item. Since the time correlation packets used to correlate the clock are transmitted in the same tracking pass as the other data, this latest data will not have the latest correlation applied. Therefore, to achieve the highest accuracy, the downlink data will need to be reprocessed after the SCLKSCET file is updated.

5.3.6 Data Management

Data collected by the spacecraft, engineering and instrument engineering and science data, is stored in files called Data Products (DPs). Some DPs are periodic that are continually being filled until they reach their 'full' criteria, which could be size or time open. The maximum file size is 4 MB. The other type of DP is a commanded DP, like an image or collection of specific, finite subsystem data. When a data product is created, it is given a priority value based on the criticality or urgency in getting the data to the ground. For example, critical engineering data has the highest priority so that the operations team can get insight into the state of the spacecraft in the event of an anomaly. Additionally, each DP has a tag that indicates whether it has been sent to the ground and is set to 'UNSENT' at the time of generation. During an HGA pass, the transmission of DPs is enabled by sequenced command and the flight software transmits the DPs in priority order, then oldest to newest based on generation time. As the DPs are sent, the tag is updated to 'SENT' and at the end of the pass a command is issued to dump the DP catalog DP. The catalog is used to compare what the spacecraft sent with what the GDS received. If there is any discrepancy, commands are generated to mark any DPs not received as 'UNSENT' and they'll be transmitted again when enabled. The DPs are not autonomously deleted so sequenced commands are generated to delete the DP that are already confirmed received on the ground.

The Psyche spacecraft supports data rates of 10, 40, 100, 250, 1k, 2k, 20k, 60k, 90k, 120k, 150k, 180k, 210k, 240k, 300k, and 360k bits per second (bps). The lower rates are typically used with the LGAs and the higher rates with the HGA. Which rate is used depends on the spacecraft antenna (LGA or HGA), the DSN antenna (34 m or 70 m), DSN antenna elevation, and distance to Earth. The lowest nominal rate with the HGA is 120k bps. During

cruise, a single rate is used for any specific HGA pass that is supportable on a 34-m antenna, even if a 70-m antenna is scheduled so late changes in the DSN schedule would not result in sequence changes. At the asteroid, the data rate needs to be maximized to return the large amount of data collected. Therefore, the ground tools are designed to adjust the data rate during each pass as the DSN antenna elevation changes. For example, the data rate needs to be lower when the antenna is at low elevation because there is more atmospheric attenuation of the signal, so if the pass started at the lowest elevation, the sequence starts with a lower data rate and then progressively increases the rate as the elevation rises up to the maximum elevation, and then back down again as the antenna moves toward the opposite horizon, a technique referred to as a “wedding cake”.

The spacecraft has three LGAs: one on the plus X face of the spacecraft, one on the minus X face, and one on the minus Z face. At the asteroid, these will be used when not in an HGA pass to collect gravity science data through navigation radiometrics. Since the spacecraft will primarily be pointed with $-X$ close to nadir for data acquisition, different LGAs will have a view of Earth at different points in the orbit. The system can be configured to transmit engineering real time data at 10 bps if the signal strength supports it. Otherwise, it will be configured for what is called ‘carrier only’ which does not have any data modulated on the signal. This configuration increases the signal strength received at Earth by a small amount. In both configurations, the DSN will transmit an uplink signal that is received by the spacecraft and used to adjust the downlink frequency to account for the motion of the spacecraft relative to Earth. This is called 2-way Doppler mode and provides the navigation team with data to determine the orbital path around the body. From that, the gravity field knowledge can be refined.

5.3.7 Momentum Management in Orbit Operations

While in orbit around Psyche, the spacecraft will be subject to gravity gradient torque, a torque present when orbiting a non-symmetrical body, which the reaction wheels need to overcome. When the spacecraft is pointing the $-X$ axis towards nadir, the gravity gradient torque is less than when the spacecraft is pointing the HGA to Earth in a fixed orientation. The spacecraft has a cold gas system consisting of thrusters that use pressurized nitrogen gas as a propellant that will be used to unload the reaction wheel momentum at the asteroid. The orbit trajectory is subject to perturbations when the thrusters fire, so these unload events must be accounted for in the trajectory design. Finalizing the operations concept for orbital operations was deferred to post launch and is still in development at the time of this writing.

5.3.8 Autonomous Fault Protection

Fault management is the system designed into the flight software and hardware to make the spacecraft robust to faults which can range from hardware failures to data corruption from solar particles to degraded or unexpected hardware performance. While the team strives to eliminate errors in the command sequences, sometimes they do happen for a range of reasons and the fault management system provides mitigations to those as well. The system consists of system level fault protection with monitors and responses to faults, as well as local fault protection. System level fault protection handles faults where normal operation cannot continue and the response is designed to resolve the fault and then reconfigure the spacecraft into a known, safe state, known as safe mode. There are three different safe modes: ‘Standby-RWA,’ ‘Safe-RWA,’ and ‘Safe-CGS.’ One of the differences between safe modes is whether they configure to use the reaction wheels (RWA) or the cold gas thrusters

Table 14 System Mode configurations for fault protection

System mode	Standby-RWA	Safe-RWA	Safe-CGS
Attitude Knowledge	3-axis inertial	2-axis sun reference	2-axis sun reference
Attitude Actuators	RWA	RWA	CGS
Communication Antenna	HGA	+X LGA	+X LGA
Uplink Rate	1 kbps	7.8125 bps	7.8125 bps
Downlink Rate	120 kbps	10 bps or carrier only	10 bps or carrier only
NS, Magnetometer state	Off	Off	Off
GRS state	DPU on, cryocooler on, HV ramped down	Off	Off

(CGS) for attitude control. The local fault protection operates at the lower level and can resolve the fault without interrupting other operations. Examples of system level monitor and response are the monitors on the electric propulsion system to ensure the thruster parameters are in the appropriate range. If they are not, the electric propulsion system is shut down and the system enters the ‘Standby-RWA’ mode which is the lowest impact safe mode. An example of local fault protection is the safety relay in the thermal subsystem. If a zone gets too hot, the safety relay is simply opened, cutting off heater power to that zone and all other operations continue.

The system modes invoked by fault protection, from least impact to greatest impact, are ‘Standby-RWA’, ‘Safe-RWA’, and ‘Safe-CGS’. The high-level spacecraft configuration is shown in Table 14. In all cases, the autonomous behaviors and sequences are terminated, electric propulsion thrusting is terminated, the Magnetometer, NS, and DSOC are powered off, and GRS high voltage (HV) is ramped down. The GRS DPU and cryocooler, if they were on, are left powered. ‘Standby-RWA’ maintains full pointing control and points the HGA to Earth and configures for 120k bps. This is the preferred response as the high data rate and pointing control enable a faster recovery. The next level of safe mode is ‘Safe-RWA’ which also turns off the star tracker(s) and the GRS DPU and cryocooler, points +X to the sun using the coarse sun sensor, and begins a 1 rev/hr rotation around the +X axis. The spacecraft is configured to use the LGA which requires a 70-m DSN antenna to receive 10 bps for some parts of the mission. This mode is used when the fault indicates that it may not be possible to safely operate in the ‘Standby-RWA’ configuration without ground intervention. The final safe mode is ‘Safe-CGS’ which does everything in ‘Safe-RWA’, plus powers off the reaction wheels and uses the cold gas thrusters for attitude control. The pointing is the same, with +X to the sun. This mode is the most drastic and used for faults where a reaction wheel failed or a deep power fault wherein as many devices as necessary are powered off.

It should be noted that the intent of system level fault protection is to configure the spacecraft into a known, safe configuration with the minimum hardware and software enabled for that state. An exception to that design is for the GRS cryocooler. Since the GRS sensor is less susceptible to radiation degradation when cold, if the cooler is turned off in a science collection phase, it will take time to determine if an anneal is required and then many days to anneal the sensor, reducing the available time to collect the science data. This resulted in the decision to leave the GRS DPU and cryocooler on in ‘Standby-RWA’ as the risk to that mode of leaving these on is small.

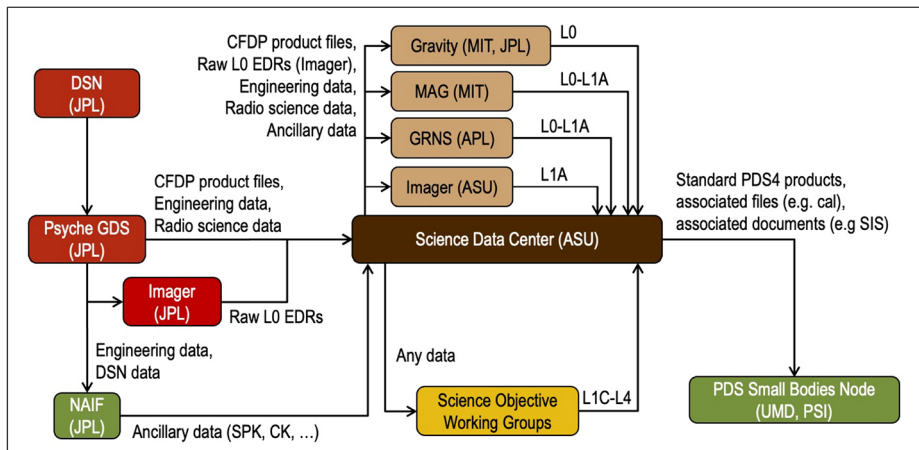


Fig. 36 Psyche science data flows from the GDS at JPL to the SDC, which coordinates all traffic of science data products between the GDS, the Psyche science team, and the PDS

6 Archiving and the Science Data Center

There are many designs that enable the analysis and processing of science data by a mission's science team. Any such design must include:

- Provision of raw data from a mission's ground data system to the science team
- Processing of raw data into standard science data products, usually by instrument-specific science teams
- Review mission-internal data products before archiving
- Archiving data products in the Planetary Data System (PDS), for use by the general science community.

In the case of the Psyche mission, we have chosen to enable this using a Science Data Center (SDC) co-located with Psyche Principal Investigator, at ASU.

The overall flow of these data and products is illustrated in Fig. 36. (In this figure, L0 refers to PDS4 Raw products, L1A refers to PDS4 Calibrated products, and L1B, L1C, L2, L3, and L4 refer to PDS4 Derived products (Planetary Data System Standards Reference 2024).)

The Psyche spacecraft downlinks science data in the form of CCSDS File Delivery Protocol (CFDP) files. While this protocol encapsulates the science data into one or more protocol data units, what we receive at JPL (and the SDC) has been unwrapped by the GDS AMPCS software into science data files (as generated by their respective instrument or the spacecraft telemetry feed, and its associated extended meta-data file that contains pertinent meta-data regarding the DSN receipt of the science file). Those, along with radio science data (originally from the DSN) and engineering data, are transferred to the SDC for pickup by the science team. The Imager team receives the science files (data file and extended meta-data file set) for each observation, in addition to an Object Description Language formatted file containing updated SCLK values, generated by MIPL. In addition, spacecraft engineering data and DSN data, especially trajectory and spacecraft attitude information, are converted to appropriate SPICE kernels (Navigation and Ancillary Information Facility 2024) and transferred to the SDC from the Navigation and Ancillary Information Facility node of the PDS.

The instrument science teams retrieve all necessary input data from the SDC to generate their standard data products at their home institutions. The results are transferred back to the SDC as PDS4 Raw and Calibrated products. Psyche's science team, often as members of Science Objective Working Groups (Sect. 5.1.1) access these data from the SDC for further processing (to PDS4 Derived products) and internal review before archiving. The SDC collects all standard science data products and packages them into the appropriate form for transfer to and archiving in the PDS. The SDC is also responsible for responding to liens identified by the PDS in external review.

6.1 Interface Between Psyche GDS and SDC

The ASU Science Data Center receives products from the Psyche GDS using the JPL developed File Exchange Interface (FEI) service. The GDS will publish instrument specific data products and requested telemetry to instrument specific file types. When those files are received at the SDC, they are ingested, cataloged, and then made available via the SDC webportal.

The ingest and catalog process updates a PostgreSQL database, running at the SDC, that contains a record for every file ingested into the SDC. This allows the SDC to track specific information about each file. Other tables aggregate the PDS4 product and its XML label, which facilitates reporting and archive generation.

The webportal allows team members to search for data products by a variety of attributes, and who can then download files of interest through the webportal (either through a web browser or the webportal ReSTful interface).

6.2 Interface Between SDC and Instrument Teams

6.2.1 Data Flow and Data Types from SDC to Instrument Teams

Instrument science teams receive their input data (as shown in Fig. 36) from the SDC. These include CFDP files of PDS4 Telemetry data, instrument and spacecraft engineering data, DSN-derived data for the Gravity Science team, and ancillary attitude and trajectory data (SPICE kernels). Each instrument operations team receives data via FEI and can also download the same data from the SDC. Once processed into PDS products, these are uploaded to the SDC either using the SDC web interface or a RESTful interface that allows large scale uploads (or downloads). For this paper, different levels of processed data are described in Table 15.

6.2.2 Return Flow of Data Products from Instrument Teams to SDC

The instrument science teams process their data into the products described in Table 16. These products are then transferred back to the SDC. The SDC then packages them into the "collections" and "bundles" described in Table 16 (Planetary Data System Standards Reference 2024). Note that the PDS4 Derived products are often generated by Science Objective Working Groups, using instrument data products picked up from the SDC and delivered back to there.

Table 15 Definitions of processing levels for science data, as used in this paper

NASA	PDS4	Description
Packet data	Telemetry	An encoded byte stream used to transfer data from one or more instruments to temporary storage where the raw instrument data will be extracted. PDS does not archive telemetry data. (These are supplied as CFDP files in the case of the Psyche mission.)
Level 0	Raw	Original data from an instrument. If compression, reformatting, packetization, or other translation has been applied to facilitate data transmission or storage, those processes will be reversed so that the archived data are in a PDS-approved archive format.
Level 1A	Partially Processed	Data that have been processed beyond the raw stage but which have not yet reached calibrated status.
Level 1B	Calibrated	Data converted to physical units, which makes values independent of the instrument.
Level 2	Derived	Results that have been distilled from one or more calibrated data products (for example, maps, gravity or magnetic fields).
Level 3		Supplementary data used to interpret observational data, such as calibration tables or tables of viewing geometry, should also be classified as derived data if not easily matched to one of the other categories.

6.3 Role of SDC in Psyche Team Review and Analysis of Psyche Data

6.3.1 Use of SDC to Access Multiple Data Sets for Combined Higher-Level Processing

Since the SDC is the clearinghouse for Psyche mission data (from instruments and several of the working groups), the SDC will be able to facilitate aggregating and delivering information for higher-level processing. The SDC will provide web mapping services (WMS) capabilities, using OpenSource software, to deliver instrument data (observation footprints, individual map-projected imaging observations, and higher level derived products from the Magnetometer, GRNS, and Imager teams). Use of the WMS capabilities would allow the team creating geologic maps to focus on their mapping task, rather than dealing with data processing and management of individual images or mosaic products generated from the Multispectral Imager. The WMS put to good use by the Lunar Reconnaissance Orbiter Camera team with their large dataset. When regional or global derived products are generated from Magnetometer and GRNS data, these can also be offered via the WMS, allowing tools that can utilize WMS resources, such as ArcPro or QGIS, to overlay multiple datasets and answer questions.

The project will also be investigating the use of tools, such as JMARS or the Small Bodies Mapping Tool (SMBT), in conjunction with the above-mentioned WMS resource, to facilitate Psyche team members' access to mission data.

6.3.2 Pre-Delivery Internal Review of Psyche Standard Data Products

Teams that generate PDS4 products from Psyche data deliver those products to the SDC. At specified intervals (weekly, monthly and prior to deliveries), reports are generated from the SDC product holdings, which can then be sent to the generating team for review. The reports have a basic listing of products (and their attributes: filename, file size, checksum, product type) as well as information as to whether that product has passed PDS4 product validation.

Table 16 The Psyche project's four science investigations archive 11 bundles containing 60 collections

Bundle	Collections	Data volume
Magnetometer	Raw (L0) Time-stamped, raw orthogonal field components	2470.44 GB
	Partially-processed (L1A) Time series of calibrated magnetic field	1897.2 GB
	Calibrated (L1B) Time series of magnetic field (s/c signature removed)	753.3 GB
	Derived (L2) Magnetic field in J2000, etc. coordinates	44.64 GB
	Housekeeping (selected spacecraft housekeeping) data	TBD
	Calibration files	TBD
	Documents	TBD
	Browse	TBD
Magnetometer - derived	Derived (L3) Spherical harmonic coefficients (including dipole moment)	10 kB
	Derived (L3) Magnetic field map	1 GB
	Documents	TBD
	Browse image(s)	TBD
Imager	Raw (L0) Raw binary images (all images for all filters)	1198.8 GB
	Calibrated (L1B) radiance & IoF images (all images for all filters)	2397.5 GB
	Housekeeping (instrument & selected spacecraft) data	3.6 GB
	Calibration files	34.0 GB
	Documents	12 MB
	Browse images	300 GB
Imager - derived	Derived (L2) Global clear filter map at ≤ 20 m/pixel	20.4 GB
	Derived (L2) Global multispectral image maps at ≤ 200 m/pixel (for filters used in sulfide mineral detection) or ≤ 500 m/pixel (for filters used for iron-bearing silicate mineral detection)	163.2 GB
	Derived (L3) Global spectral parameter maps (3 parameters)	5.9 GB
	Documents	12 MB
	Browse images	TBD
Cartography	Derived (L3) Global topographic map (DTM), derived by SPG	2.3 GB
	Derived (L4) Global geologic unit map	3 GB
	Documents	
	Browse	TBD
GRS	Raw (L0) Time series of raw GRS counts	21.4 GB
	Calibrated (L1B) Time series of corrected calibrated gamma-ray spectra	21.4 GB
	Instrument Housekeeping data	1.5 GB
	Calibration files and geometry correction factors	<5 GB
	Documents	<10 MB
	Browse	<10 MB
GRS - derived	Derived (L3) elemental abundance map	<10 MB
	Derived (L3) Fe, Ni, Si, K, S, Al, Ca, Th, U average abundances	
	Documents	TBD
	Browse image(s)	<10 MB

Table 16 (Continued)

Bundle	Collections	Data volume
NS	Raw (L0) Time series of raw NS counts	7.4 GB
	Calibrated (L1B) Time series of corrected calibrated neutron spectra	7.4 GB
	Instrument Housekeeping data	1.1 GB
	Calibration files and geometry correction factors	<500 MB
	Documents	TBD
NS - derived	Derived (L3) elemental abundance map	<10 MB
	Derived (L3) neutron count rates	TBD
	Documents	TBD
	Browse	<10 MB
Radio Science	Raw (L0) Raw Radio Metric Tracking Data (TRK-2-34)	43 GB
	Raw (L0) Media Calibration File (TRK-2-23)	6.7 MB
	Raw (L0) Weather Data File (TRK-2-24)	TBD
	Raw (L0) Spacecraft Mass History File	<1 MB
	Raw (L0) Spacecraft Small-Forces File	67 MB
	Housekeeping (selected spacecraft) data	TBD
	Calibration files	TBD
	Documents	TBD
	Browse	TBD
Radio Science - derived	Derived (L3) Gravity field spherical harmonic coefficients	<1 MB
	Derived (L3) Spacecraft ephemeris (SPK file)	<1 GB
	Derived (L3) Tessellated plates shape model, derived by SPC (DSK)	<1 GB
	Documents	TBD
	Browse	TBD

Teams can review and cross-correlate with their holdings, to ensure the SDC has a complete set of products for a defined time period. This process is especially important prior to PDS4 deliveries, to make sure no issues relating to products are found during bundle/collection validation.

6.4 Creating Collections and Bundles at the SDC

The generation of archive bundles begins several months prior to the actual PDS publish date, to allow ample time for review of products, generation of the archive, and validation of the bundles. As stated in the previous section, the process begins with a review of data holdings, based on the report generated for the upcoming release.

Once the reports have been reviewed, and any issues identified and resolved, the SDC team can begin the process of generating an archive bundle and its collections. This task is performed using an automated process that queries the SDC database, creates the directory structure, and then populates the directories with the appropriate files (new files for delivery). Once the directory has been created and populated, dynamically created files such as manifest, bundle inventory, and collection inventory are generated from the newly created bundle. Once a bundle is successfully created, validation of that bundle is performed using the PDS tool VALIDATE, with the appropriate parameters and inputs. The report that is generated from the invocation is captured, as a record of the successful generation and validation

of the bundle. Validation reports are ingested into the SDC, and available to the Psyche team for review. After validation, the bundle can then be transferred to the Small Bodies Node, for review by the SBN team and eventual integration into any existing released data bundles.

6.5 Advantages of Data Processing and Archiving Design

Data processing is carried out by each instrument team for their respective instrument, in order to convert science data and telemetry into PDS4 data products. As these science data products and PDS4 products are all stored at the SDC, the SDC team has coordinated with the instrument teams, the Data Archive working group, and PDS representatives on data product formats, PDS4 products, and archive bundle structures. The centralization of the PDS4 archiving allows teams to focus on their operations and processing and not be distracted by developing PDS4 expertise to support archiving functionality. ASU has a long and proven history of delivering data products to the PDS as exemplified by three instances: the Data Node for the Lunar Reconnaissance Orbiter Camera archive (Imaging Node), the THEMIS instrument archive (Imaging Node), and the TES instrument archive (Geosciences Node). The SDC expanded upon ASU's foundation and provides a unified interface for the Psyche team to access data products, and a single interface for the SBN to deal with for deliveries. The SDC team works with each of the teams creating products to advise on labels, and collection and bundle structures. The SDC team is responsible for the mission bundle and mission dictionary which the other Psyche archives reference.

7 Summary

Since 2011, the Psyche mission has grown from an idea formulated in a paper to a spacecraft with four science investigations and a successful technology demonstration on its way to explore a metallic world. The primary science objective, to determine whether Psyche is the remnant core of a planetesimal, is accomplished by observations from a series of circular mapping orbits.

The use of SEP enables the spacecraft to traverse interplanetary space and multiple different orbits about a massive, irregularly shaped, previously unexplored solar system object. Although commonly used for Earth-orbiting satellites, SEP is still relatively uncommon for deep space applications. The low-thrust mission design developed by the Dawn team and inherited by Psyche could be replicated to investigate other novel targets in the solar system.

The project developed a strong set of goals, objectives, and requirements during the proposal phase and thereafter avoided adding scope or changing the high-level approach to mapping the asteroid. Despite two changes in launch date and a significant change in arrival conditions, the science implementation plan outlined in the Concept Study Report is remarkably similar to the one detailed in this paper. The modular nature of the science orbits provides the flexibility needed to adjust the science orbit timeline to arrival conditions without changing the science observation strategies. Future surprises from the asteroid, the spacecraft, or the instruments could force changes to these plans, but the architecture provides a stable baseline from which to adjust to new challenges.

Acknowledgements The authors thank the entire Psyche team, composed of dedicated individuals whose work in many disciplines over many years has brought this project from proposal to deep space operations. We would like to specifically thank Klaus-Dieter Matz at DLR for CKVIEW results, Bruce Bills for an earlier version of the Psyche seasons graphic, David Seal for an earlier version of the mission overview graphic, and Laura Ratliff, Ellen Lamel, and Alison Johnson for help with structurally organizing and copyediting

the manuscript. The flight and ground preparation for the DSOC technology demonstration owes a depth of gratitude to NASA HQ STMD/TDM, SOMD/SCaN and SMD/Discovery Program Management for guidance and support.

Author Contribution All authors wrote parts of the paper and/or commented on previous versions of the manuscript.

Funding Information The research was carried out at the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration (80NM0018D0004) and supported by NASA contract NNM16AA09, “Psyche: Journey to a Metal World.” Work with CKVIEW was supported by DLR.

Declarations

Competing Interests The authors have no competing interests to declare. CAP, LTET, and JFB are guest editors of this collection, but were not involved in the peer review of this article.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article’s Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article’s Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

- Badolati E (2007) Gasparis, Annibale de. In: Hockey T, et al (eds) The biographical encyclopedia of astronomers. Springer, New York, p 408. https://doi.org/10.1007/978-0-387-30400-7_502
- Bell JF III, Ravine MA, Caplinger MA, Schaffner JA, Brylow SM, Clark MJ, Peckham DA, Otjens PT, Price GJ, Rowell T, Ravine JW, Laramie JD, Juergens RC, Morgan W, Parker AG, Williams DA, Winhold A, Dibb S, Cisneros E, Walworth M, Zigo H, Warner N, Bates-Tarasewicz H, Amiri N, Polanskey C, Mastrodemos N, Park RS, Alonge NK, Jaumann R, Binzel RP, McCoy TJ, Martin MG, Arthur PA (2025) The Psyche multispectral imager investigation: characterizing the geology, topography, and multispectral properties of a metal-rich world *Space Sci Rev*, 221:47. <https://doi.org/10.1007/s11214-025-01169-3>
- Bell JF, Davis DR, Hartmann WK, Gaffey MJ (1989) Asteroids: the big picture. In: Binzel RP, Gehrels T, Matthews MS (eds) Asteroids II. Univ. Arizona Press, Tucson, pp 921–943
- Belton MJS, Chapman CR, Klaasen KP, Harch AP, Thomas PC, Veverka J, McEwen AS, Pappalardo RT (1996) Galileo’s encounter with 243 Ida: an overview of the imaging experiment. *Icarus* 120:1. <https://doi.org/10.1006/icar.1996.0032>
- Binzel RP, Bus SJ, Xu S, Sunshine J, Burbine TH, Neely AW, Brown RW (1995) Rotationally resolved spectra of asteroid 16 Psyche. *Icarus* 117:443–445
- Binzel RP, Rivkin AS, Bus SJ, Sunshine JM, Burbine TH (2001) MUSES-C target asteroid 1998 SF36: a reddened ordinary chondrite. *Meteorit Planet Sci Suppl* 36:A20-21
- Biswas A, Srinivasan M, Andrews K, Velasco A, Alerstam E, Allmaras J, Wollman E, Meenehan S, Wright M, Rogalin R (2024) Overview of the Deep Space Optical Communications (DSOC) technology demonstration. *Proc SPIE* 13355:133550J
- Borson DM, Robinson BS, Murphy DV, Burianek DA (2014) Overview and results of the Laser Lunar Communication Demonstration. In: Proceedings of the SPIE. 105240U-1-11
- Bowman CDD, Elkins-Tanton LT, Talamante A, Bell JF III, Cisneros E, Cook A, Frieman JD, Gainor D, Hunziker J, Khan S, Lawler CR, Maschino J, McCoy TJ, Nessi K, Oran R, Seal D, Simon A, Singh R, Tolbert CM, Valentine K, Weiss B, Wenkert DD, Williams DA (2023) Mission to Psyche: including undergraduates and the public on the journey to a metal world. *Space Sci Rev* 219:25. <https://doi.org/10.1007/s11214-023-00967-x>

- Buck BR, Allen GD, Duerr EK, McIntosh KA, Moynihan ST, Shukla VN, Wang JD (2020) Photon counting camera for the NASA deep space optical communication demonstration on the Psyche mission. In: Proceedings of the SPIE. 1097809-1-10
- Burks MT, Drury OB, Goldsten JO, Hagler LB, Heffern LE, Hines NR, Kim G-B, Lawrence DJ, Nelson KE, Peplowski PN, Yokley ZW (2020) GeMini: a high-resolution, low-resource, gamma-ray spectrometer for planetary science applications. *Space Sci Rev* 216:115
- Carporzen L, Weiss BP, Elkins-Tanton LT, Shuster DL, Ebel D, Gattacceca (2011) Magnetic evidence for a partially differentiated carbonaceous chondrite parent body. *Proc Natl Acad Sci USA* 108(16):6386–6389. www.pnas.org/lookup/suppl/doi:10.1073/pnas.1017165108/-DCSupplemental
- CCSDS Blue Book (2019) Optical Communications Coding and Synchronization, 142.0-B-1; and Optical Communications Physical Layer 141.0-B-1
- Chapman CR (1976) Asteroids as meteorite parent-bodies: the astronomical perspective. *Geochim Cosmochim Acta* 40:701–719
- Chapman CR, Salisbury JW (1973) Comparisons of meteorite and asteroid spectral reflectivities. *Icarus* 19:507–522
- Chapman CR, Morrison D, Zellner B (1975) Surface properties of asteroids: a synthesis of polarimetry, radiometry, and spectrophotometry. *Icarus* 25:104–130
- Dailey JM, Dinu M, Prego R, Kolchmayer J, Berry M, Monte A, Engelberth J, Malaikal A, Piccirilli A, Le Grange J, Ahrens R, Sochor T, Jacques J, Wright MW (2019) High output power laser transmitter for high-efficiency deep-space optical communications. In: Proceedings of the SPIE. 109100M-1-12
- de Gasparis A (1852) Discovery of a new planet. *Mon Not R Astron Soc* 12:125
- de Soria-Santacruz M, Soriano M, Quintero O, Wong F, Hart S, Kokorowski M, Bone B, Solish B, Trofimov D, Bradford E, Raymond C, Narvaez P, Keys C, Lord P, Ream J, Oran R, Weiss BP, Russell C, Ascrizzi K, Elkins-Tanton L (2020) An approach to magnetic cleanliness for the Psyche mission. In: 2020 IEEE Aerospace Conference, Big Sky, MT. <https://doi.org/10.1109/aero47225.2020.9172801>
- Deutsch L (2014) DSN future: a user perspective. In: DSN 50th anniversary symposium. https://descanso.jpl.nasa.gov/dsn50th/DSN_Symposium_2-20-14/DSN_50th_symposium_full.pdf
- Dibb SD, Asphaug E, Bell JF, Binzel RP, Bottke WF, Cambioni S, Christoph JM, Elkins-Tanton LT, Jaumann R, Lawrence DJ, Oran R, O'Rourke JG, Polanskey C, Weiss BP, Wieczorek M, Williams DA (2024) A post-launch summary of the science of NASA's Psyche mission. *AGU Adv* 5:e2023AV001077. <https://doi.org/10.1029/2023AV001077>
- Dollfus A, Mandeville J-C (1977) The M-type asteroids and the origin of iron meteorites. *Meteoritics* 12:206–207
- Driscoll D, Zellers B, Schomacker J, Carro J, Roberts WT, Macdonald D, Guergian J, Klipstein W (2020) Deep Space Optical Communications (DSOC) beam expander design and engineering. In: Proceedings of the SPIE. 112720H-1-10
- Elkins-Tanton LT, Weiss BP (eds) (2017) *Planetesimals: early differentiation and consequences for planets*. Cambridge University Press, Cambridge
- Elkins-Tanton LT, Weiss BP, Zuber MT (2011) Chondrites as samples of differentiated planetesimals. *Earth Planet Sci Lett* 305:1–10
- Elkins-Tanton LT, Asphaug E, Bell JF III, Bercovici H, Bills B, Binzel RP, Bottke WF, Dibb S, Lawrence DJ, Marchi S, McCoy TJ, Oran R, Park RS, Peplowski PN, Polanskey CA, Prettyman TH, Russell CT, Schaefer L, Weiss BP, Wieczorek MA, Williams DA, Zuber MT (2020) Observations, meteorites, and models: a pre-flight assessment of the composition and formation of (16) Psyche. *J Geophys Res* 125. <https://doi.org/10.1029/2019JE006296>
- Elkins-Tanton LT, Asphaug E, Bell JF III, Bills B, Bottke W, Jun I, Lawrence D, Marchi S, McCoy T, Merayo J, Oran R, Park R, Peplowski P, Prettyman T, Raymond C, Weiss B, Wieczorek M, Zuber M (2022) Distinguishing the origin of asteroid (16) Psyche. *Space Sci Rev* 218:17. <https://doi.org/10.1007/s11214-022-00880-9>
- Evans S, Taber W, Drain T, Smith J, Wu H-C, Guevara M, Sunseri R, Evans J (2018) MONTE: the next generation of mission design and navigation software. *CEAS Space J* 10:79–86. <https://doi.org/10.1007/s12567-017-0171-7>
- Farnocchia D, Fuentes-Munoz O, Park RS, Baer J, Chesley SR (2024) Mass, density, and radius of asteroid (16) Psyche from high-precision astrometry. *Astron J* 168:21. <https://doi.org/10.3847/1538-3881/ad50ca>
- Ferrari F, Tanga P (2022) Interior of top-shaped asteroids with cohesionless surface. *Icarus* 378:114914. <https://doi.org/10.1016/j.icarus.2022.114914>. ISSN 0019-1035
- Gehrels T (1970) Photometry of asteroids. In: Dollfus A (ed) *Surfaces and interiors of planets and satellites*. Academic Press, San Diego, pp 317–375
- Goldsten JO, Rhodes EA, Boynton VW, Feldman WC, Lawrence DJ, Trombka JJ, Smith DM, Evans LG, White J, Madden NW, Berg PC, Murphy GA, Gurnee RS, Strohbehn K, Williams BD, Schaefer ED,

- Monaco CA, Cork CP, Del Eckels J, Miller WO, Burks MT, Hagler LB, Deteresa SJ, Witte MC (2007) The MESSENGER gamma-ray and neutron spectrometer. *Space Sci Rev* 131:339–391. <https://doi.org/10.1007/s11214-007-9262-7>
- Goossens S, Smith DE (2023) Gravity degree-depth relationship using point mass spherical harmonics. *Geophysics J Int* 23:1878. <https://doi.org/10.1093/gji/ggad036>
- Goossens SJ, Smith DE, Zuber MT, Head JW (2025) Shallow crustal density in the lunar south polar region. *Planet Sci J* 6:191. <https://doi.org/10.3847/PSJ/adf05f>
- Hanus J, Viikinkoski M, Marchis F, Durech J, Kaasalainen M, Delbo M, Herald D, Frappa E, Hayamizu T, Kerr S, Preston S, Timerson B, Dunham D, Talbot J (2017) Volumes and bulk densities of forty asteroids from ADAM shape modeling. *Astron Astrophys* 601:A114. <https://doi.org/10.1051/0004-6361/201629956>
- Hardersen PS, Gaffey MJ, Abell PA (2005) Near-IR spectral evidence for the presence of iron-poor orthopyroxenes on the surfaces of six M-type asteroids. *Icarus* 175:141
- Hurley K, et al (2010) The Third Interplanetary Network. Paper presented at Conference of the Gamma Ray Bursts, Annapolis, MD, Nov 01–04
- Jaumann R, Bell JF III, Polanskey CA, Raymond CA, Aspaugh E, Bercovici D, Bills BR, Binzel R, Bottke W, Christoph JM, Marchi S, Neesemann A, Otto K, Park RS, Preusker F, Roatsch T, Williams DA, Wiczorek MA, Zuber MT (2022) The Psyche topography and geomorphology investigation. *Space Sci Rev* 218:7. <https://doi.org/10.1007/s11214-022-00874-7>
- Kaasalainen M, Torppa J, Piironen J (2002) Models of twenty asteroids from photometric data. *Icarus* 159:369–395
- Konopliv AS, Asmar SW, Park RS, Bills BG, Centinello F, Chamberlin AB, Ermakov A, Gaskell RW, Rambaux N, Raymond CA, Russell CT, Smith DE, Tricarico P, Zuber MT (2014) The Vesta gravity field, spin pole and rotation period, landmark positions, and ephemeris from the Dawn tracking and optical data. *Icarus* 240:103–117. <https://doi.org/10.1016/j.icarus.2013.09.005>
- Lawrence DJ, Goldsten JO, Peplowski PN, Prettyman TH, Yokley ZW, Elkins-Tanton LT (2020) Preparing to measure (16) Psyche's elemental composition: interpretation of Psyche neutron measurements. In: 51st Lunar and Planetary Science Conference. Abstract #1973
- Lawrence DJ, Goldsten JO, Peplowski PN, Yokley ZW, Beck AW, Burks M, Elkins-Tanton LT, Jun I, McCoy T, Prettyman TH (2021) Planetary Neutron Spectroscopy for Metal-rich Compositions: Development of Analysis Framework for Measurements at the Asteroid (16) Psyche. ESS Open Archive. <https://doi.org/10.1002/essoar.10506738.1>
- Lawrence DJ, Goldsten JO, Peplowski PN, Burks MT, Cheng S, Cully MJ, Effron JM, Elkins-Tanton LT, Espiritu RC, Fix SG, Graziano MB, Hoffer EM, Jun I, Kim G-B, Hines NR, LeBlanc MT, Livingstone EM, Marcotte KM, McCoy TJ, Polanskey CA, Sreekantamurthy M, Yokley ZW (2025) The Psyche gamma-ray and neutron spectrometer. *Space Sci Rev* 221:78. <https://doi.org/10.1007/s11214-025-01201-6>
- Leinweber HK, Russell CT, Torkar K, Zhang TL, Angelopoulos V (2008) An advanced approach to finding magnetometer zero levels in the interplanetary magnetic field. *Meas Sci Technol* 19(5)
- Magri C, Nolan MC, Ostro SJ, Giorgini JD (2007) A radar survey of main-belt asteroids: Arecibo observations of 55 objects during 1999–2003. *Icarus* 186:126–151
- Marchi S, Asphaug E, Bell JF III, Bottke WF, Jaumann R, Park RS, Polanskey CA, Prettyman TH, Williams DA, Binzel R, Oran R, Weiss B, Russell CT (2022) Determining the relative cratering ages of regions of Psyche's surface. *Space Sci Rev* 218:24. <https://doi.org/10.1007/s11214-022-00891-6>
- Matz K-D, Roatsch T (2018) CKVIEW – a multi-mission planning tool. In: SpaceOps conference, p AIAA 2018-2392. <https://doi.org/10.2514/6.2018-2392>
- McCord TB, Gaffey MJ (1974) Asteroids: surface composition from reflection spectroscopy. *Science* 186:352–355
- McCoy TJ, Dibb SD, Peplowski PN, Maurel C, Bercovici HL, Corrigan CM, Bell JF III, Weiss BP, Lawrence DJ, Wenkert DD, Prettyman TH, Elkins-Tanton LT (2022) Deciphering redox state for a metal-rich world. *Space Sci Rev* 218:6. <https://doi.org/10.1007/s11214-022-00872-9>
- McFadden LA, Skillman DR, Memarsadeghi N, Li J-Y, Joy SP, Polanskey CA, Rayman MD, Sykes MV, Tricarico P, Palmer E, O'Brien DP, Mottola S, Carsenty U, Mutchler M, McLean B, Schröder SE, Mas-trodes N, Schiff C, Keller HU, Nathues A, Gutiérrez-Marques P, Raymond CA, Russell CT (2015) Vesta's missing moons: comprehensive search for natural satellites of Vesta by the Dawn spacecraft. *Icarus* 257:207–216
- McFadden L, Skillman DR, Memarsadeghi N, Carsenty U, Schröder SE, Li J-Y, Mottola S, Mutchler M, McLean B, Joy SP, Polanskey CA, Rayman MD, Fieseler PD, Sykes MV, Nathues A, Gutiérrez-Marques P, Keller HU, Raymond CA, Russell CT (2018) Dawn mission's search for satellites of Ceres: intact protoplanets don't have satellites. *Icarus* 316:191–204
- Merayo JMG, Jørgensen JL, Friis-Christensen E, Brauer P, Primdahl F, Jørgensen PS, Allin TH, Denver T (2008) The swarm magnetometry package. In: Sandau R, Röser H-P, Valenzuela A (eds) *Small satellites for Earth observation*. Springer, Berlin, pp 143–151. https://doi.org/10.1007/978-1-4020-6943-7_13

- Moision B, Hamkins J (2003) Deep-space, optical communications downlink budget: modulation and coding. In: IPN progress report 42-154. https://ipnpr.jpl.nasa.gov/progress_report/42-154/154K.pdf
- Morrison D (1974) Radiometric diameters and albedos of 40 asteroids. *Astrophys J* 194:203–212
- Nakamura T, Noguchi T, Tanaka M, Zolensky ME, Kimura M, Tsuchiyama A, Nakato A, Ogami T, Ishida H, Uesugi M, Yada T, Shirai K, Fujimura A, Okazaki R, Sandford SA, Ishibashi Y, Abe M, Okada T, Ueno M, Mukai T, Yoshikawa M, Kawaguchi J (2011) Itokawa dust particles: a direct link between S-type asteroids and ordinary chondrites. *Science* 333(6046):1113–1116. <https://doi.org/10.1126/science.1207758>. PMID: 21868667
- National Academies of Sciences, Engineering, and Medicine (2011) Vision and voyages for planetary science in the decade 2013–2022. The National Academies Press, Washington, 398 pp. <https://doi.org/10.17226/13117>
- National Academies of Sciences, Engineering, and Medicine (2022) Origins, worlds, and life: a decadal strategy for planetary science and astrobiology 2023–2032. The National Academies Press, Washington, 736 pp. <https://doi.org/10.17226/26522>
- Navigation and Ancillary Information Facility (2024). <https://naif.jpl.nasa.gov/naif/sitemap.html>
- Ockert-Bell ME, Clark BE, Shepard MK, Isaacs RA, Cloutis EA, Fornasier S, Bus SJ (2010) The composition of M-type asteroids: synthesis of spectroscopic and radar observations. *Icarus* 210:674–692
- Oh D, Collins S, Randolph T, Vanelli C, Tilley S (2014) Feasibility of All-Electric Three Axis Momentum Management for Deep Space Small Body Rendezvous. AIAA Paper 2014-3906, July 2014. <https://doi.org/10.2514/6.2014-3906>
- Oh DY, Collins S, Drain T, Hart W, Imken T, Larson K, Marsh D, Muthulingham D, Snyder J, Trofimov D, Elkins-Tanton L, Johnson I, Lord P, Pirkel Z (2019) Development of the Psyche mission for NASA's discovery program. In: 36th international electric propulsion conference, Vienna, Austria. IEPC-2019-192
- Oh D, Collins S, Goebel D, Hart W, Imken T, Larson K, Maxwell J, Martin M, Polanskey C, Snyder J, Weise T, Elkins-Tanton L, Lord P (2022) Preparation for launch of the Psyche spacecraft for NASA's discovery program. In: 37th international electric propulsion conference, Boston, MA. IEPC-2022-452
- Oh D, Alonge N, Bairstow S, Biswas A, Chesko C, Goebel D, Imken T, Na-Nakornpanom A, Nicholas A, Snyder JS, Johnson I, Linggi P, Lord P, Malone S, Pirkel Z, Ream J, Weiss B, Lawrence D, Peplowski P, Goldsten J, Burks M, Bell JF (2025b) Launch and Initial Checkout of the Psyche Spacecraft. AIAA J Spacec Rockets. In press
- Oh DY, Collins S, Drain T, Goebel D, Hart W, Imken T, Larson K, Maxwell J, Marsh D, Martin M, Muthulingam D, Snyder JS, Trofimov D, Weise T (2025a) Design of the Psyche Spacecraft for NASA's Discovery Program. *Space Sci Rev* 221
- Ortega CA, Polanskey CA, Llopis M, Lawler CR, Rosemurgy PC, Alonge EK, Elkins-Tanton LT, Dailis ML (2021) Psyche science planning with the science opportunity analyzer. In: 16th international conference on space operations. Virtual Edition, 3 - 5 May 2021
- Ostro SJ, Campbell DB, Shapiro II (1985) Mainbelt asteroids: dual-polarization radar observations. *Science* 229:442–446
- Paper Elemental Blog (2021) Astropost: Astronomical Objects, Their Symbols, and Their Namesakes. <http://paperelemental.blogspot.com/2021/07/astrosymbols.html>
- Parcher DW, Whiffen GJ (2011) Dawn statistical maneuver design for Vesta operations. In: AIAA/AAS astrodynamics specialist conference, New Orleans, Louisiana. AAS 2011-180
- Park RS, Konopliv AS, Bills BG, Rambaux N, Castillo-Rogez JC, Raymond CA, Vaughan AT, Ermakov AI, Zuber MT, Fu RR, Topley MJ, Russell CT, Nathues A, Preusker F (2016) A partially differentiated interior for (1) Ceres deduced from its gravity field and shape. *Nature* 537:515–517. <https://doi.org/10.1038/nature18955>
- Park RS, Vaughan AT, Konopliv AS, Ermakov AI, Mastrodemos N, Castillo-Rogez JC, Joy SP, Nathues A, Polanskey CA, Rayman MD, Riedel JE, Raymond CA, Russell CT, Zuber MT (2019) High-resolution shape model of Ceres from stereophotoclinometry using Dawn imaging data. *Icarus* 319:812–827. <https://doi.org/10.1016/j.icarus.2018.10.024>
- Park RS, Konopliv AS, Ermakov AI, Castillo-Rogez JC, Fu RR, Hughson KHG, Prettyman TH, Raymond CA, Scully JEC, Sizemore HG, Sori MM, Vaughan AT, Mitri G, Schmidt BE, Russell CT (2020) Evidence of non-uniform crust of Ceres from Dawn's high-resolution gravity data. *Nat Astron* 4:748–755. <https://doi.org/10.1038/s41550-020-1019-1>
- Pieters CM, Noble SK (2016) Space weathering on airless bodies. *J Geophys Res* 121:1865–1884. <https://doi.org/10.1002/2016JE005128>
- Planetary Data System Standards Reference (2024). <https://pds.nasa.gov/datastandards/documents/sr/>
- Polanskey CA, Joy SP, Raymond CA (2011) Dawn science planning, operations and archiving. In: Russell C, Raymond C (eds) The Dawn mission to minor planets 4 Vesta and 1 Ceres. Springer, New York. https://doi.org/10.1007/978-1-4614-4903-4_17

- Polanskey CA, Joy SP, Raymond CA (2012) Efficacy of the Dawn Vesta science plan. In: AIAA 2012-1275915. SpaceOps 2012 Conference. June 2012. <https://doi.org/10.2514/6.2012-1275915>
- Polanskey CA, Joy SP, Raymond CA, Rayman MD (2014) Architecting the Ceres science plan. In: AIAA 2014-1720. SpaceOps 2014 Conference. May 2014. <https://doi.org/10.2514/6.2014-1720>
- Polanskey CA, Joy SP, Raymond CA, Rayman MD (2016) Dawn Ceres mission: science operations performance. In: AIAA 2016-2442. SpaceOps 2016 Conference. <https://doi.org/10.2514/6.2016-2442>
- Polanskey CA, Marsh DM, Moore R, Park RS, de Soria-Santacruz Pich M, Wenkert D, Elkins-Tanton LT, Williams DA, Jaumann R, Lawrence DJ, Russell CT (2018) Psyche science operations concept: maximize reuse to minimize risk. In: AIAA 2018-2703. 2018 SpaceOps Conference. May 2018. <https://doi.org/10.2514/6.2018-2703>
- Prettyman TH, Mittlefehldt DW, Yamashita N, Lawrence DJ, Beck AW, Feldman WC, McCoy TJ, McSween HY, Toplis MJ, Titus TN, Tricarico P, Reedy RC, Hendricks JS, Forni O, Le Corre L, Li J-Y, Mizzon H, Reddy V, Raymond CA, Russell CT (2012) Elemental mapping by Dawn reveals exogenic H in Vesta's regolith. *Science* 338:242–246. <https://doi.org/10.1126/science.1225354>
- Prettyman TH, Yamashita N, Toplis MJ, McSween HY, Schorghofer N, Marchi S, Feldman WC, Castillo-Rogez J, Forni O, Lawrence DJ, Ammannito E, Ehlmann BL, Sizemore HG, Joy SP, Polanskey CA, Rayman MD, Raymond CA, Russell CT (2017) Extensive water ice within Ceres' aqueously altered regolith: evidence from nuclear spectroscopy. *Science* 355:55–59. <https://doi.org/10.1126/science.aah6765>
- Prettyman TH, Yamashita N, Ammannito E, Ehlmann BL, McSween HY, Mittlefehldt DW, Marchi S, Schorghofer N, Toplis MJ, Li J-Y, Pieters CM, Castillo-Rogez JC, Raymond CA, Russell CT (2019) Elemental composition and mineralogy of Vesta and Ceres: distribution and origins of hydrogen-bearing species. *Icarus* 318:42–55. <https://doi.org/10.1016/j.icarus.2018.04.032>
- Prettyman TH, Mittlefehldt DW, Asphaug EI, Binzel RP, Courville SW, Jun I, Lawrence DJ, Marchi S, Merayo JMG, McCoy TJ, Weiss BP, Elkins-Tanton LT (2025) The Psyche Light Elements Investigation. *Space Sci Rev* 221
- Prindahl F (1979) The fluxgate magnetometer. *J Phys E, Sci Instrum* 12:241–253
- Pugh M, Kuperman I, Aguirre F, Mojaradi H, Spurgers C, Kobayashi M, Sartorius E, Jedrey T (2017) The universal space transponder: a next generation software defined radio. In: IEEE Aerospace Conference. 978-1-5090-1613-6/17
- Rayman MD (2020) Lessons to the Dawn mission to Ceres and Vesta. *Acta Astronaut* 176:233–237. <https://doi.org/10.1016/j.actaastro.2020.06.023>
- Rayman MD, Williams SN (2002) Design of the first interplanetary electric propulsion mission. *J Spacecr Rockets* 39:4. <https://doi.org/10.2514/2.3848>
- Rayman MD, Fraschetti TC, Raymond CA, Russell CT (2007) Coupling of system resource margins through the use of electric propulsion: implications in preparing for the Dawn mission to Ceres and Vesta. *Acta Astronaut* 60:930–938. <https://doi.org/10.1016/j.actaastro.2006.11.012>
- Raymond CA, Jaumann R, Nathues A, Sierks H, Roatsch T, Preusker F, Scholten F, Gaskell RW, Jorda L, Keller HU, Zuber MT, Smith DE, Mastrodemos N, Mottola S (2011) The Dawn topography investigation. *Space Sci Rev* 163:487–510. <https://doi.org/10.1007/s11214-011-9863-z>
- Ream JB, Weiss BP, Oran R, Raymond CA, Polanskey CA, Wenkert D, Elkins-Tanton LT, Hart RA, Russell CT, Merayo JMG (2022) Magnetic gradiometry using frequency-domain filtering. *Meas Sci Technol* 33:1
- Rogalin R, Grigorian E, Cho D, Tsao P, Richard N, Cheng M, Eckernach J, Okino C, Tkacenko A, Towfic Z, Xie H, Srinivasan M (2021) A high dynamic-range photon-counting receiver for deep space optical communication. In: Proceedings of the SPIE. 117680J-1-16
- Russel CT, Raymond CA, Coradini A, McSween HY, Zuber MT, Nathues A, De Sanctis MC, Jaumann Kono-pliv RAS, Preusker F, Asmar SW, Park RS, Gaskell R, Keller HU, Mottola S, Roatsch T, Scully JEC, Smith DE, Tricarico P, Toplis MJ, Christensen UR, Feldman WC, Lawrence DJ, McCoy TJ, Prettyman TH, Reedy RC, Sykes ME, Titus TN (2012) Dawn at Vesta: testing the protoplanetary paradigm. *Science* 336:684
- Russell CT, Raymond CA, Ammannito E, Buczkowski DL, De Sanctis MC, Hiesinger H, Jaumann R, Kono-pliv AS, McSween HY, Nathues A, Park RS, Pieters CM, Prettyman TH, McCord TB, McFadden LA, Mottola S, Zuber MT, Joy SP, Polanskey C, Rayman MD, Castillo-Rogez JC, Chi PJ, Combe JP, Ermakov A, Fu RR, Hoffmann M, Jia YD, King SD, Lawrence DJ, Li J-Y, Marchi S, Preusker F, Roatsch T, Ruesch O, Schenk P, Villarreal MN, Yamashita N (2016) Dawn arrives at Ceres: exploration of a small, volatile-rich world. *Science* 353(6303):1008–1010
- Sanchez JA, Reddy V, Shepard MK, Cristina T, Cloutis EA, Takir D, Conrad A, Kiddell C, Applin D (2017) Detection of rotational spectral variation on the M-type asteroid (16) Psyche. *Astron J* 153:29
- Shepard MK, Clark BE, Nolan MC, Howell ES, Magri C, Giorgini JD, Benner LAM, Ostro SJ, Harris AW, Warner B, Pray D, Pravac P, Fauerbach M, Bennett T, Klotz A, Behrend R, Correia H, Coloma J, Casulli S, Rivkin A (2008) A radar survey of M- and X-class asteroids. *Icarus* 195:184

- Shepard MK, Richardson J, Taylor PA, Rodriguez-Ford LA, Conrad A, de Pater I, Adamkovics M, de Kleer K, Males JR, Morzinski KM, Close LM, Kaasalainen M, Viikinkoski M, Timerson B, Reddy V, Magri C, Nolan MC, Howell ES, Benner LAM, Giorgini JD, Warner BD, Harris AW (2017) Radar observations and shape model of asteroid 16 Psyche. *Icarus* 281:388–403
- Shepard MK, de Kleer K, Cambioni S, Taylor PA, Virkki AK, Rivera-Valentin EG, Rodriguez Sanchez-Vahamonde C, Fernanda Zambrano-Marin L, Magri C, Dunham D, Moore J, Camarca M (2021) Asteroid 16 Psyche: shape, features, and global map. *Planet Sci J* 2:125
- Sherwood B, McCleese D (2013) JPL innovation foundry. *Acta Astronaut* 89:236–247. <https://doi.org/10.1016/j.actaastro.2013.04.020>
- Snyder JS, Lenhuito G, Frieman JD, Haag TW, Mackey JA (2020) Effects of background pressure on SPT-140 Hall thruster performance. *J Propuls Power* 36:5. <https://doi.org/10.2514/1.B37702>. <http://arc.aiaa.org>
- Srinivasan M, Rogalin R, Lay N, Shaw M, Tkacenko A (2017) Downlink receiver algorithms for deep space optical communications. In: *Proceedings of the SPIE*. 100960A-1-11
- Takir D, Reddy V, Sanchez J, Shepard M, Emery J (2017) Detection of water and/or hydroxyl on asteroid (16) Psyche. *Astron J* 153:31
- Tholen DJ (1984) Asteroid Taxonomy from Cluster Analysis of Photometry. Ph.D. dissertation, Univ. Arizona, Tucson, 150 pp.
- Veverka J, Belton M, Klaasen K, Chapman C (1994) Galileo's encounter with 951 Gaspra: overview. *Icarus* 107(1):2–17. <https://doi.org/10.1006/icar.1994.1002>
- Veverka J, Robinson M, Thomas P, Murchie S, Bell JF III, Izenberg N, Chapman C, Harch A, Bell M, Yeomans DK, et al (2000) NEAR at Eros: imaging and spectral results. *Science* 289:2088–2097. <https://doi.org/10.1126/science.289.5487.2088>
- Viikinkoski M, Vernazza P, Hanuš J, Le Coroller H, Tazhenova K, Carry B, Marsset M, Drouard A, Marchis F, Fetic R, Fusco T, Ďurech J, Birlan M, Berthier J, Bartczak P, Dumas C, Castillo-Rogez J, Cipriani F, Colas F, Ferrais M, Grice J, Jehin E, Jorda L, Kaasalainen M, Kryszczyńska A, Lamy P, Marciniak A, Michalowski T, Michel P, Pajuelo M, Podlowska-Gac E, Santana-Ros T, Tanga P, Vachier F, Vigan A, Warner B, Witasse O, Yang B (2018) (16) Psyche: a mesosiderite-like asteroid? *Astron Astrophys* 619:L3
- Walsh KJ, et al (2022) Near-zero cohesion and loose packing of Bennu's near subsurface revealed by spacecraft contact. *Sci Adv* 8:eabm6229. <https://doi.org/10.1126/sciadv.abm6229>
- Weiss BP, Elkins-Tanton LT (2013) Differentiated planetesimals and the parent bodies of chondrites. *Annu Rev Earth Planet Sci* 41:529–560. <https://doi.org/10.1146/annurev-earth-040610-133520>
- Weiss BP, Merayo JMG, Ream JB, Oran R, Brauer P, Cochrane CJ, Cloutier K, Elkins-Tanton LT, Jørgensen JL, Maurel C, Park RS, Polanskey CA, de Soria Santacruz-Pich M, Raymond CA, Russell CT, Wenkert D, Wieczorek MA, Zuber MT (2023) The Psyche magnetometry investigation. *Space Sci Rev* 219:22. <https://doi.org/10.1007/s11214-023-00965-z>
- Whiffen GJ (2006) Mystic: implementation of the static dynamic optimal control algorithm for high-fidelity, low-thrust trajectory design. In: *AIAA/AAS astrodynamics specialist conference*, Keystone, Colorado. AIAA 2006-6741
- Whiffen GJ, et al (2016) Dawn's mission at Ceres. In: *AIAA/AAS astrodynamics specialist conference*, Long Beach, California. AIAA 2016-5375
- Winhold A, Bell JF III, Caplinger MA, Brylow S, Ravine M, Kubacki T, Dahl-Hansen R, Walworth M, Dibb S, Zhao A, Rodea Y, Cisneros E, Williams D, Warner N, Bates-Tarasewicz H, Monacelli B, Mastrodomos N (2025) Pre-flight and in-flight calibration of the Psyche Multispectral Imagers. *Earth Space Sci*. To be submitted
- Wollman EE, Allmaras JP, Beyer AD, Korzh B, Frasca S, Runyan MC, Lautaro N, Farr WH, Marsili F, Briggs RM, Miles GJ, Shaw MD (2024) SNSPD-based detector system for NASA's deep space optical communications project. *Opt Express* 32(27):48185–48198
- Yokley ZW, Peplowski PN, Goldsten JO, Lawrence DJ, Jun I, Prettyman TH, Elkins-Tanton LT (2019) Neutron spectroscopy on a metal world: tailoring the Psyche neutron spectrometer. In: *50th lunar and planetary science conf. Abstract #1295*
- Zellner B, Gradie J (1976) Polarimetric evidence for the albedos and compositions of 94 asteroids. *Astron J* 81:262–280
- Zuber MT, Smith DE, Lehman DH, Hoffman TL, Asmar SW, Watkins MM (2013) Gravity Recovery and Interior Laboratory (GRAIL): mapping the lunar interior from crust to core. *Space Sci Rev* 178:3–24. <https://doi.org/10.1007/s11214-012-9952-7>
- Zuber MT, Park RS, Elkins-Tanton LT, Bell JF III, Bruvold KN, Bercovici D, Bills BR, Binzel RP, Jaumann R, Marchi S, Polanskey CA, Raymond CA, Roatsch T, Wang CC, Weiss BP, Wenkert D, Wieczorek MA (2022) The Psyche gravity investigation. *Space Sci Rev* 218:57. <https://doi.org/10.1007/s11214-0>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Carol A. Polanskey¹  · Linda T. Elkins-Tanton^{2,3} · James F. Bell III² · Eleanor K. Alonge¹ · Sarah H. Bairstow¹ · Richard P. Binzel⁴ · Abhijit Biswas¹ · Luke Bury^{1,5} · Ernest Cisneros² · Dongsuk Han¹ · Insoo Jun¹ · William M. Klipstein¹ · David J. Lawrence⁶ · Timothy J. McCoy⁷ · Nickolaos Mastrodemos¹ · José M.G. Merayo⁸ · Sarah K. Noble⁹ · David Y. Oh¹ · Rona Oran⁴ · Carolyn A. Ortega¹ · Ryan S. Park¹ · Patrick N. Peplowski⁶ · Thomas H. Prettyman¹⁰ · Marc D. Rayman¹ · Jodie B. Ream⁴ · Thomas Roatsch¹¹ · Timothy M. Weise¹ · Daniel D. Wenkert¹ · Benjamin P. Weiss⁴ · Maria T. Zuber⁴

✉ C.A. Polanskey
Carol.A.Polanskey@jpl.nasa.gov

¹ Jet Propulsion Laboratory/California Institute of Technology, Pasadena, CA, USA

² Arizona State Univ., School of Earth & Space Exploration, Tempe, AZ, USA

³ Present address: Space Sciences Laboratory, University of California, Berkeley, Berkeley, CA, USA

⁴ Massachusetts Institute of Technology, Cambridge, MA, USA

⁵ Present address: Advanced Space, LLC, Westminster, CO, USA

⁶ Johns Hopkins Applied Physics Laboratory, Laurel, MD, USA

⁷ Smithsonian Institution, Washington, DC, USA

⁸ DTU Space, Technical University of Denmark, Lyngby, Denmark

⁹ NASA Headquarters, Washington, DC, USA

¹⁰ Planetary Science Institute, Tucson, AZ, USA

¹¹ German Aerospace Center (DLR), Berlin, Germany